



第7章

注意力机制与Transformer

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

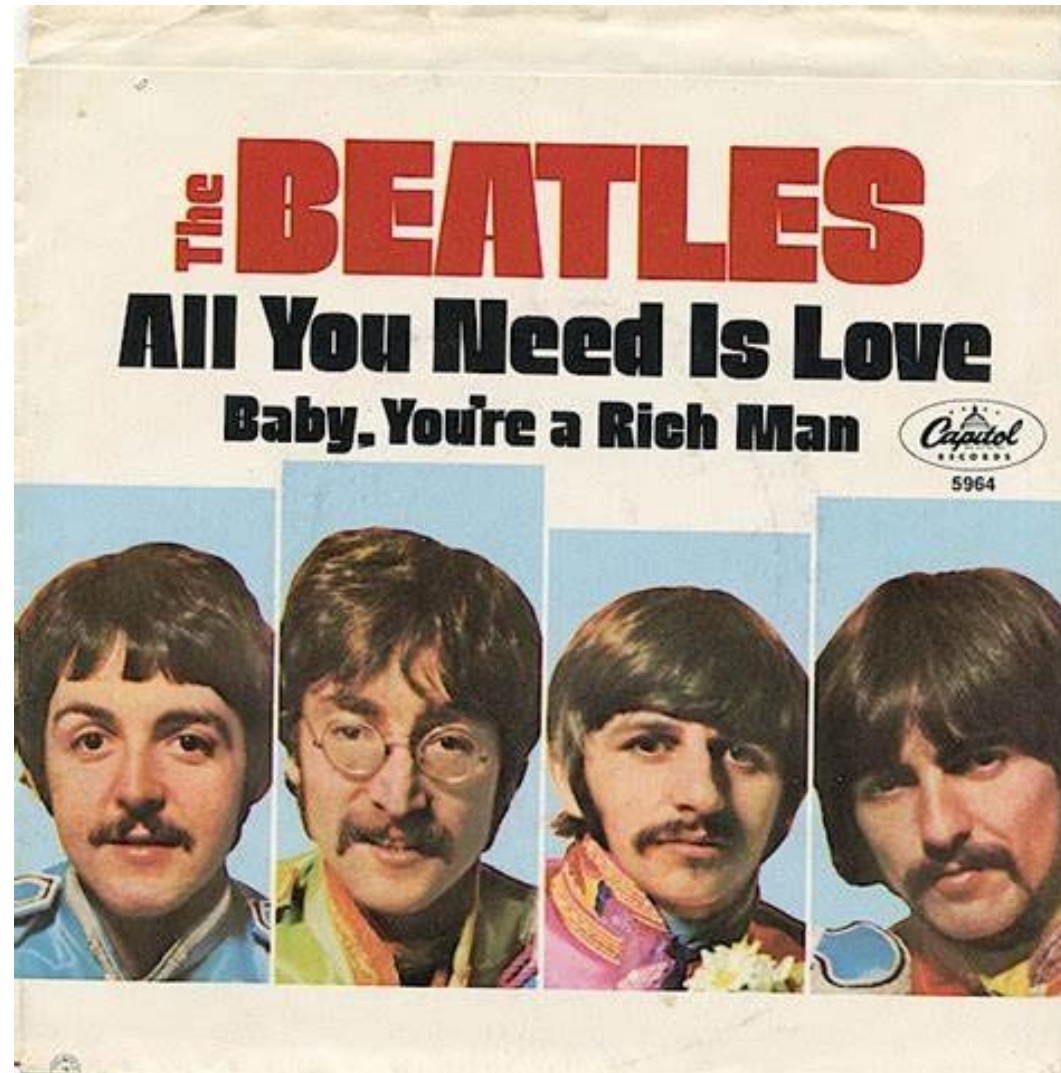
Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions





Transformer



 OpenAI  Microsoft

INTRODUCING



GPT-4



人类的注意力



大脑中的注意力

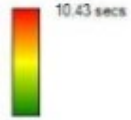
- 人脑通过**注意力**来解决**信息超载**问题。

- 人脑每个时刻接收的外界输入信息非常多，包括来源于视觉、听觉、触觉的各种各样的信息。
- 单就视觉来说，眼睛每秒钟都会发送千万比特的信息给视觉神经系统。



大脑中的注意力

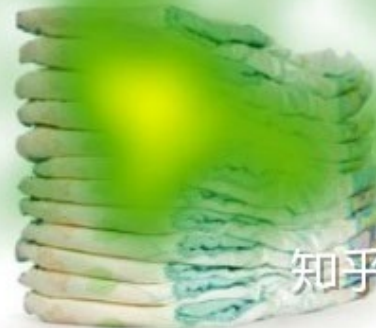
Participant filter: All



Engineered for the
most sensitive skin.

For extra protection, add the natural oils and moisture
from the baby's skin.

It's the unique high-absorbency natural-blend cotton
padding that provides cotton-soft, extra thick, gel-free protection
for baby's sensitive skin. The chlorine-free materials and
polymer is non-toxic and non-irritating. Clinically
proven, it's a pediatrician recommended for babies with allergies
and sensitive skin.



TM

知乎 @Microstrong

大脑中的注意力

- **聚焦式注意力 (Focus Attention)**

- 自上而下的有意识的注意力
- 当一个人在吵闹的鸡尾酒会上和朋友聊天时，尽管周围噪音干扰很多，他还是可以听到朋友的谈话内容，而忽略其他人的声音。

- **基于显著性的注意力 (Saliency-Based Attention)**

- 自下而上的无意识的注意力
- 如果未注意到的背景声中有重要的词（比如他的名字），会马上注意到。



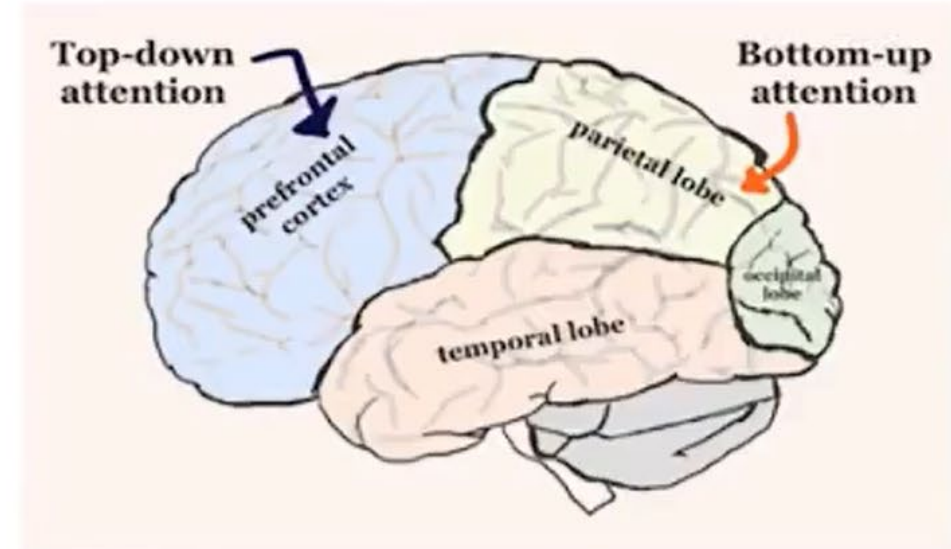
FROM ATTENTION TO CONSCIOUSNESS

C-word not taboo anymore in cognitive neuroscience

Global Workspace Theory

(Baars 1988++, Dehaene 2003++)

- Bottleneck of conscious processing
 - **WHY A BOTTLENECK?**
- Selected item is broadcast, stored in short-term memory, conditions perception and action
- System 2-like sequential processing, conscious reasoning & planning & imagination
- Can only run 1 simulation at a time, unlike a movie, only few abstract concepts involved at each step



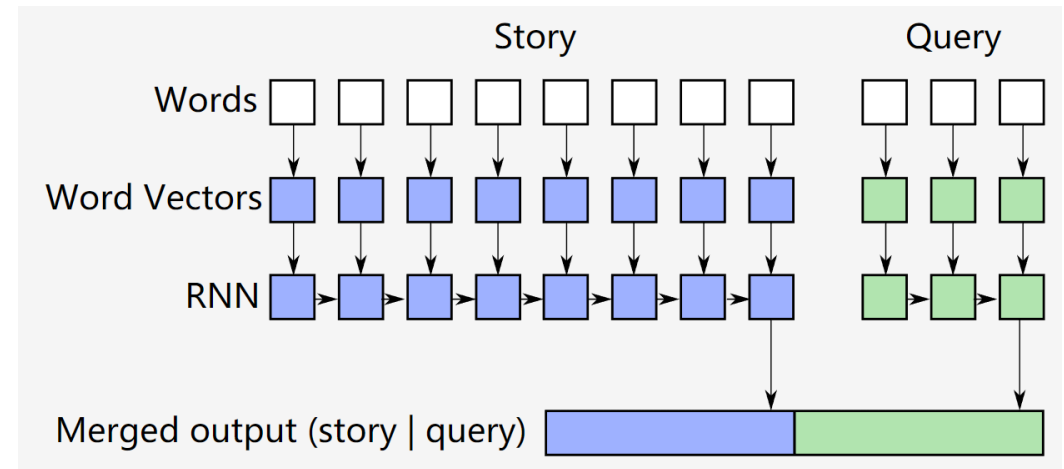
大脑中的注意力



所有黑色牌的数字之和为多少？

Facebook bAbi tasks

- 1 John moved to the bedroom.
- 2 Mary grabbed the football there.
- 3 Sandra journeyed to the bedroom.
- 4 Sandra went back to the hallway.
- 5 Mary moved to the garden.
- 6 Mary journeyed to the office.
- 7 Where is the football? office



例子：阅读理解

The first recorded travels by Europeans to China and back date from this time. The most famous traveler of the period was the Venetian Marco Polo, whose account of his trip to "Cambaluc," the capital of the Great Khan, and of life there astounded the people of Europe. The account of his travels, *Il milione* (or, *The Million*, known in English as the *Travels of Marco Polo*), appeared about the year 1299. Some argue over the accuracy of Marco Polo's accounts due to the lack of mentioning the Great Wall of China, tea houses, which would have been a prominent sight since Europeans had yet to adopt a tea culture, as well the practice of foot binding by the women in capital of the Great Khan. Some suggest that Marco Polo acquired much of his knowledge **through contact with Persian traders** since many of the places he named were in Persian.

How did some suspect that Polo learned about China instead of by actually visiting it?

Answer: **through contact with Persian traders**

神经网络中的注意力机制



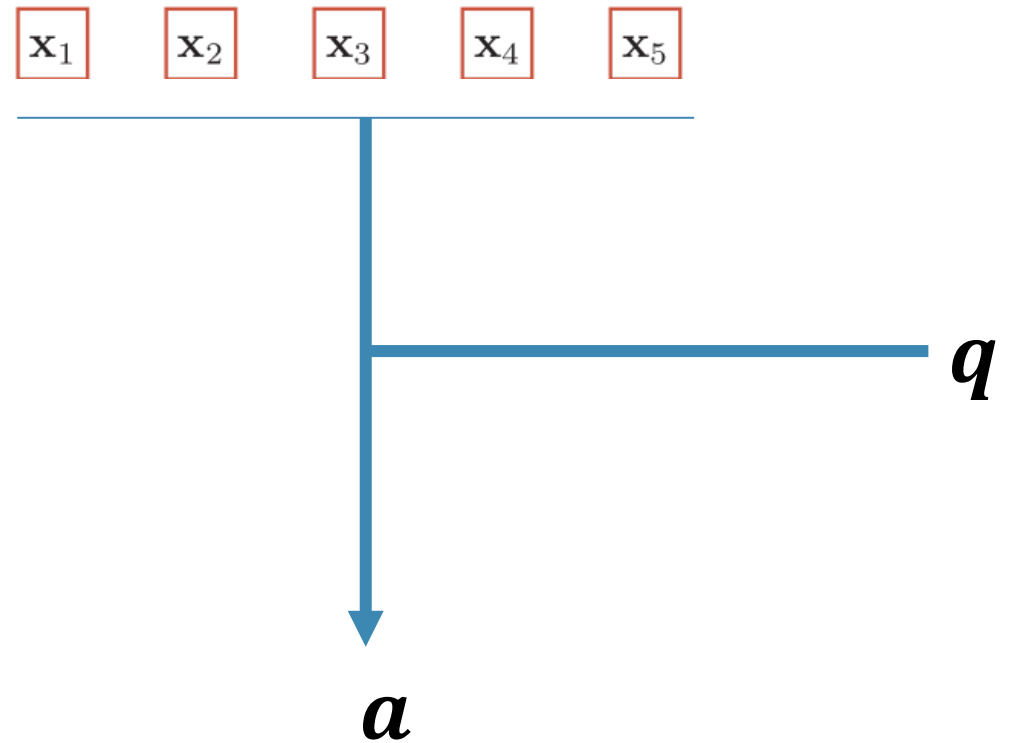
通用近似定理

- 由于优化算法和计算能力的限制，神经网络在实践中很难达到通用近似的能力。
 - 网络不能太复杂（参数太多）
- 如何提高网络能力
 - 局部连接
 - 权重共享
 - 汇聚操作
 - ?

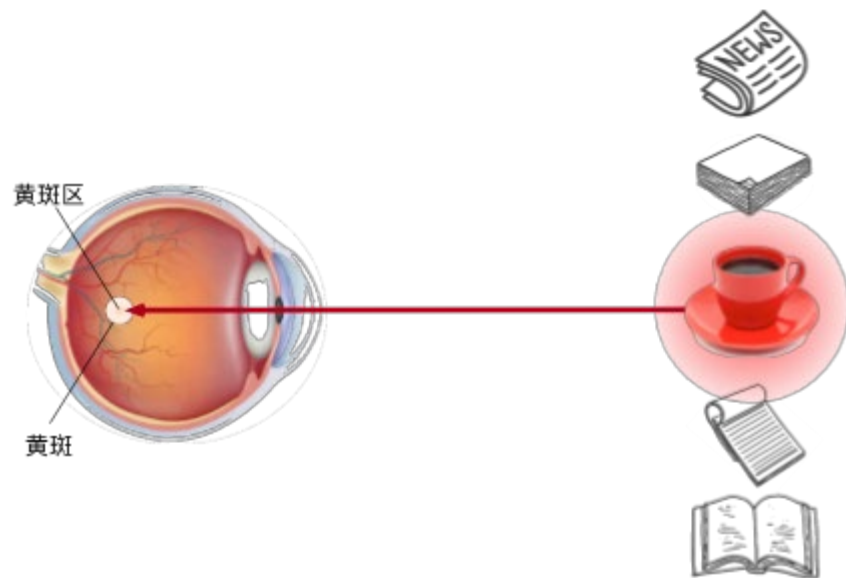
注意力机制



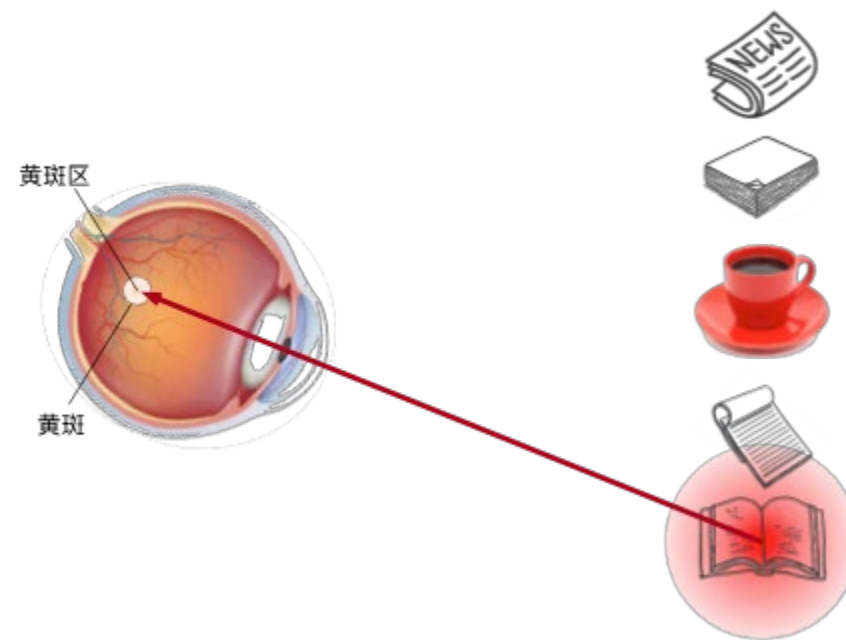
Try to find 8 less than 8 seconds



注意力机制



非自主性提示/非意志线索
(Key)

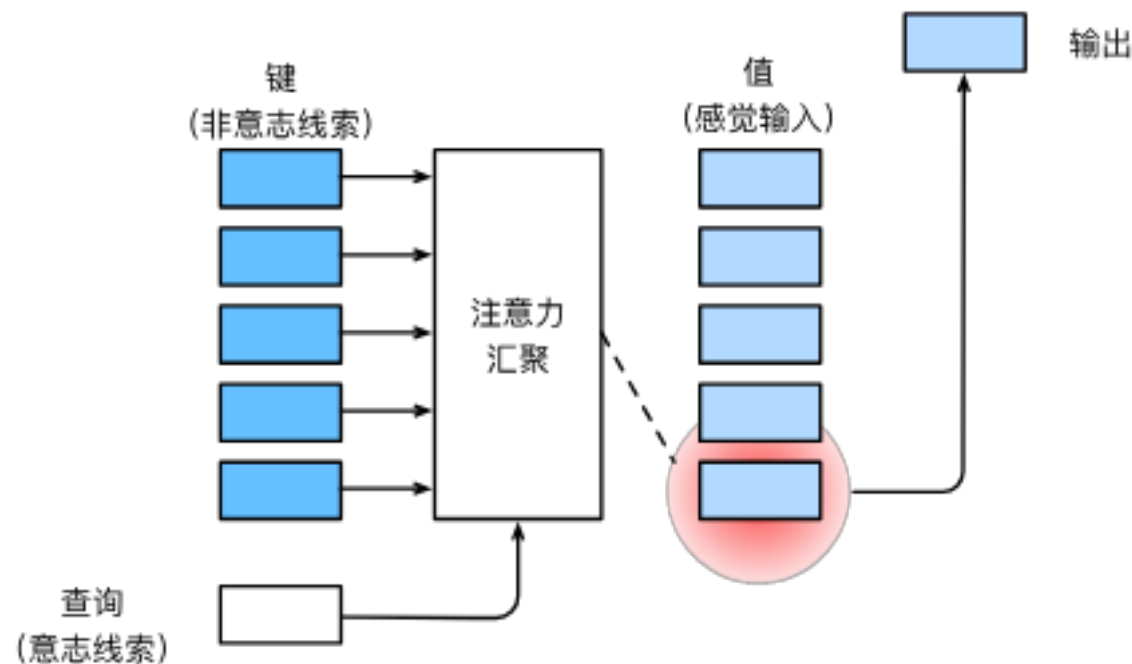


自主性提示/意志线索
(Query)

感官输入
(Value)

注意力机制

- 全连接、卷积、池化等都只考虑了非意志线索；
- 注意力机制则显式地考虑了意志线索：
 - 意志线索：查询 (Query)
 - 非意志线索：键 (Key)
 - 感官输入：值 (Value)
 - 每个输入是值 (Value) 和键 (Key) 组成的对；
 - 通过注意力池化层有偏向性地选择某些输入。



非参注意力池化

- 给定数据: $(x_i, y_i), i = 1, \dots, n$

- 平均池化是最简单的方案: $f(x) = \frac{1}{n} \sum_i y_i$

- 更好的方案: Nadaraya-Watson核回归 (1960s)

$$f(x) = \sum_{i=1}^n \frac{K(x - x_i)}{\sum_{j=1}^n K(x - x_j)} y_i$$

- 思想: 找跟 x 相近的数据

例如使用高斯核:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$$



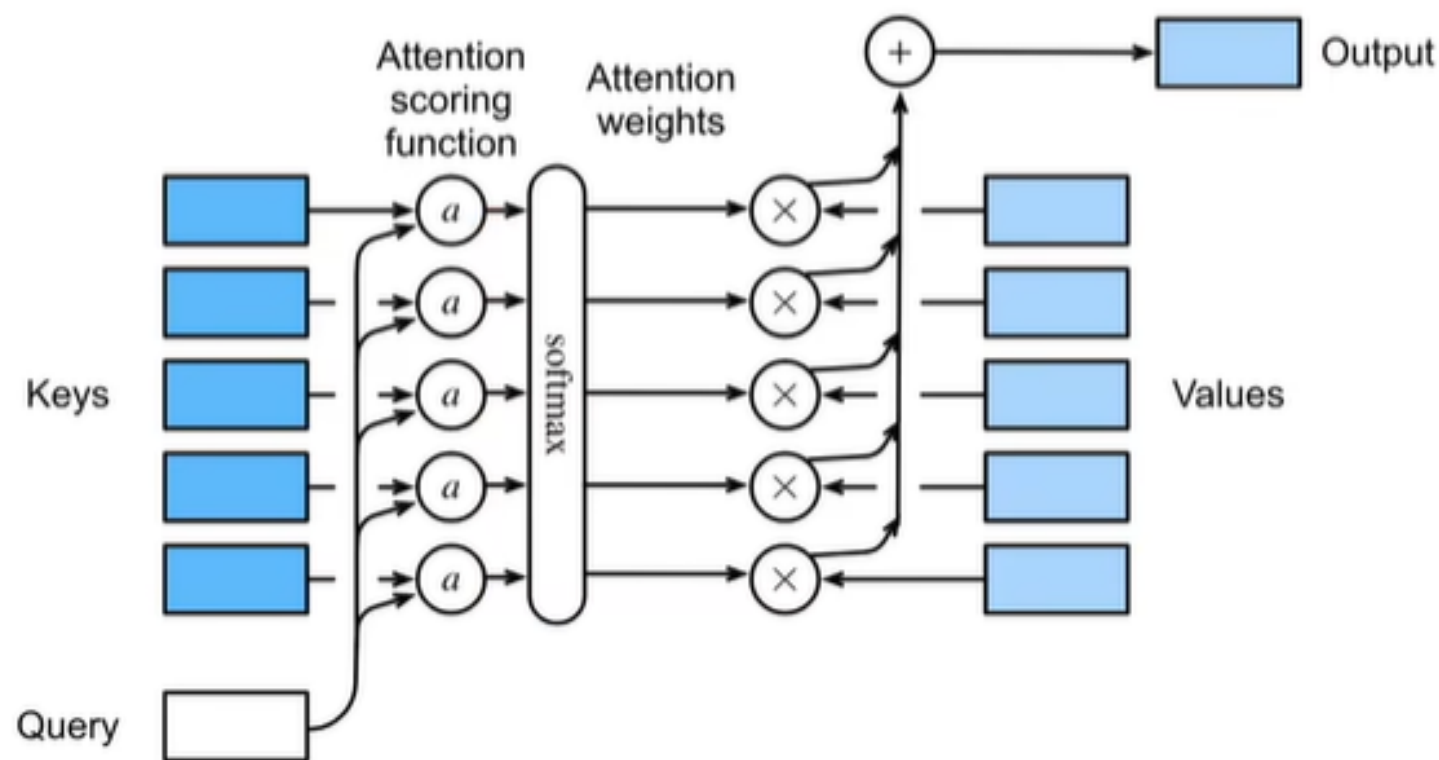
$$\begin{aligned} f(x) &= \sum_{i=1}^n \frac{\exp\left(-\frac{1}{2}(x - x_i)^2\right)}{\sum_{j=1}^n \exp\left(-\frac{1}{2}(x - x_j)^2\right)} y_i \\ &= \sum_{i=1}^n \text{softmax}\left(-\frac{1}{2}(x - x_i)^2\right) y_i \end{aligned}$$

注意力分数

$$f(x) = \sum_i \alpha(x, x_i) y_i = \sum_{i=1}^n \text{softmax} \left(-\frac{1}{2}(x - x_i)^2 \right) y_i$$

归一化前：注意力分数

归一化后：注意力权重



注意力模型

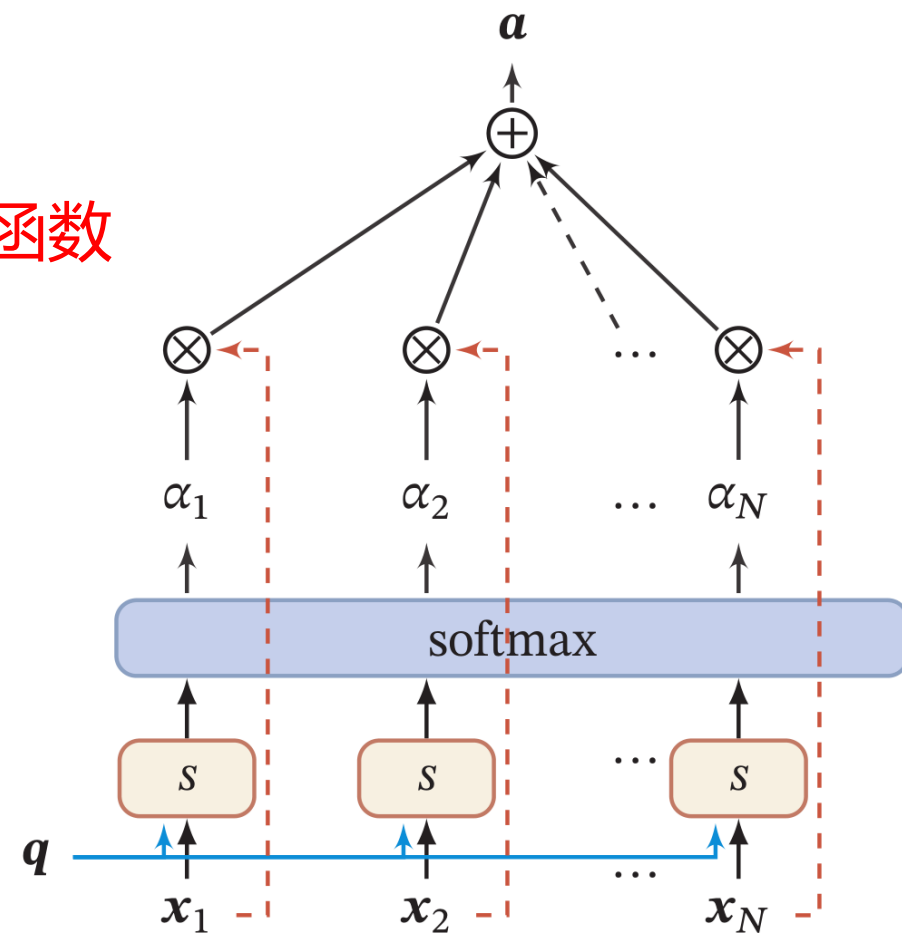
- 软性注意力机制 (soft attention mechanism)

- 注意力机制可以分为两步

- 计算注意力分布 α , $\alpha_n = p(z = n | \mathbf{X}, \mathbf{q})$
 $= \text{softmax}(s(\mathbf{x}_n, \mathbf{q}))$ $s(\mathbf{x}_n, \mathbf{q})$ 打分函数
 $= \frac{\exp(s(\mathbf{x}_n, \mathbf{q}))}{\sum_{j=1}^N \exp(s(\mathbf{x}_j, \mathbf{q}))}$

- 根据 α 来计算输入信息的加权平均。

$$\begin{aligned} \text{att}(\mathbf{X}, \mathbf{q}) &= \sum_{n=1}^N \alpha_n \mathbf{x}_n, \\ &= \mathbb{E}_{z \sim p(z | \mathbf{X}, \mathbf{q})} [\mathbf{x}_z] \end{aligned}$$



注意力模型

- 注意力打分函数

加性模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{x} + \mathbf{U}\mathbf{q}),$$

点积模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{q},$$

缩放点积模型

$$s(\mathbf{x}, \mathbf{q}) = \frac{\mathbf{x}^\top \mathbf{q}}{\sqrt{D}},$$

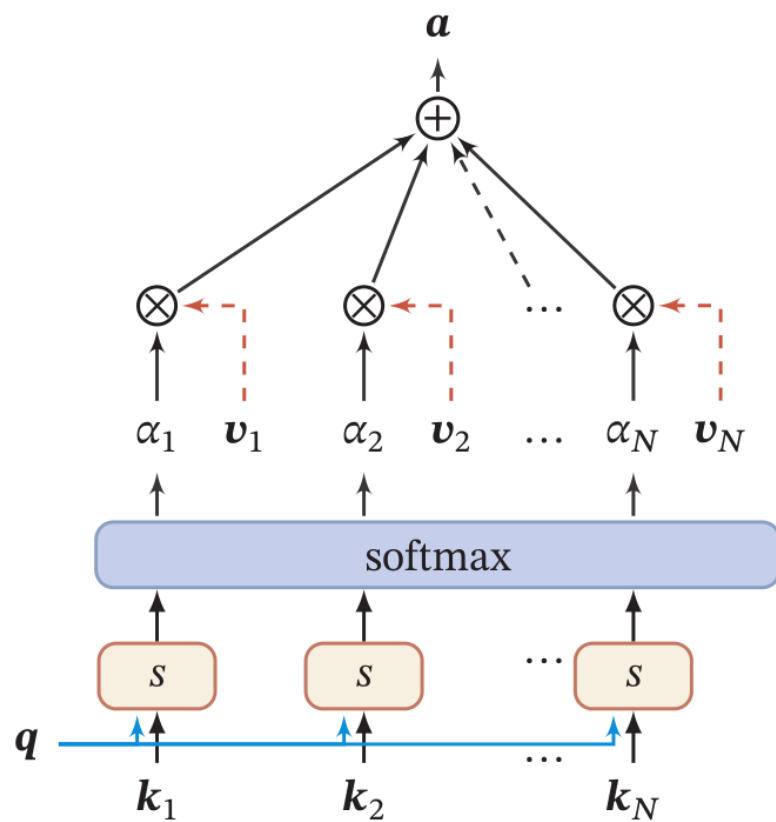
双线性模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{W} \mathbf{q},$$

注意力模型

● 注意力机制的变体

- 硬注意力 (hard attention)
- 键值对注意力 (key-value pair attention)



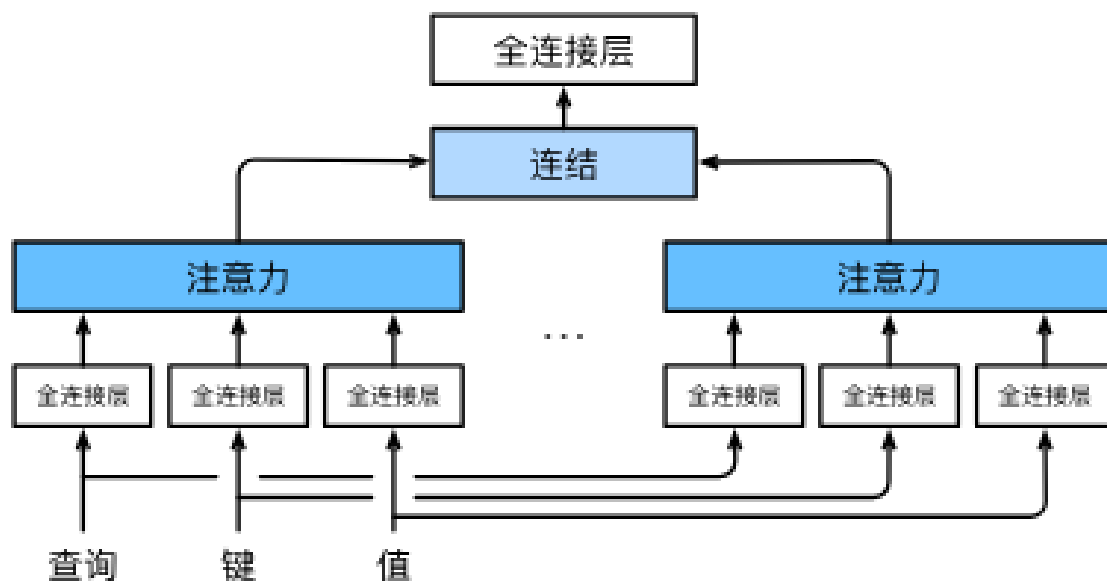
用 $(K, V) = [(k_1, v_1), \dots, (k_N, v_N)]$ 表示
N个输入信息

$$\begin{aligned} \text{att}((K, V), q) &= \sum_{n=1}^N \alpha_n v_n, \\ &= \sum_{n=1}^N \frac{\exp(s(k_n, q))}{\sum_j \exp(s(k_j, q))} v_n \end{aligned}$$

注意力模型

● 注意力机制的变体

- 硬注意力 (hard attention)
- 键值对注意力 (key-value pair attention)
- 多头注意力 (multi-head attention)



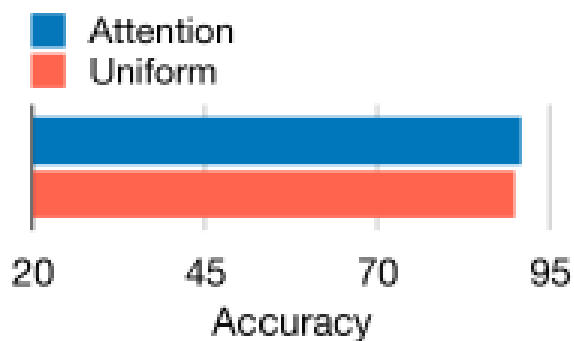
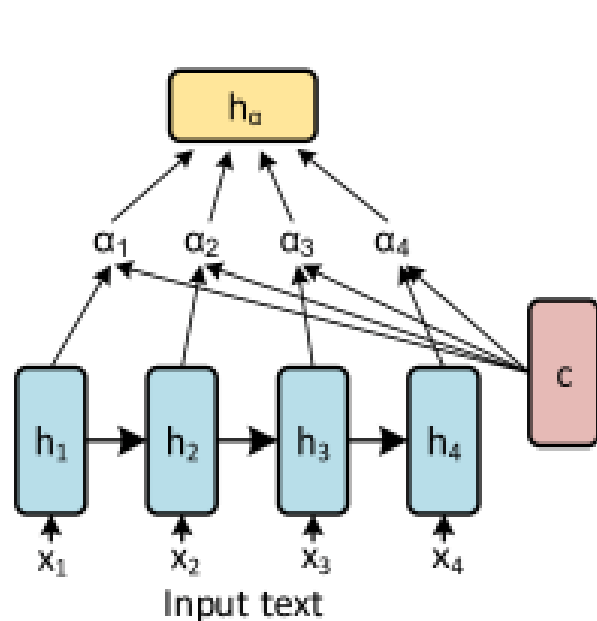
利用多个查询 $Q = [q_1, \dots, q_M]$, 来并行地从输入信息中选取多组信息. 每个注意力关注输入信息的不同部分.

$$\text{att}((K, V), Q) = \text{att}((K, V), q_1) \oplus \dots \oplus \text{att}((K, V), q_M)$$

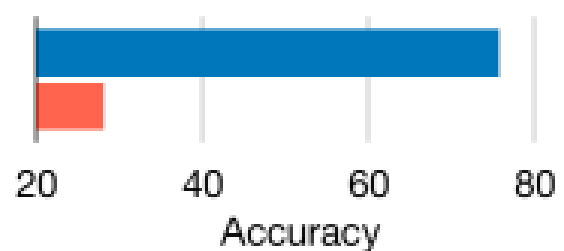
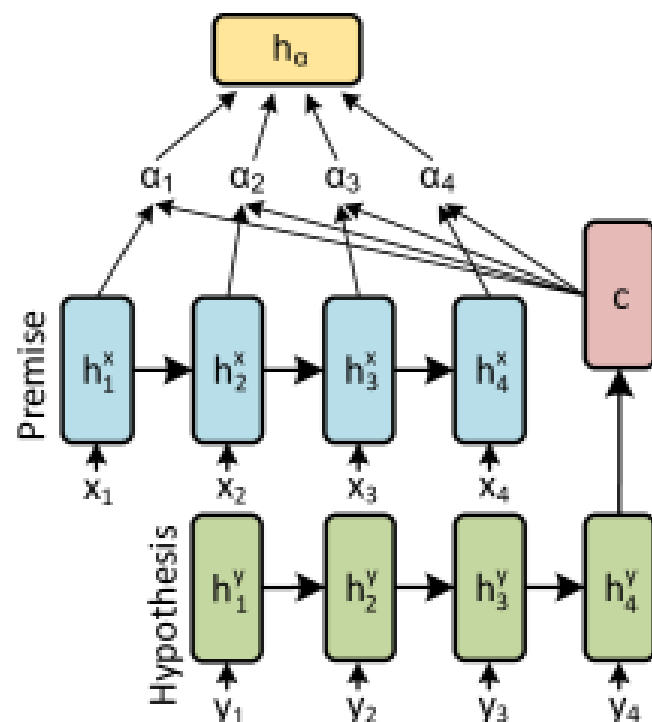
注意力机制的应用



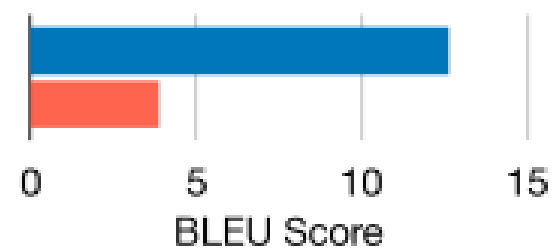
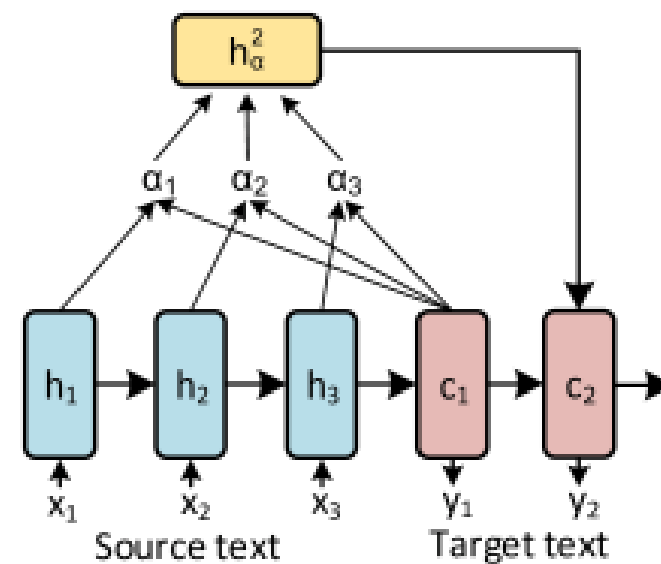
一些常见的应用



(a) Text Classification

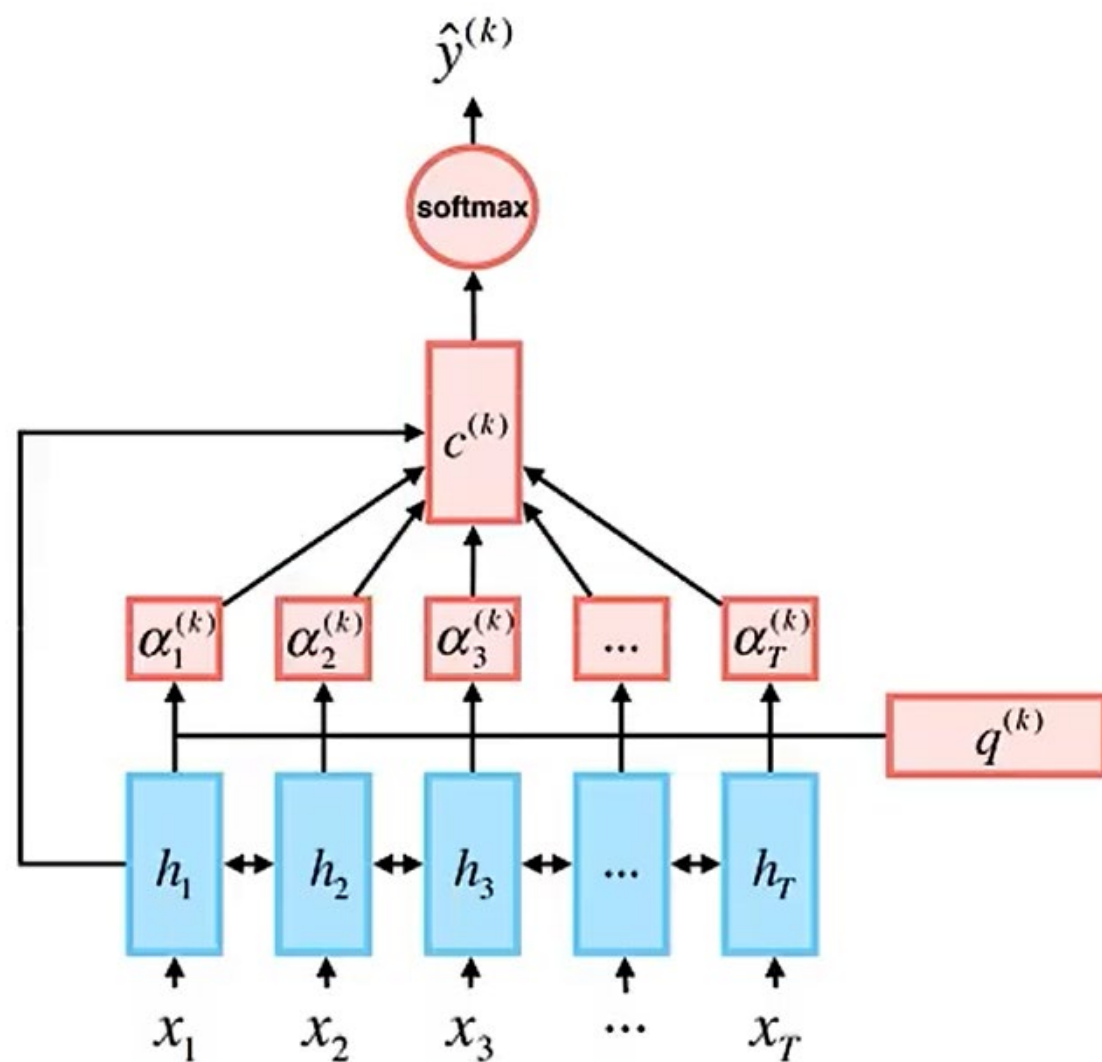


(b) Natural Language Inference



(c) Neural Machine Translation

The infantile cart is easy to use.



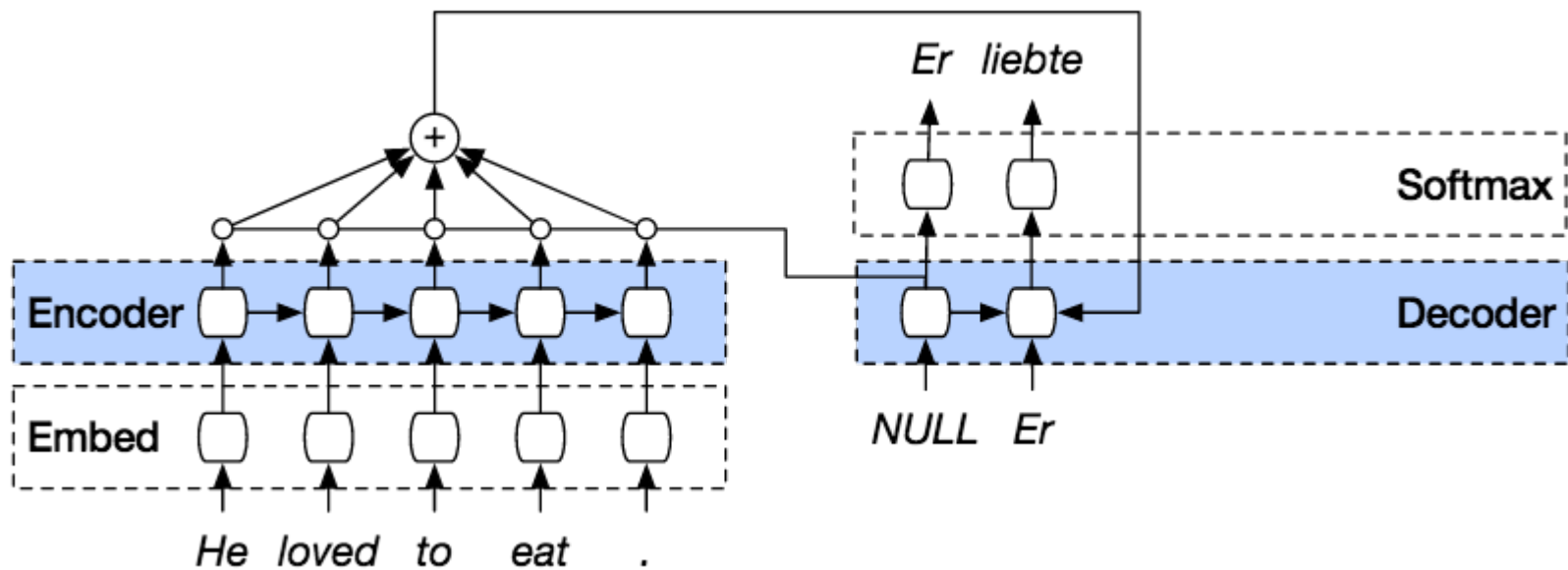
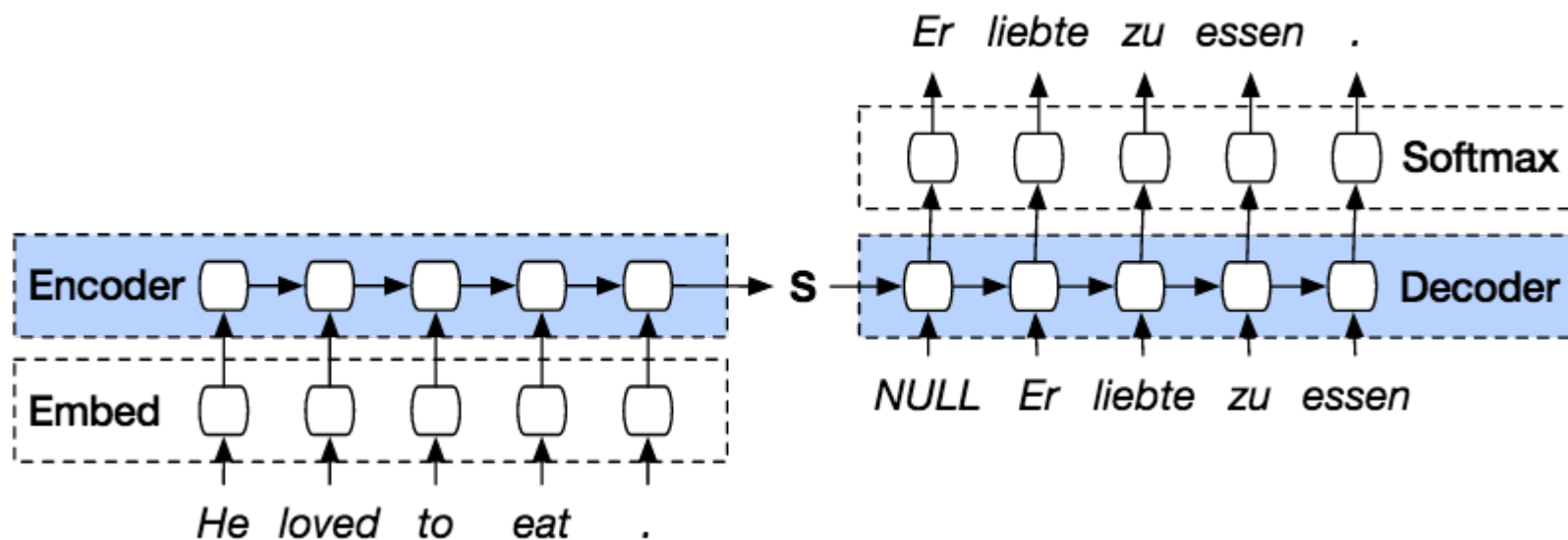
I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

(a) Attention of task “Books” in SA-MTL, Output: Negative

I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

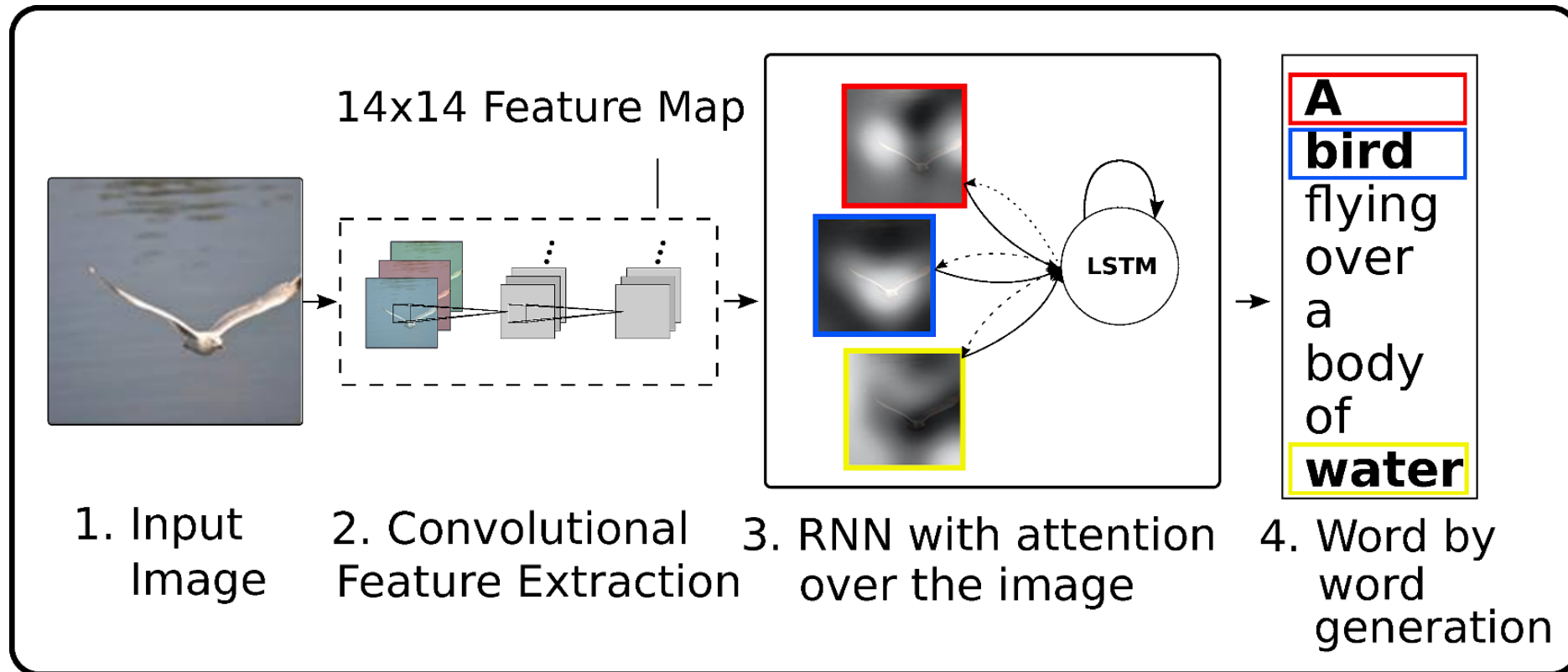
(b) Attention of task “DVD” in SA-MTL, Output: Positive

机器翻译

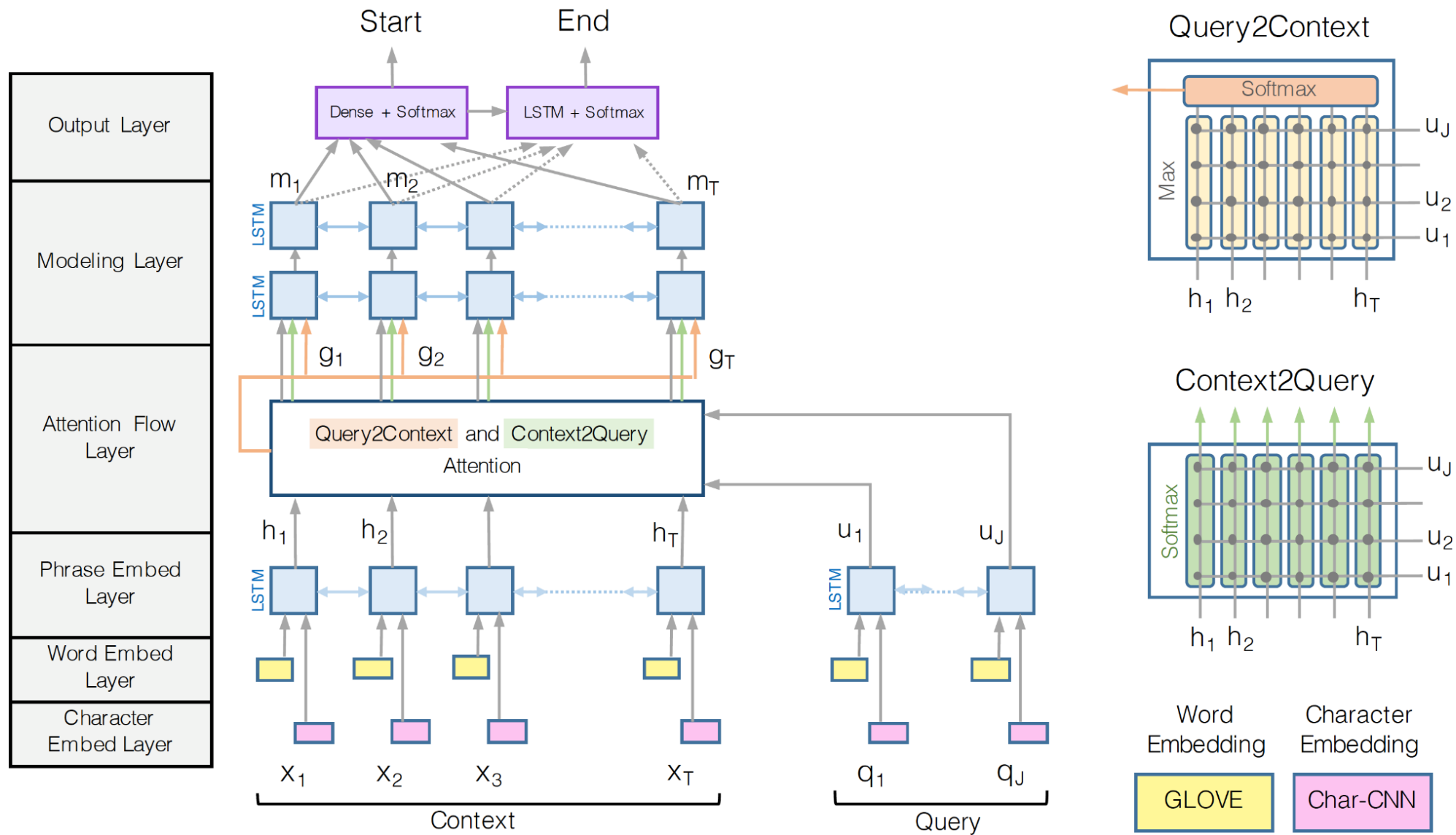


. la maison de Léa <end> .

Image Caption



阅读理解

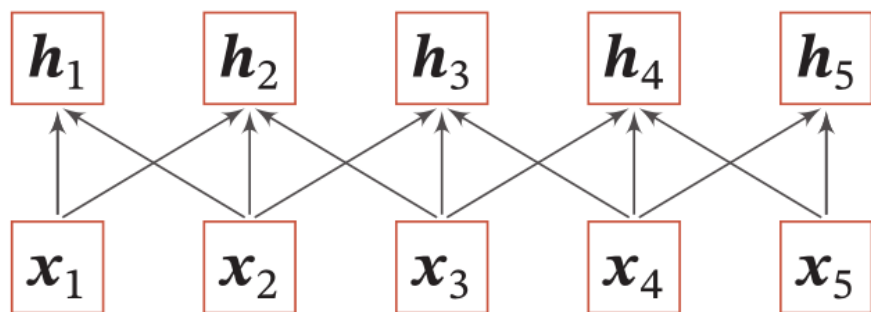


自注意力模型

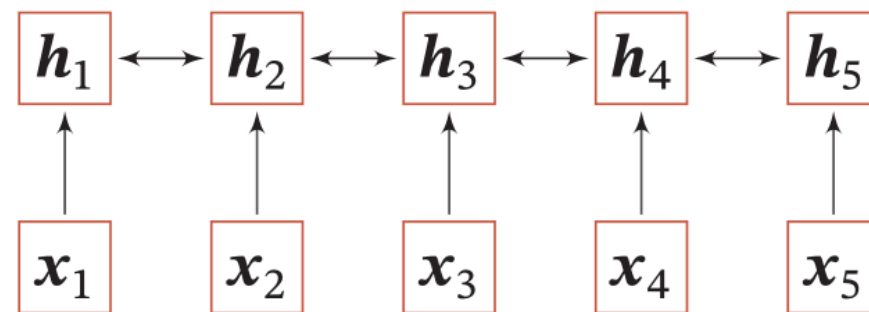


自注意力模型

- 当使用神经网络来处理一个变长的向量序列时，我们通常可以使用卷积网络或循环网络进行编码来得到一个相同长度的输出向量序列。



(a) 卷积网络



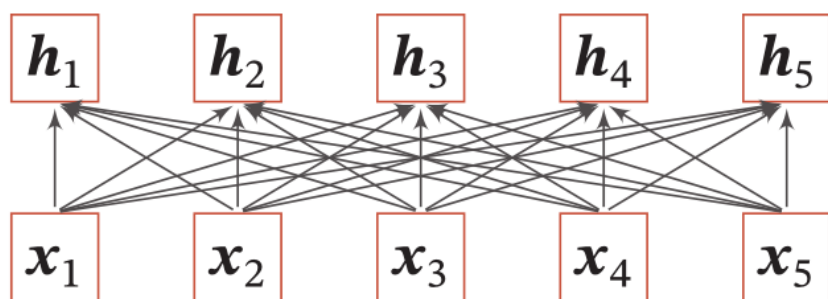
(b) 双向循环网络

- 但它们只建模了输入信息的局部依赖关系。

自注意力模型

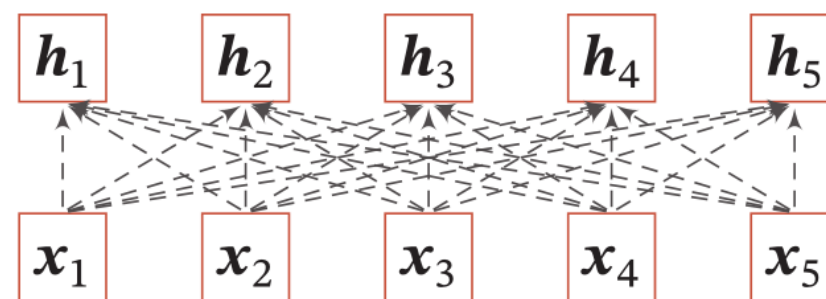
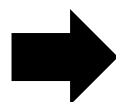
- 如何建立非局部 (Non-local) 的依赖关系

- 全连接?



(a) 全连接模型

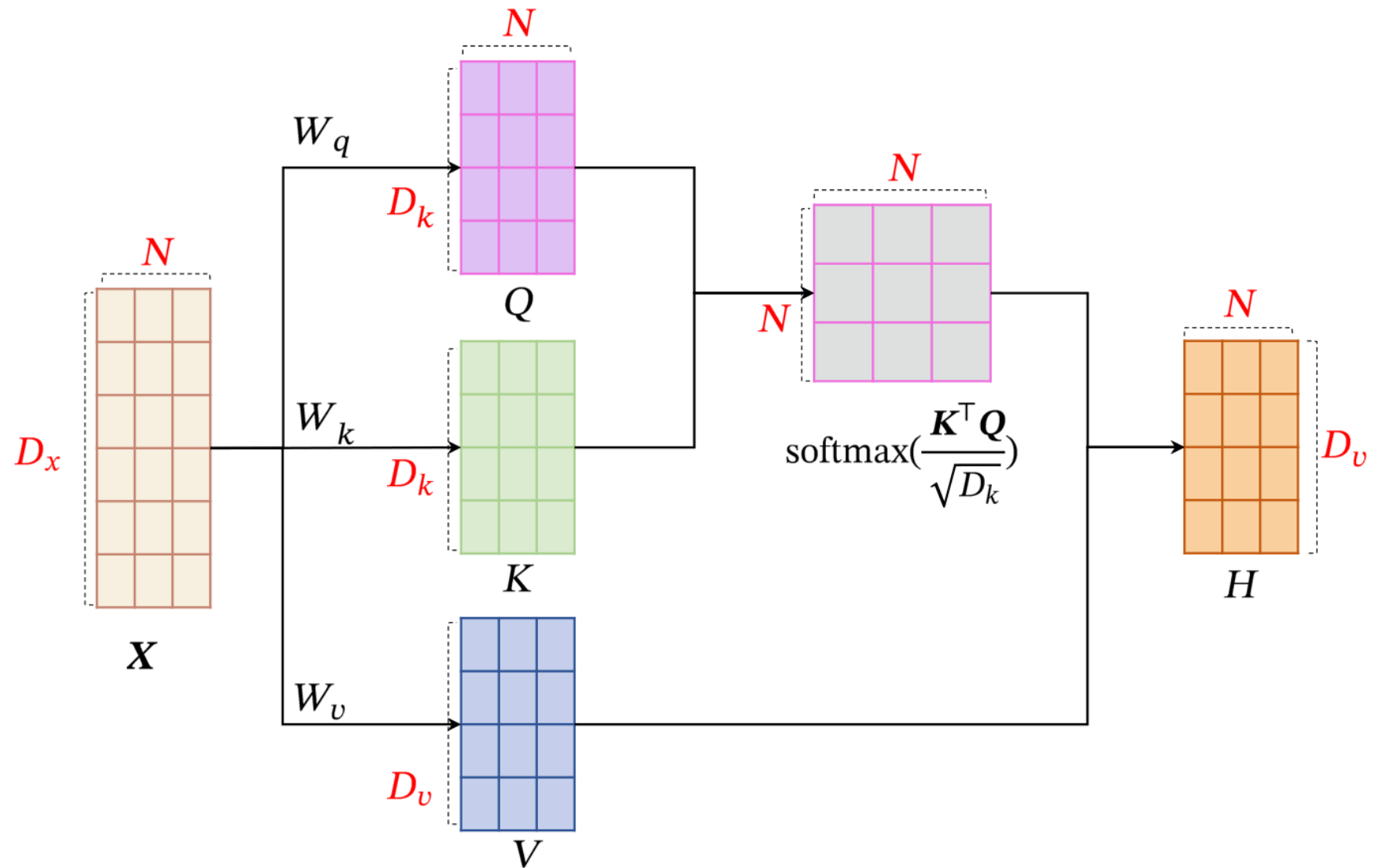
无法处理变长问题



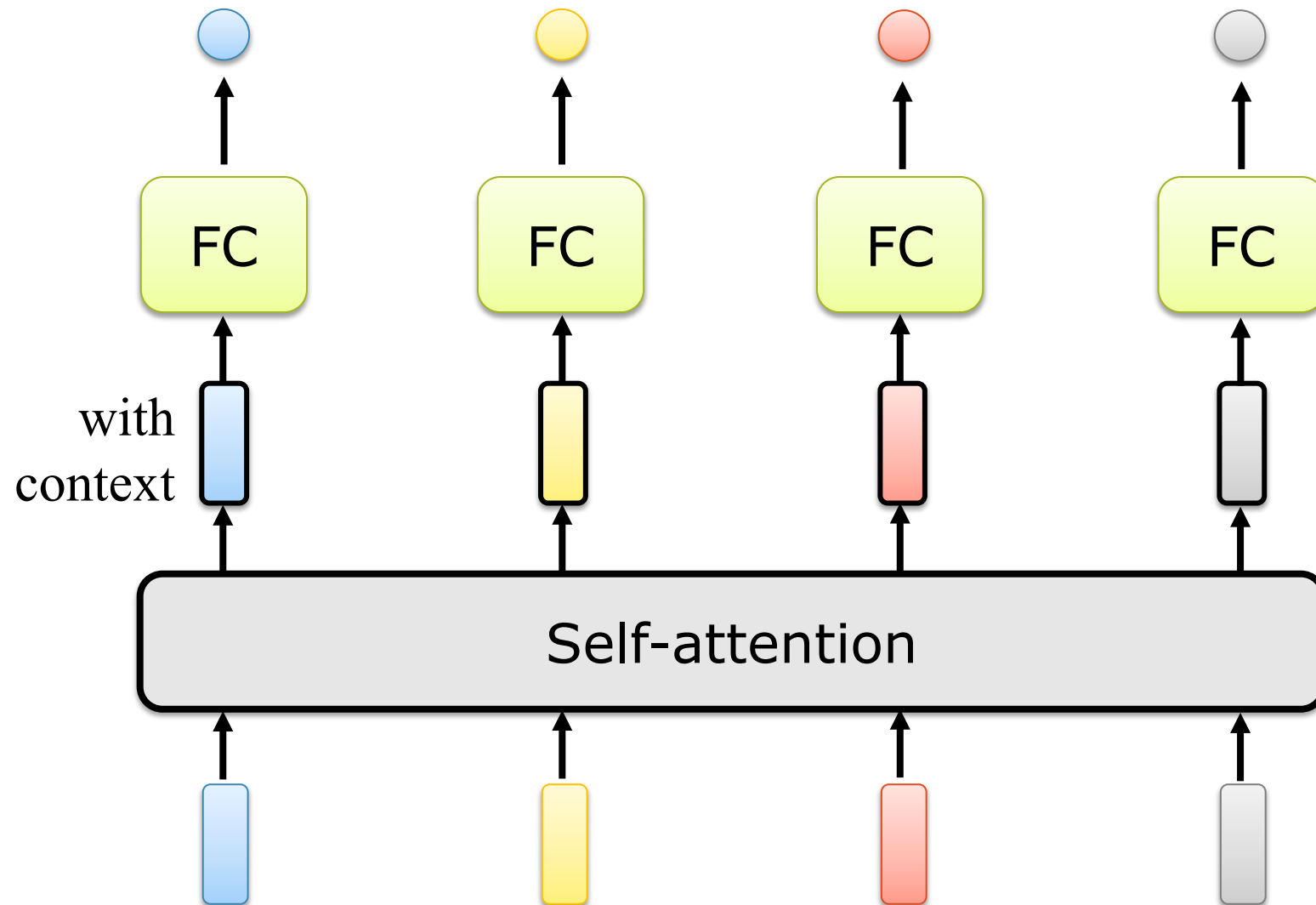
(b) 自注意力模型

连接权重 α_{ij} 由注意力机制动态生成

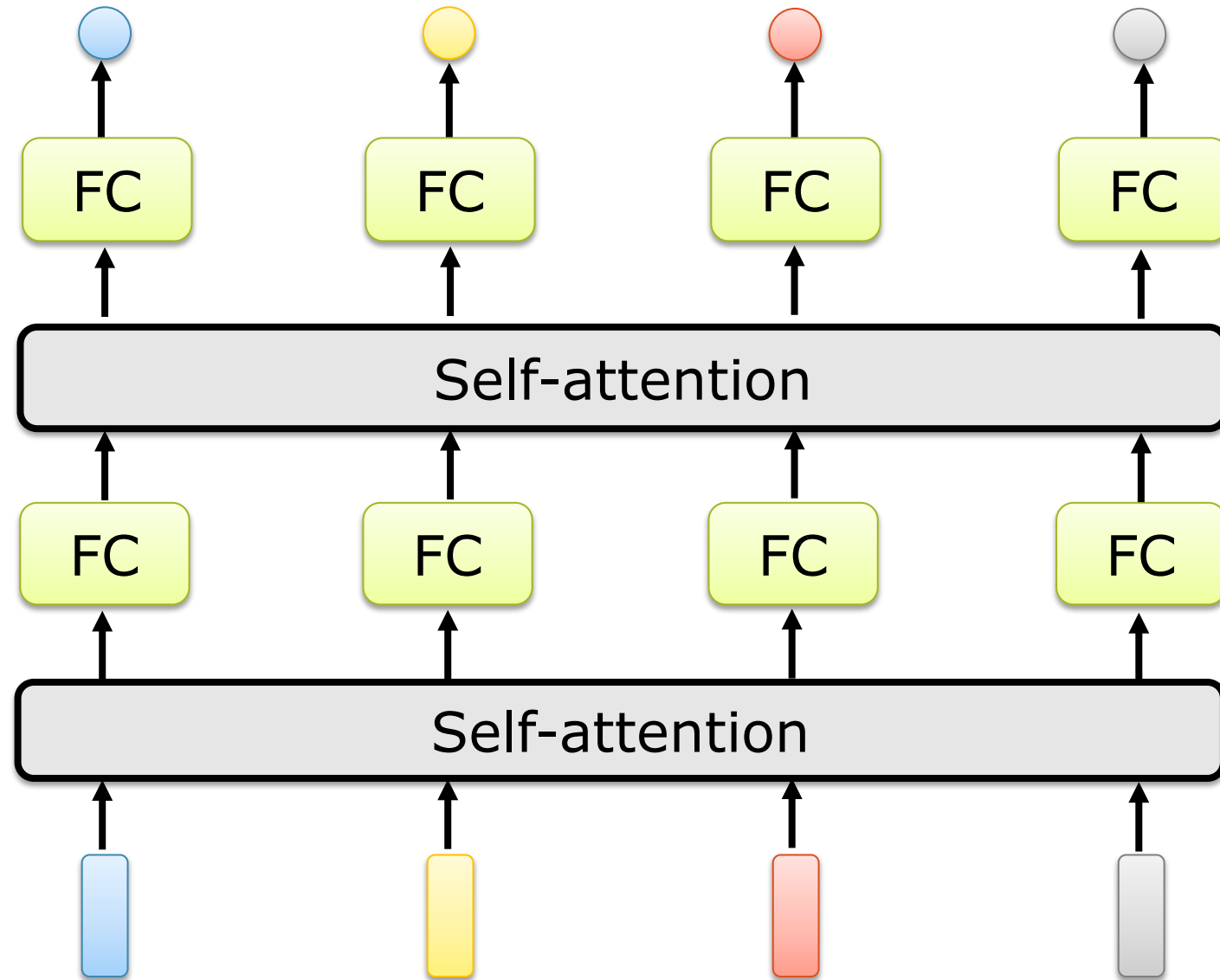
QKV模式 (Query-Key-Value)



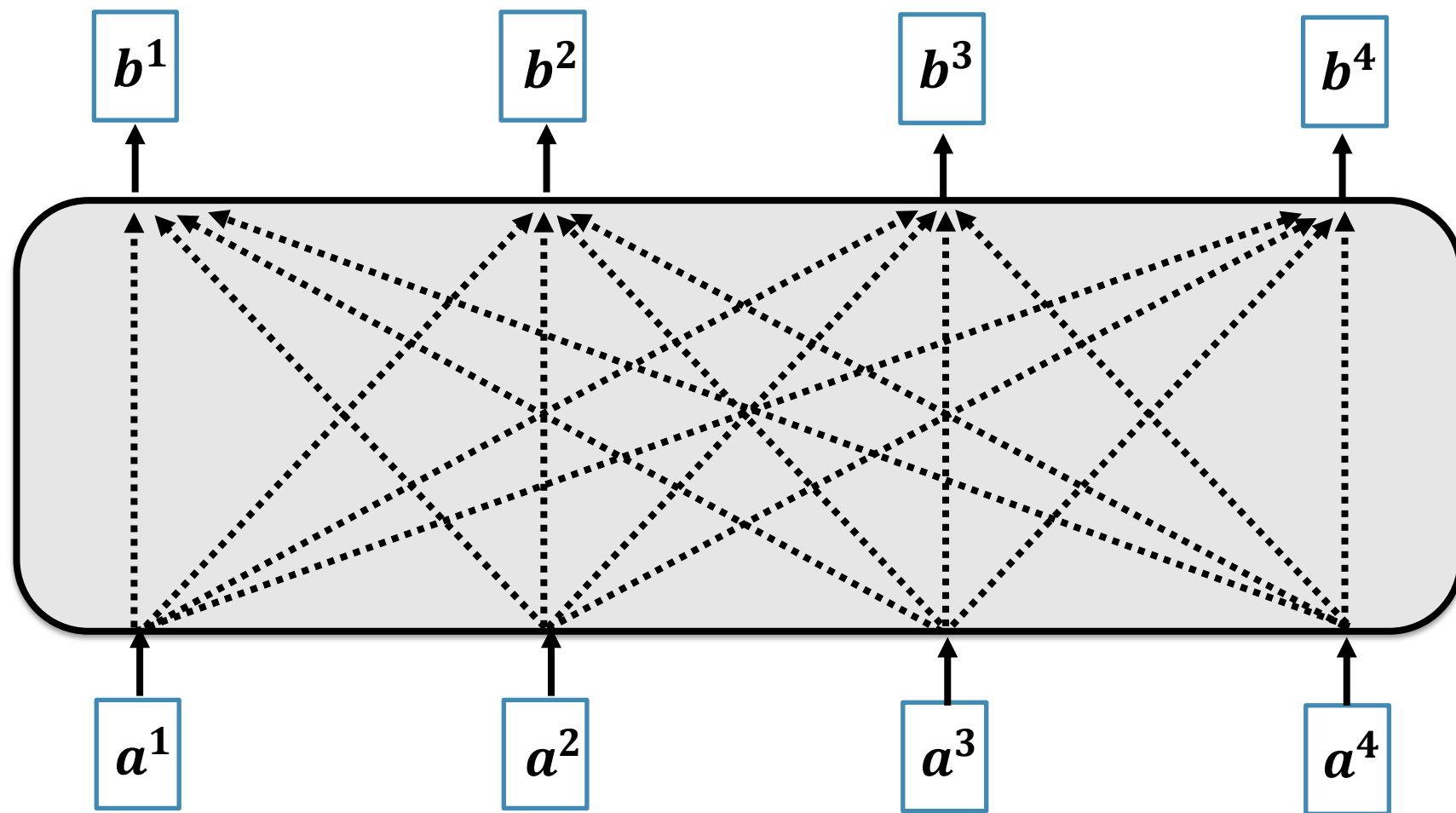
自注意力模型



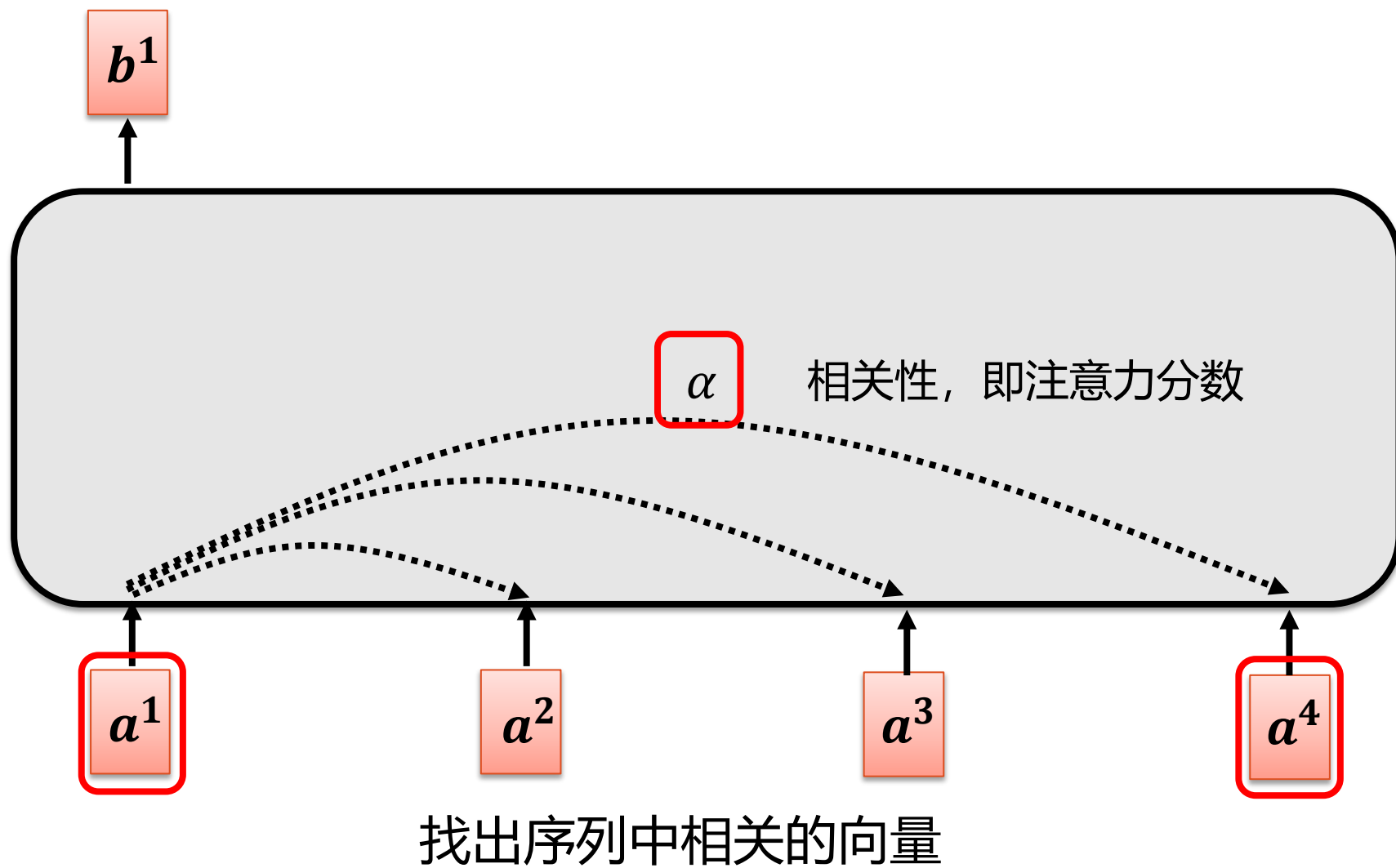
自注意力模型



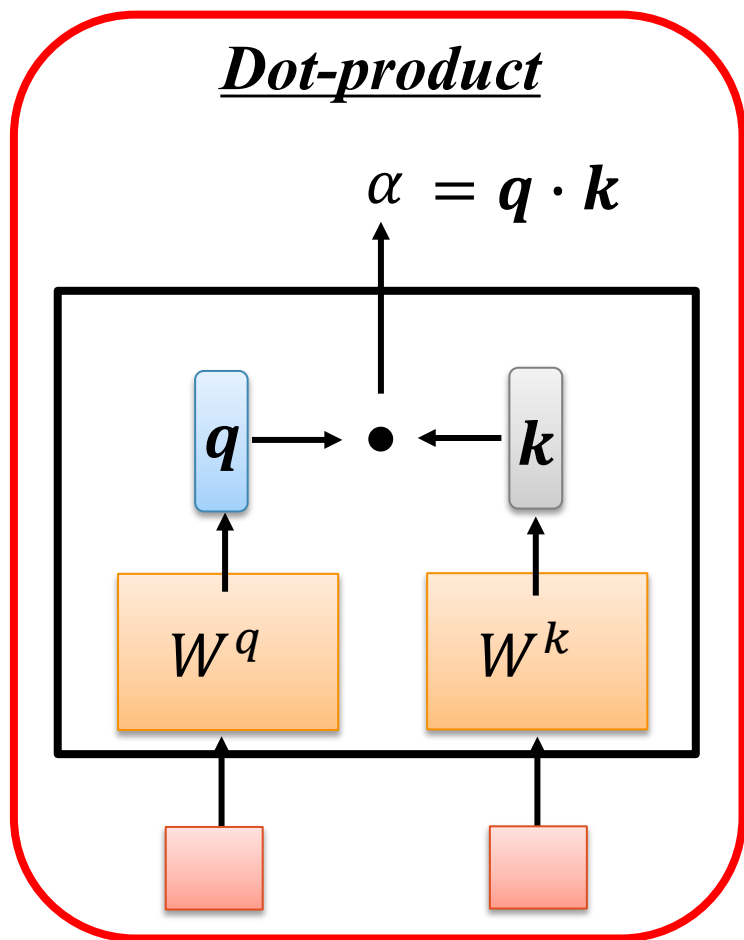
自注意力模型



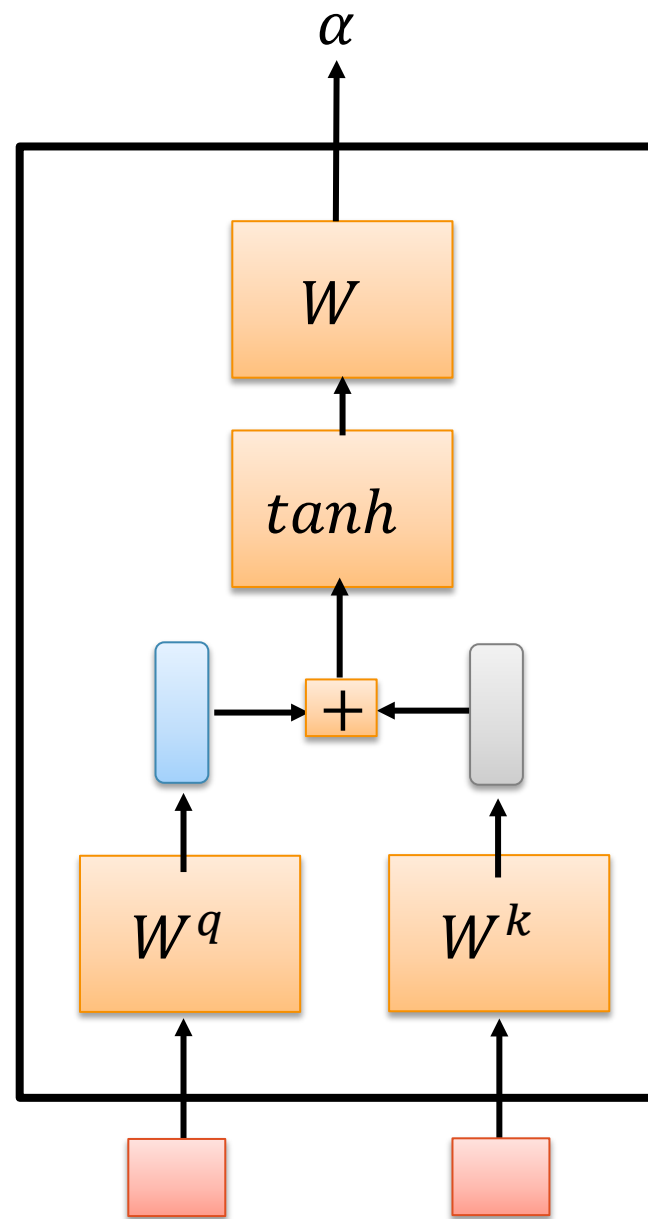
自注意力模型



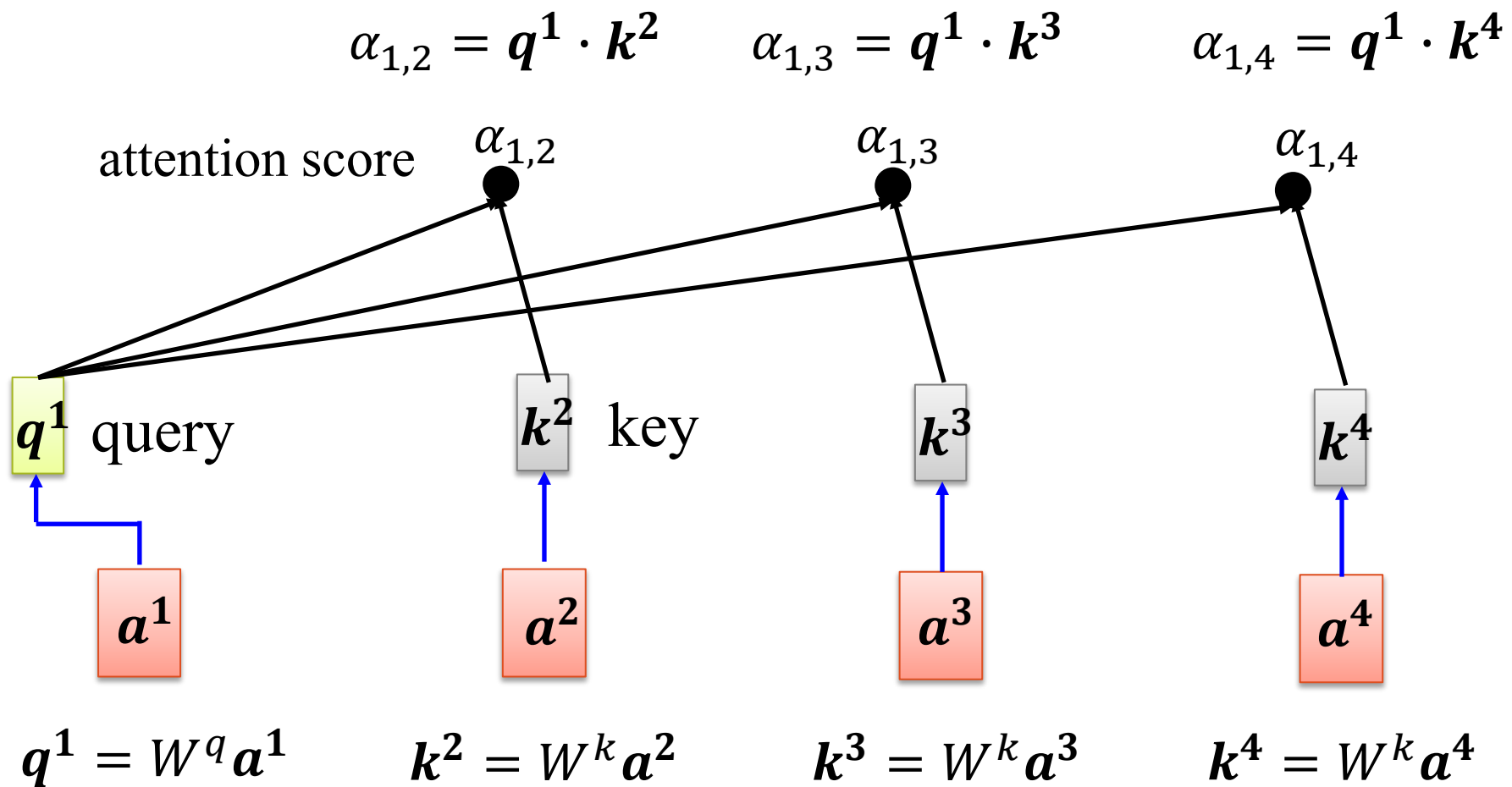
自注意力模型



Additive

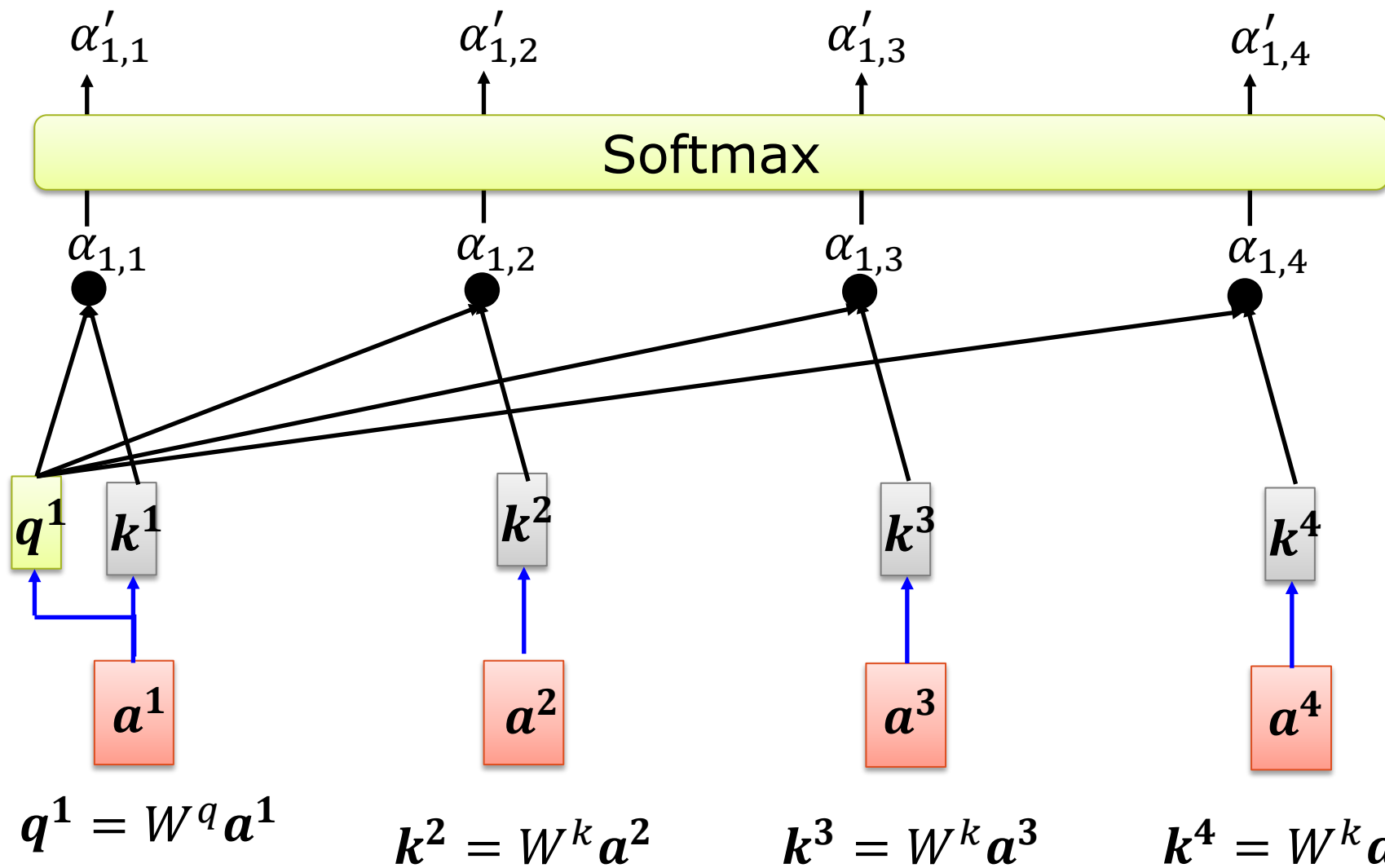


自注意力模型

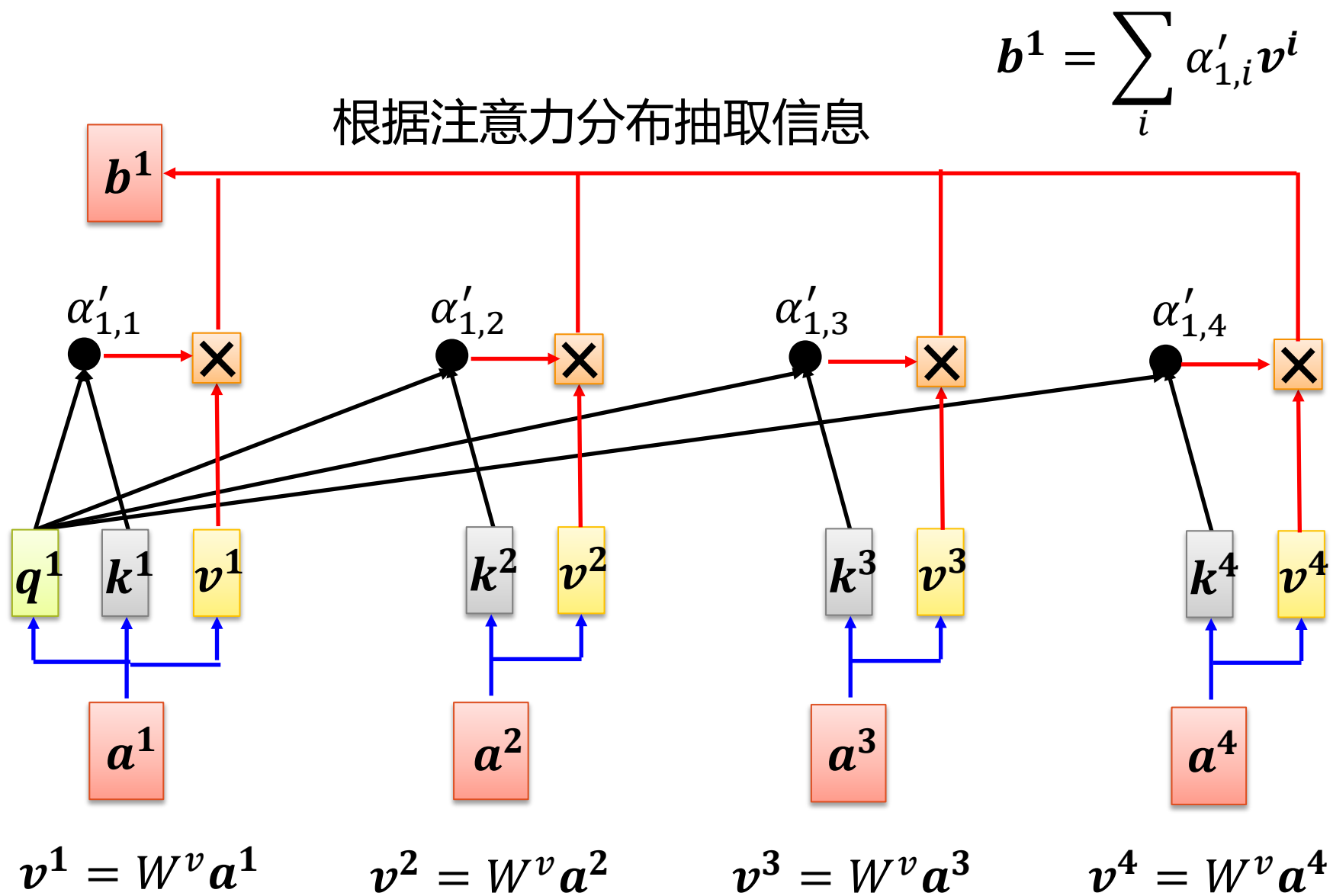


自注意力模型

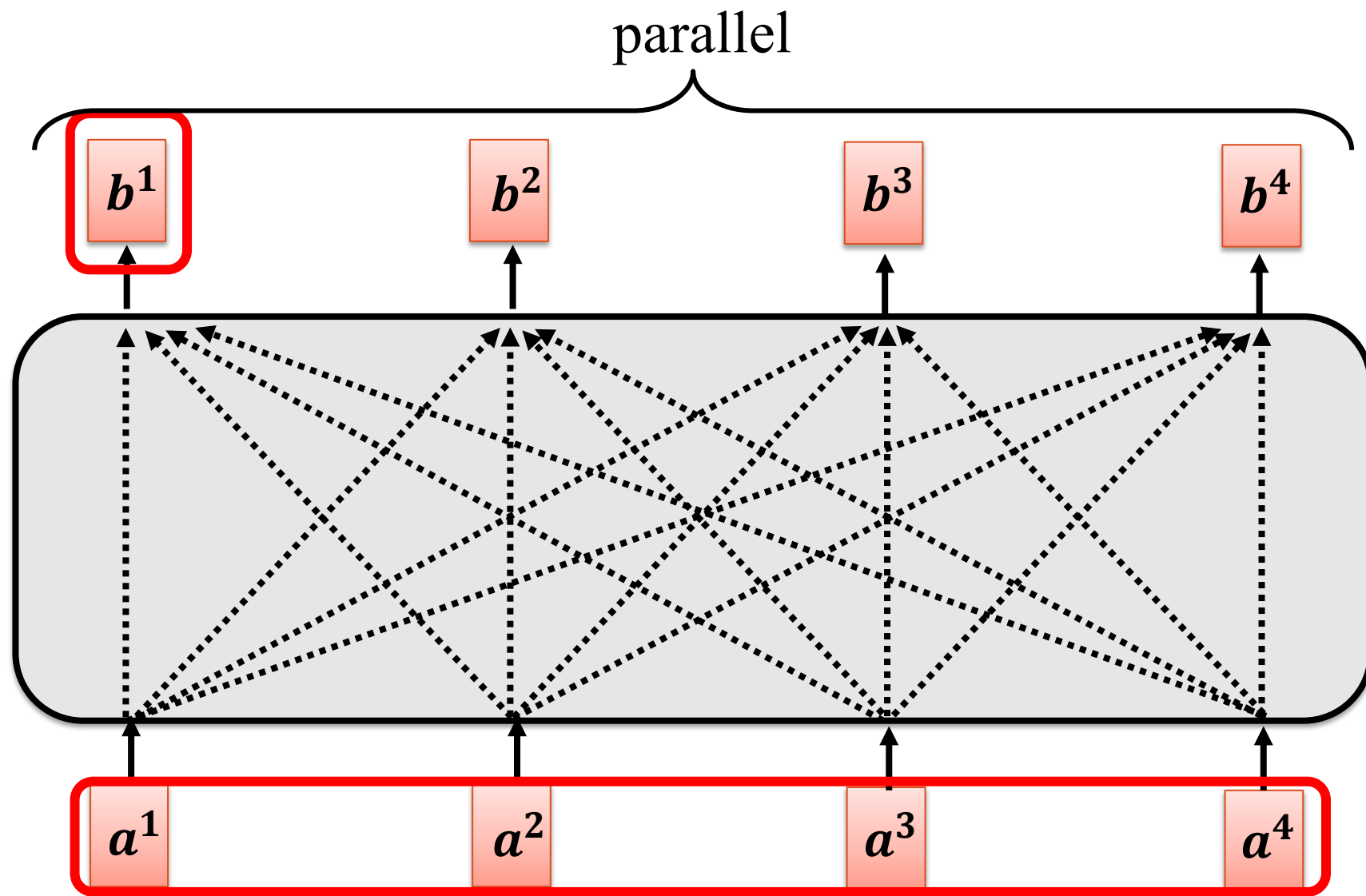
$$\alpha'_{1,i} = \exp(\alpha_{1,i}) / \sum_j \exp(\alpha_{1,j})$$



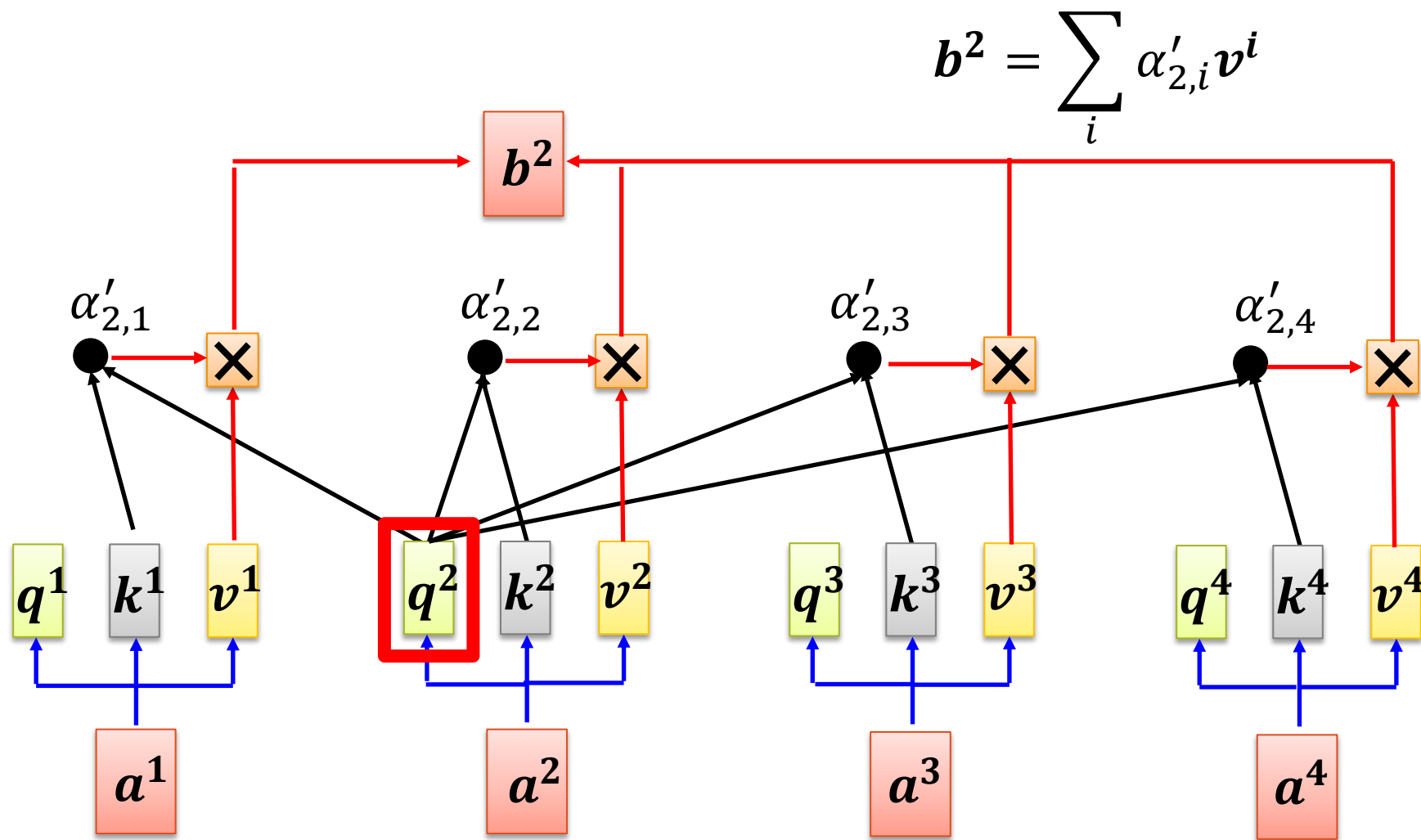
自注意力模型



自注意力模型



自注意力模型

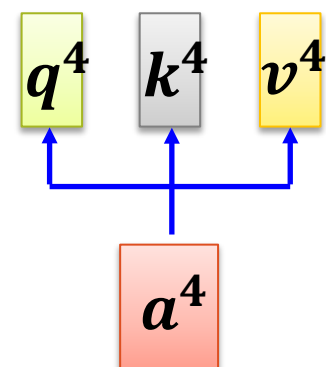
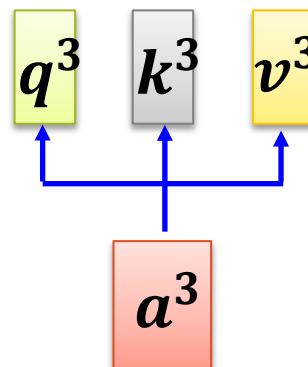
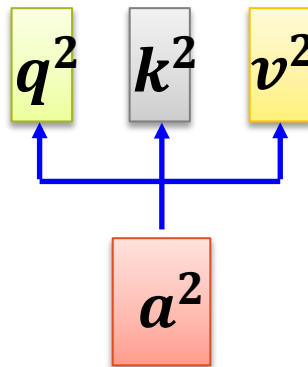
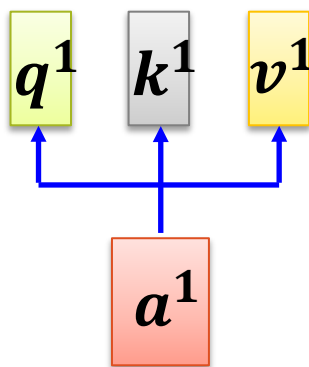


自注意力模型

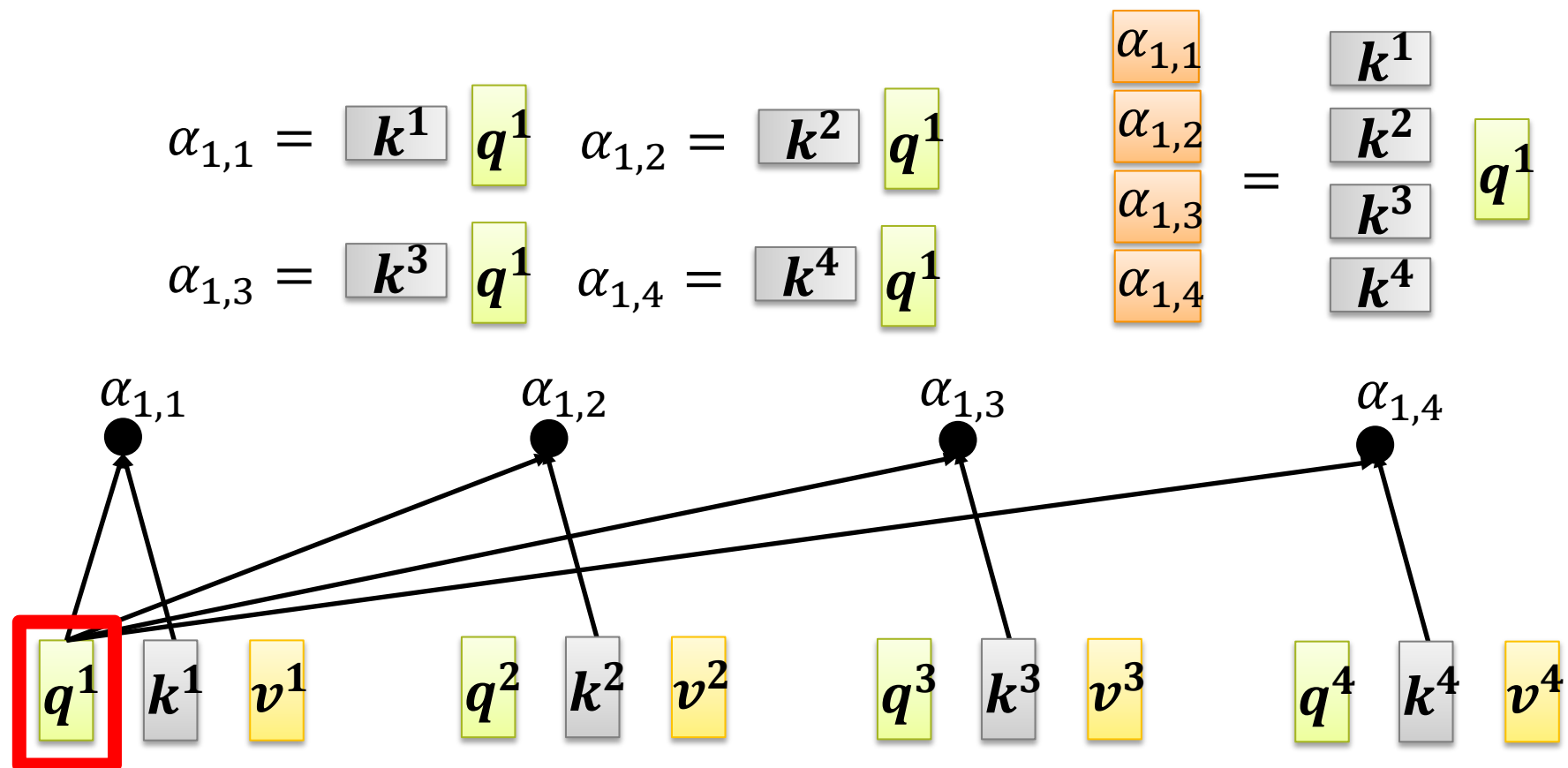
$$q^i = W^q a^i \quad \begin{matrix} q^1 & q^2 & q^3 & q^4 \\ \hline Q \end{matrix} = \begin{matrix} W^q & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ \hline I \end{matrix} \end{matrix}$$

$$k^i = W^k a^i \quad \begin{matrix} k^1 & k^2 & k^3 & k^4 \\ \hline K \end{matrix} = \begin{matrix} W^k & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ \hline I \end{matrix} \end{matrix}$$

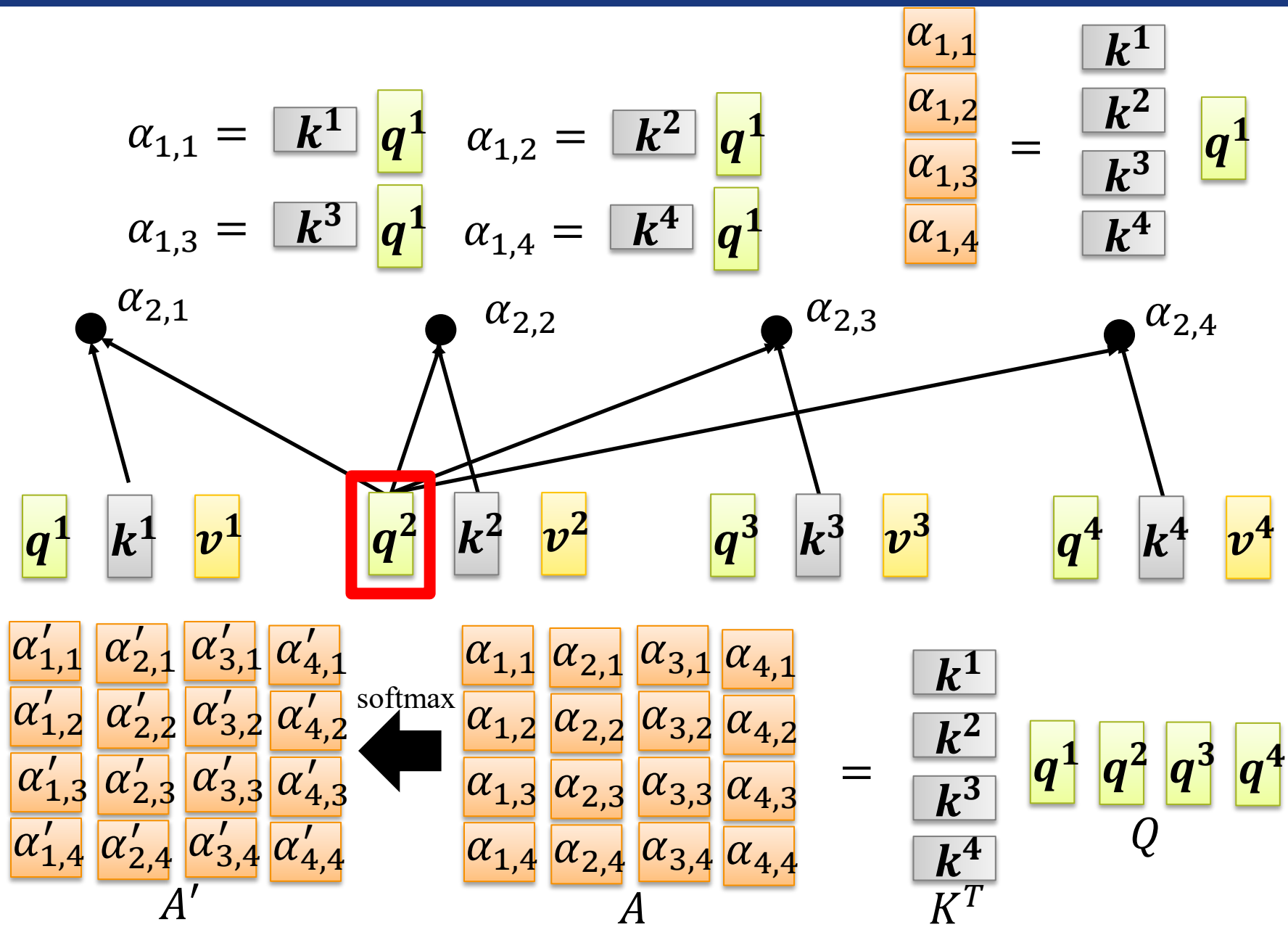
$$v^i = W^v a^i \quad \begin{matrix} v^1 & v^2 & v^3 & v^4 \\ \hline V \end{matrix} = \begin{matrix} W^v & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ \hline I \end{matrix} \end{matrix}$$



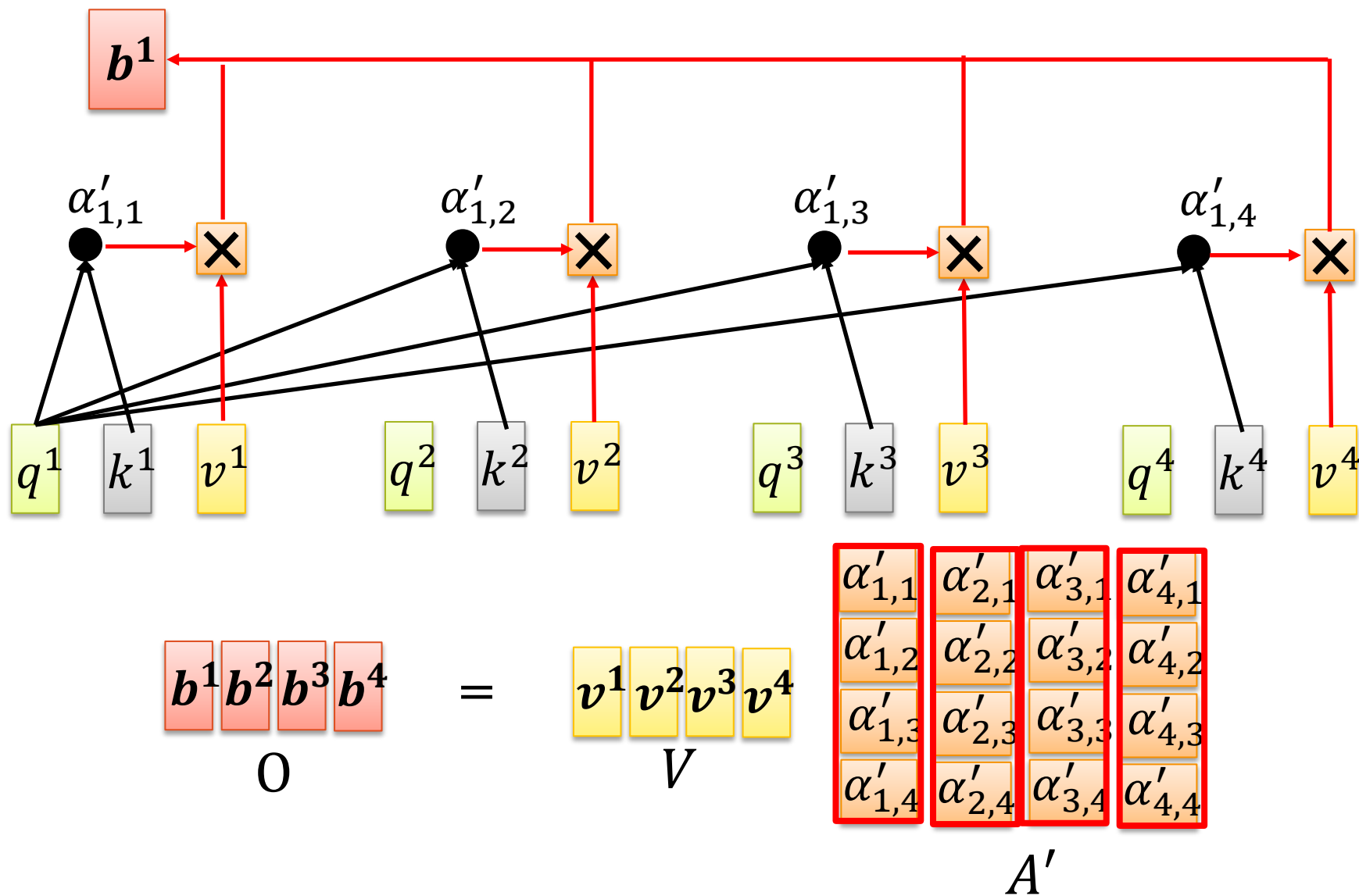
自注意力模型



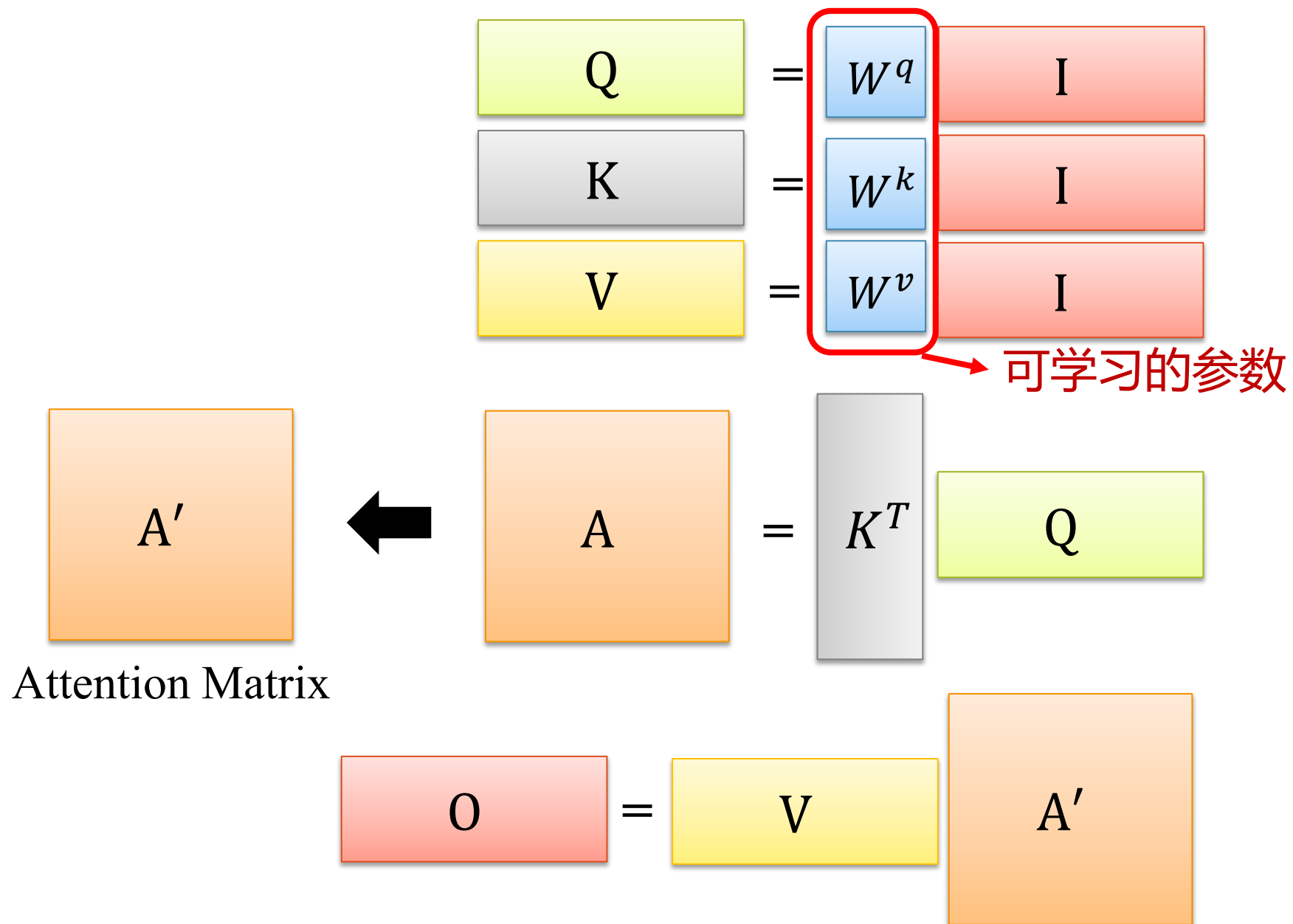
自注意力模型



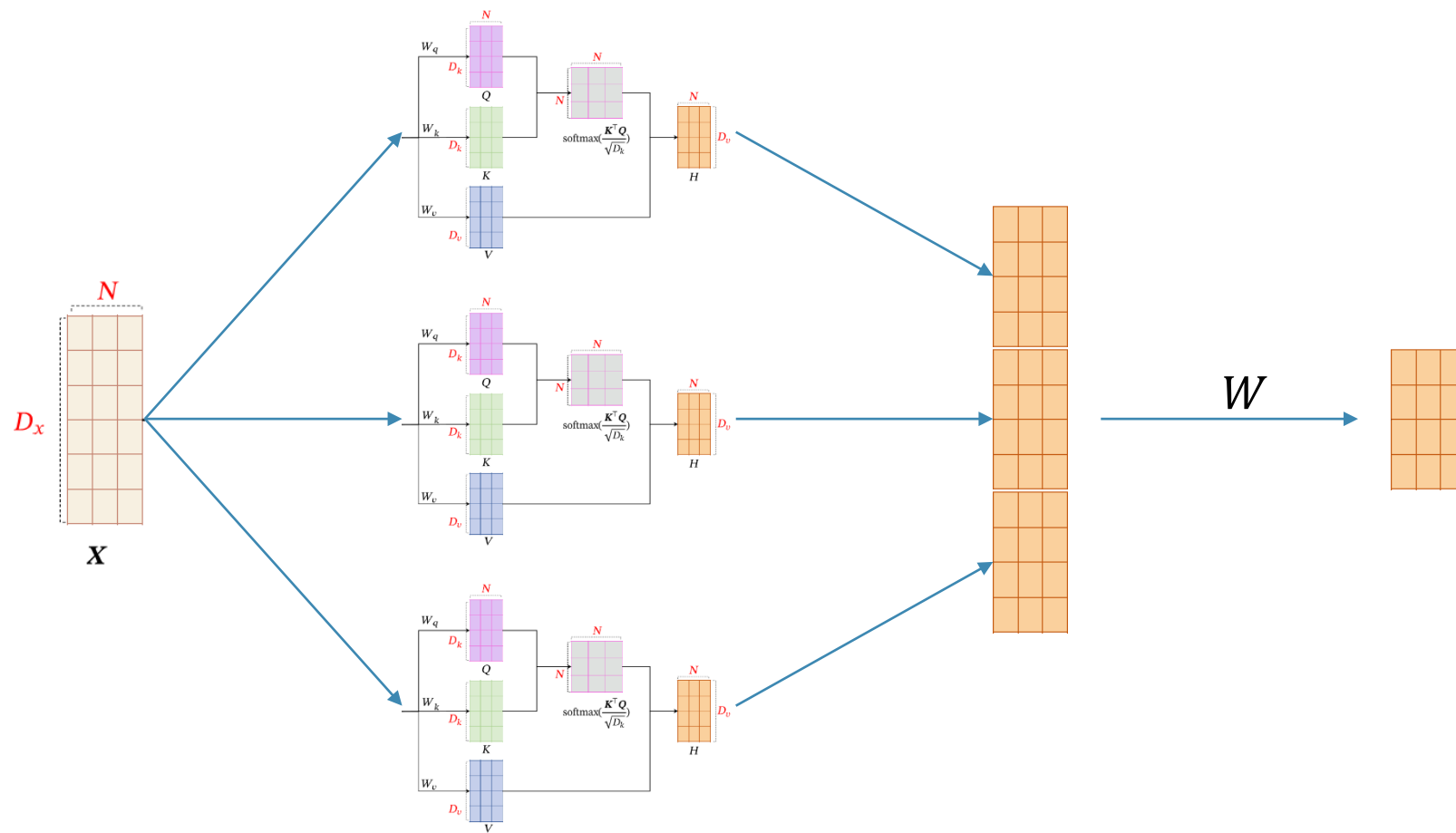
自注意力模型



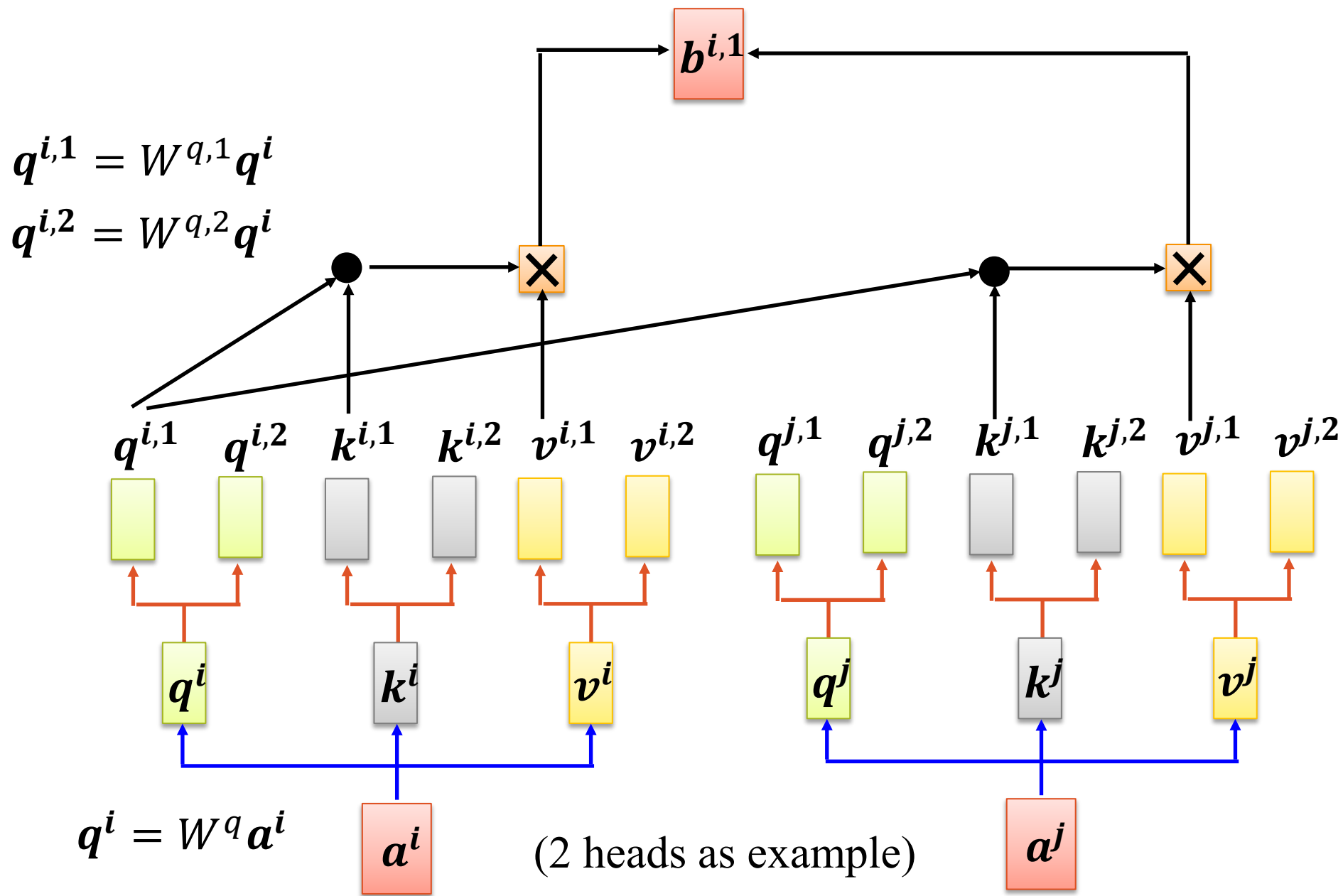
自注意力模型



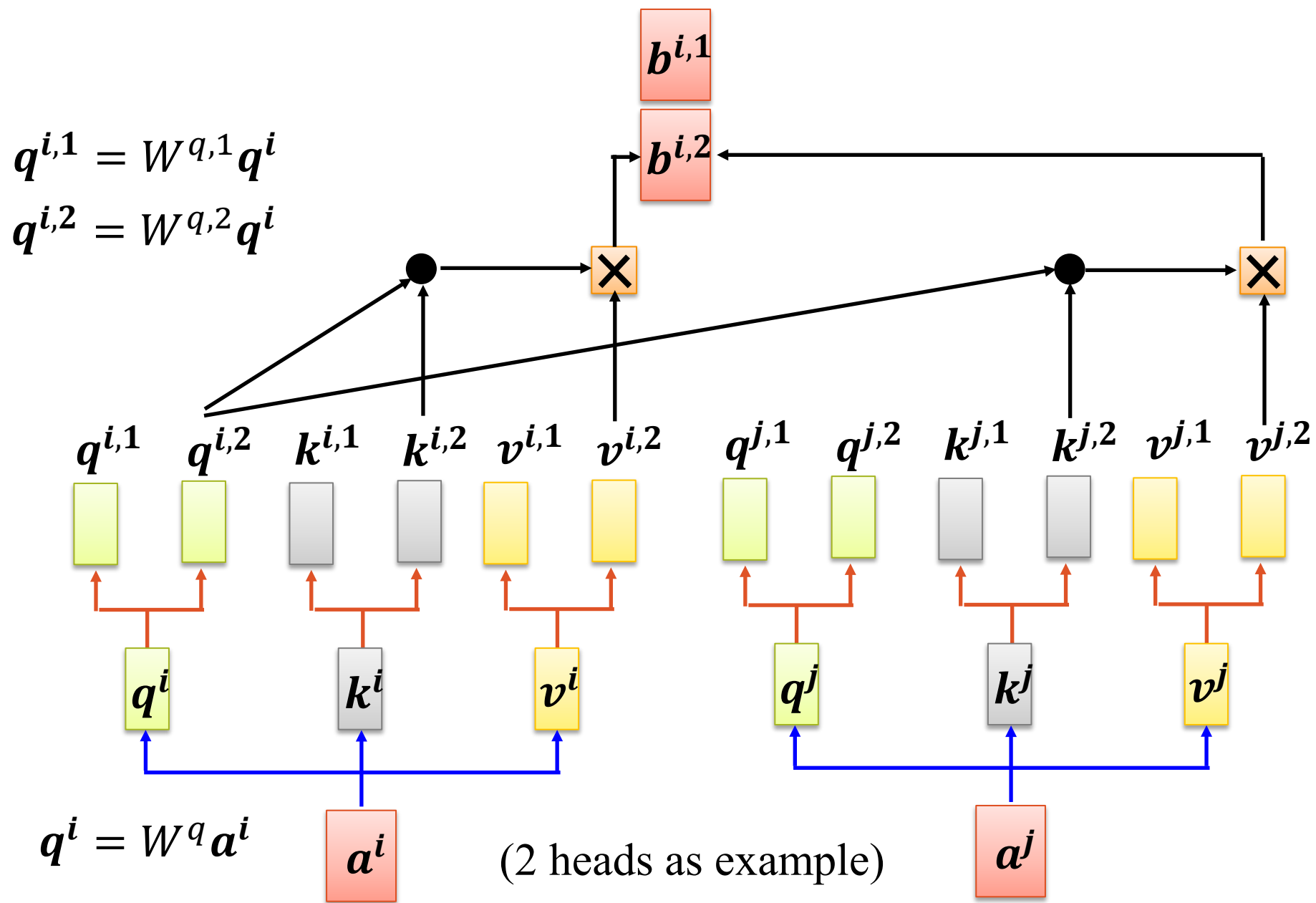
多头 (multi-head) 自注意力模型



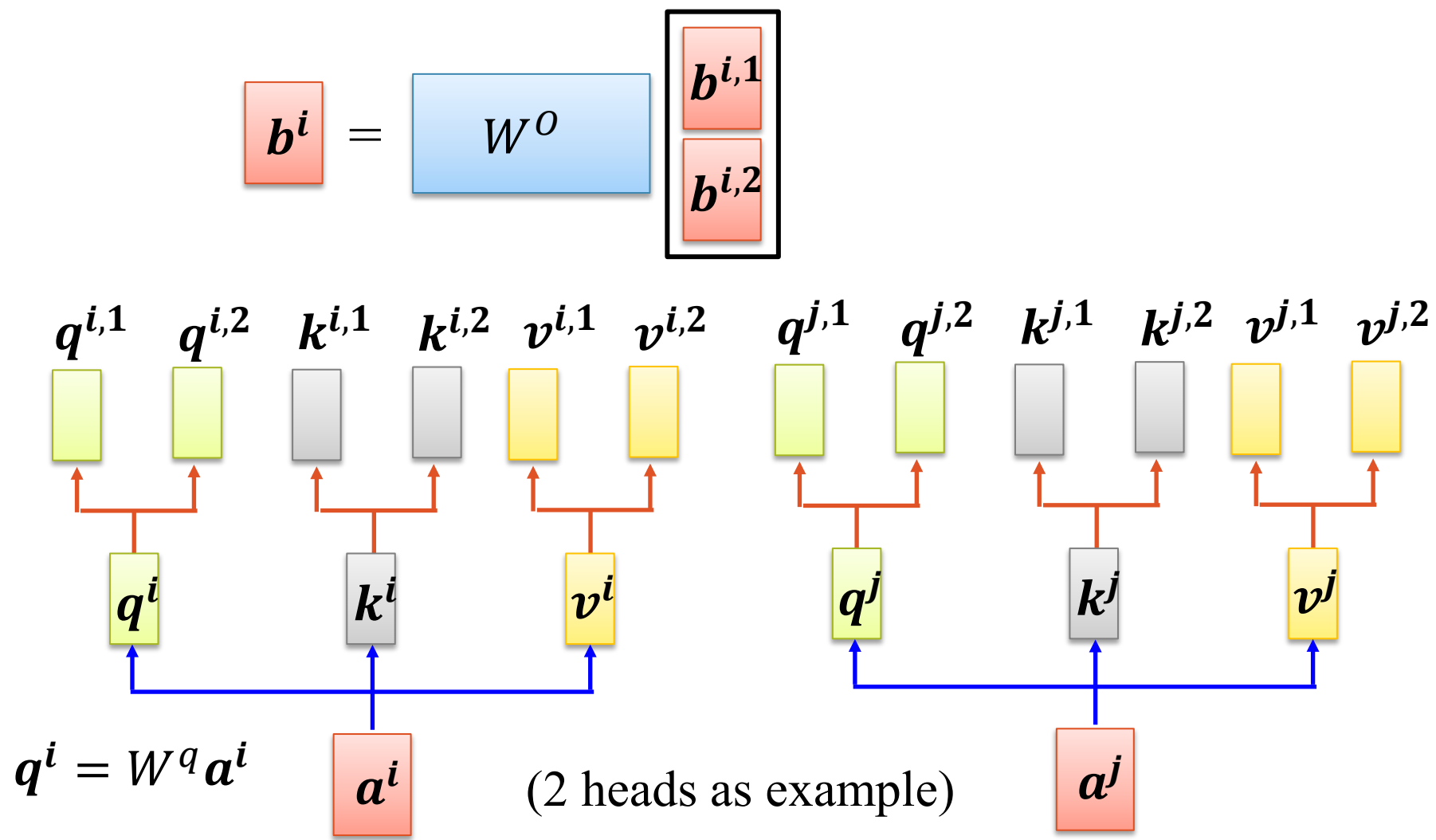
多头自注意力



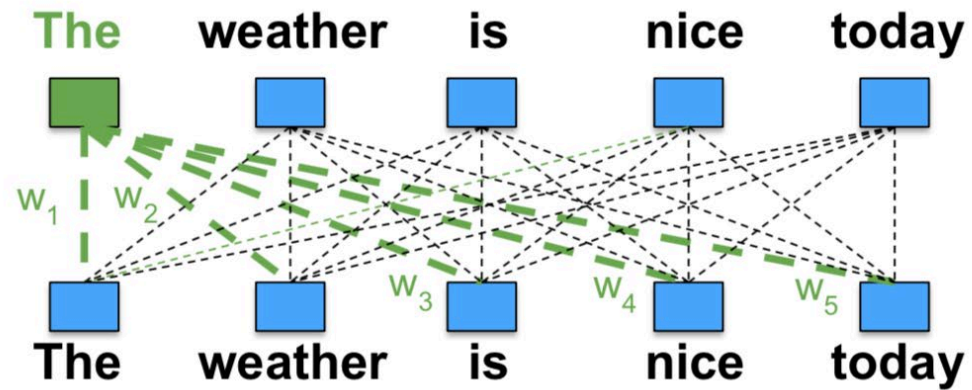
多头自注意力



多头自注意力



自注意力示例



$$w_1, w_2, w_3, w_4, w_5 = \text{softmax} \left(\begin{bmatrix} 0.6 & 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.6 & 0.2 & 0.9 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.1 & 0.1 & 0.1 \\ 0.8 & 0.1 & 0.8 & 0.4 & 0.6 \end{bmatrix} \right)$$

The The weather is nice today

$$\begin{bmatrix} 1.8 \\ 2.3 \\ 0.4 \end{bmatrix} = w_1 \times \begin{bmatrix} 0.6 \\ 0.2 \\ 0.8 \end{bmatrix} + w_2 \times \begin{bmatrix} 0.2 \\ 0.3 \\ 0.1 \end{bmatrix} + w_3 \times \begin{bmatrix} 0.9 \\ 0.1 \\ 0.8 \end{bmatrix} + w_4 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.4 \end{bmatrix} + w_5 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.6 \end{bmatrix}$$

The The weather is nice today

自注意力模型的形式化表示

- 输入序列为 $X = [x_1, \dots, x_N] \in \mathbb{R}^{D_x \times N}$

- 首先生成三个向量序列

$$Q = W_q X \in \mathbb{R}^{D_k \times N},$$

$$K = W_k X \in \mathbb{R}^{D_k \times N},$$

$$V = W_v X \in \mathbb{R}^{D_v \times N},$$

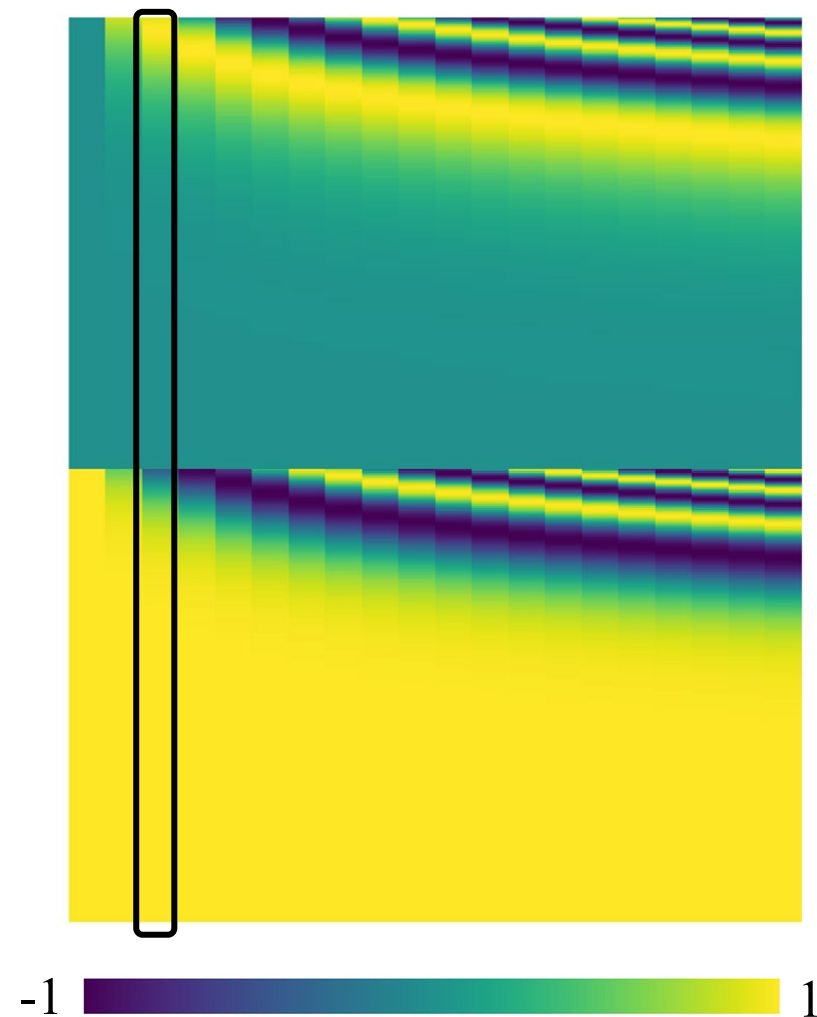
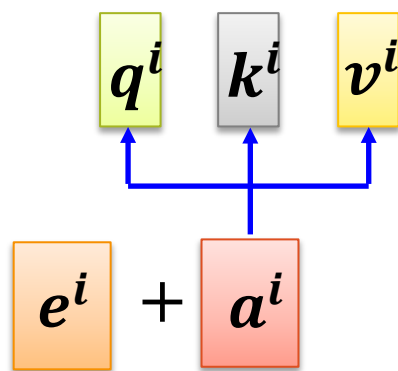
- 计算 h_n $h_n = \text{att}((K, V), q_n)$

- 如果使用缩放点积来作为注意力打分函数，输出向量序列可以简写为

$$H = V \text{softmax}\left(\frac{K^\top Q}{\sqrt{D_k}}\right),$$

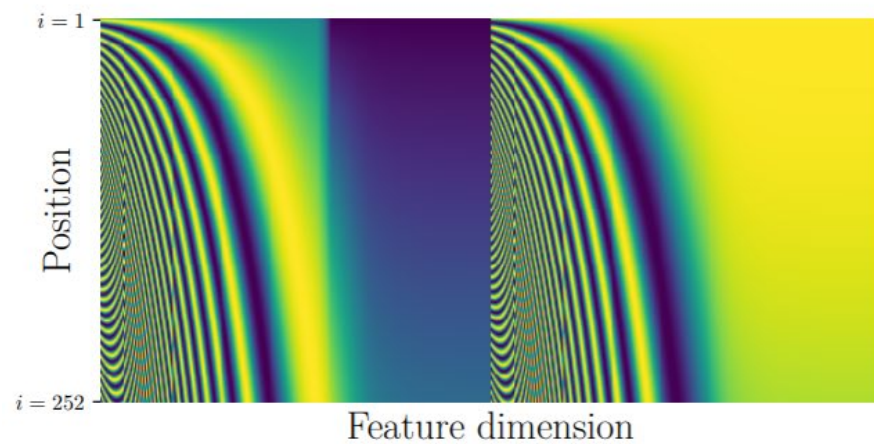
位置编码

- 自注意力机制没有位置信息
- 解决办法：为每个位置设置一个位置向量 e^i
- 把位置向量跟原始向量相加
- 如何设计位置编码？
 - 人工设计
 - 从数据中学习

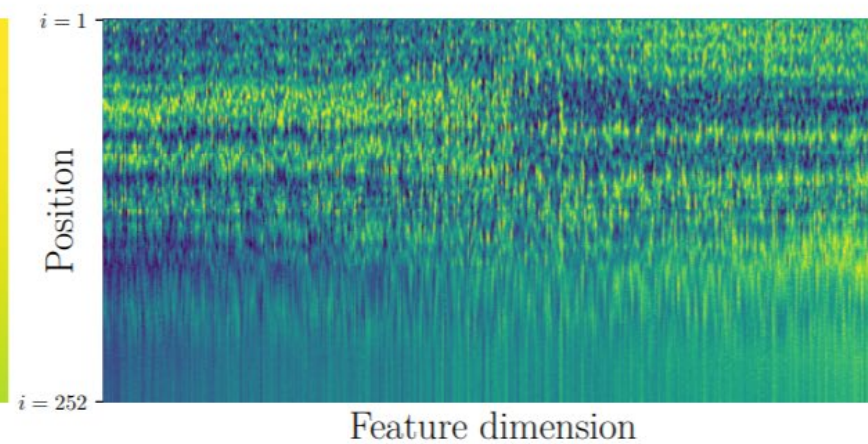


位置编码

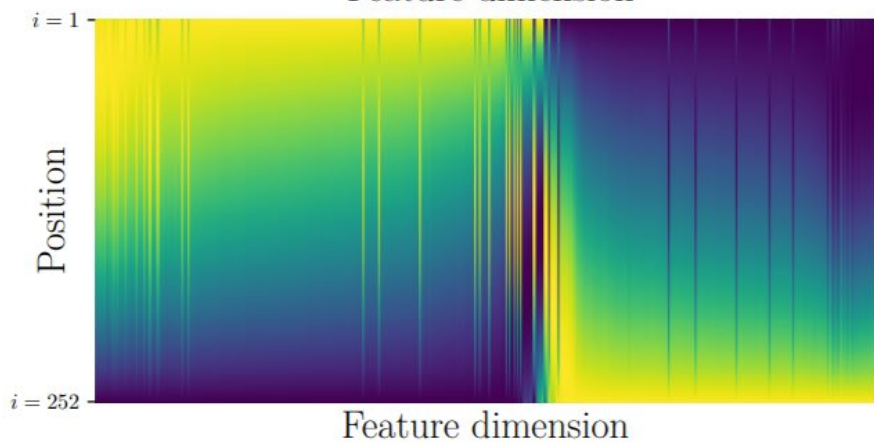
(a) Sinusoidal



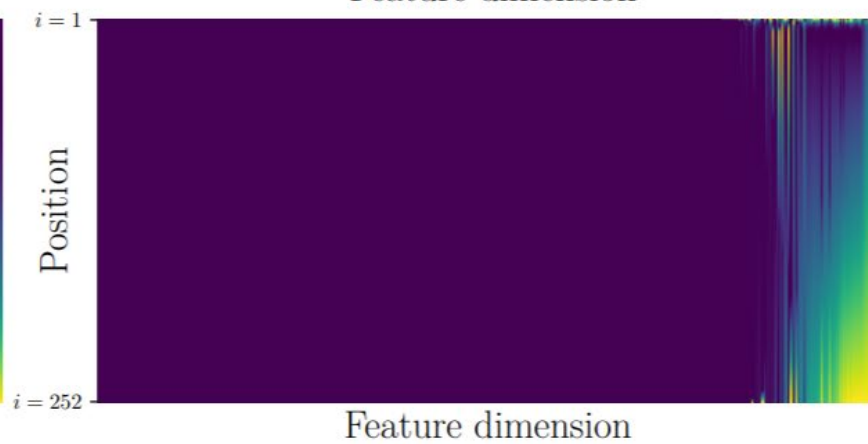
(b) Position embedding



(c) FLOATER



(d) RNN





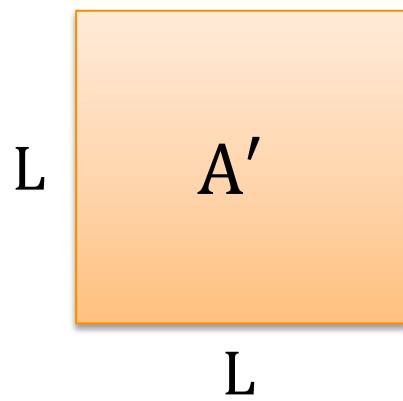
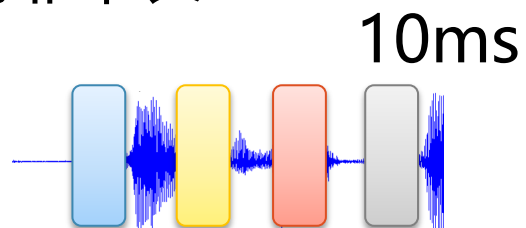
Transformer



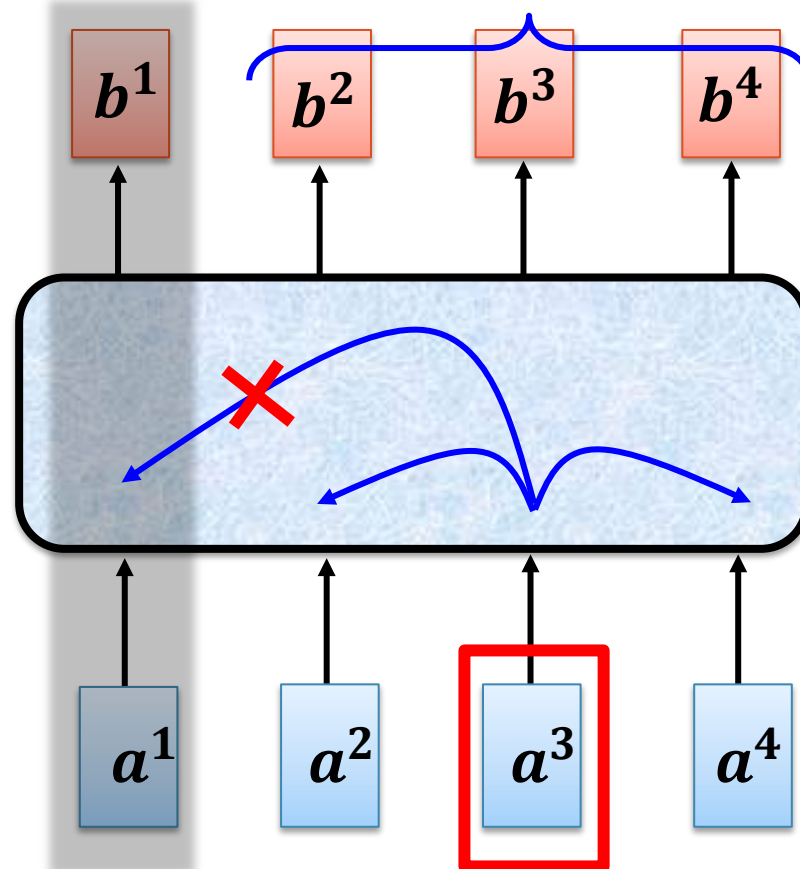
BERT

自注意力的应用

语音序列非常长



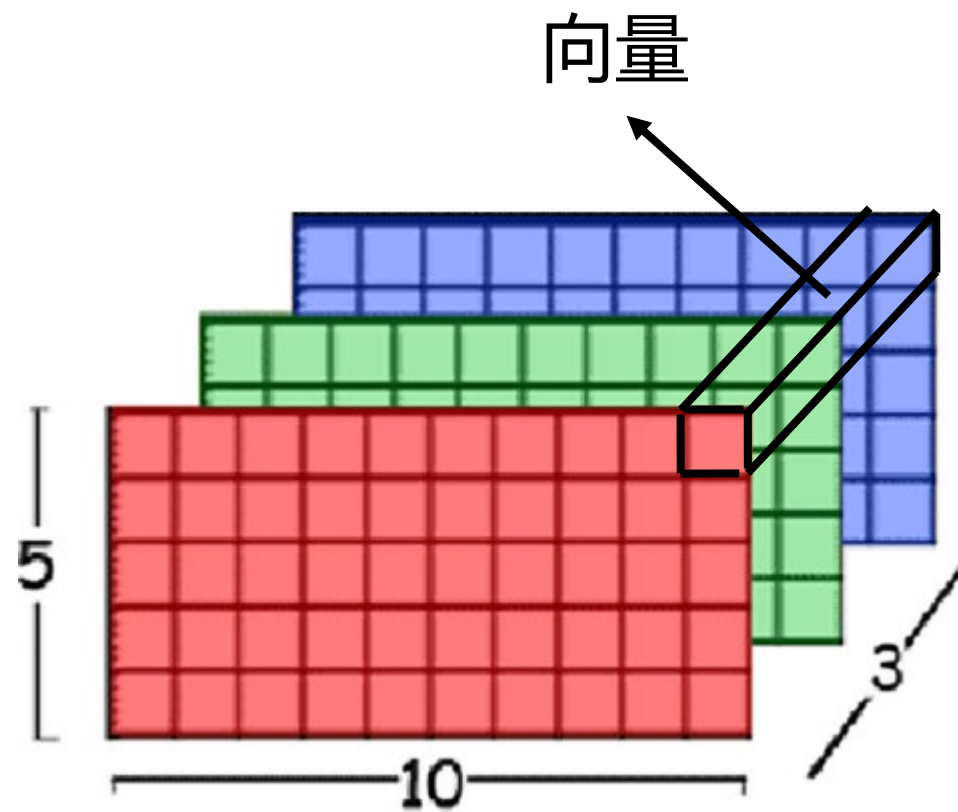
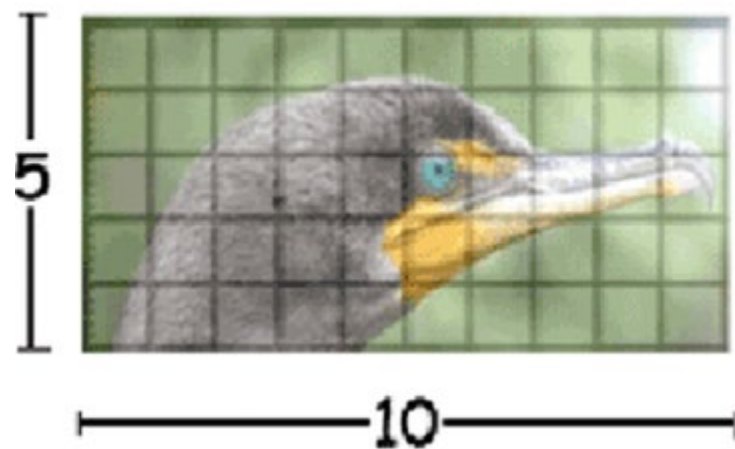
取一定范围的注意力



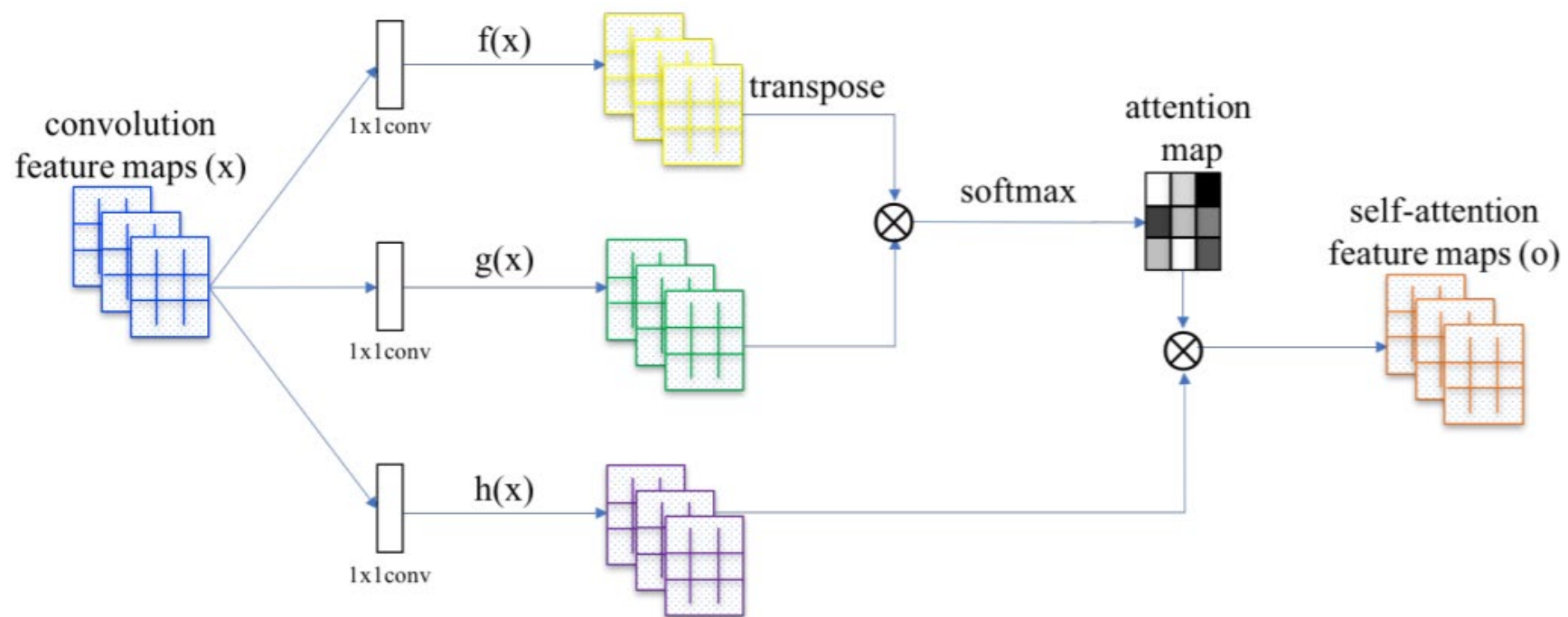
Truncated Self-attention

自注意力的应用

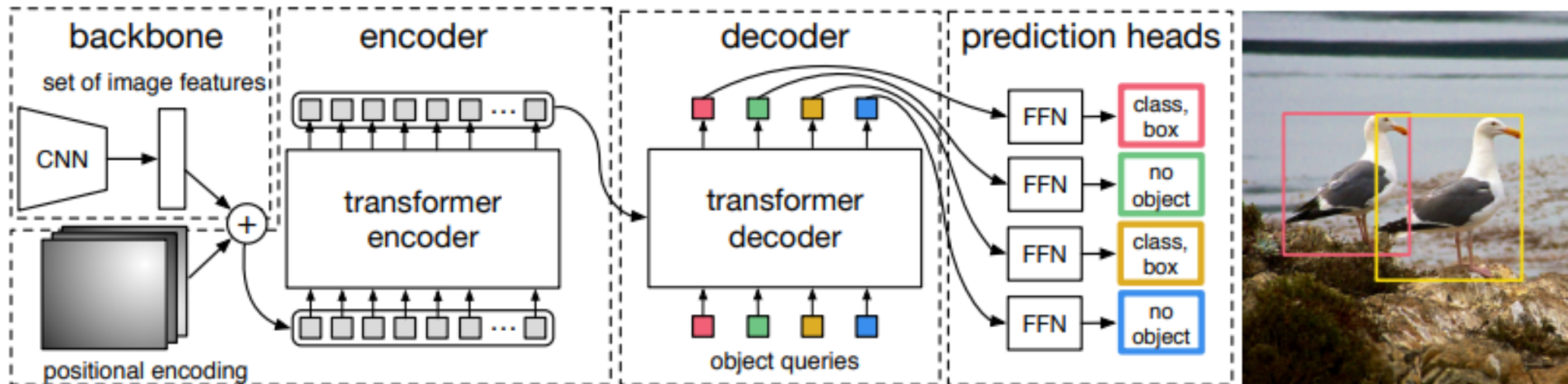
把图片看成向量的集合



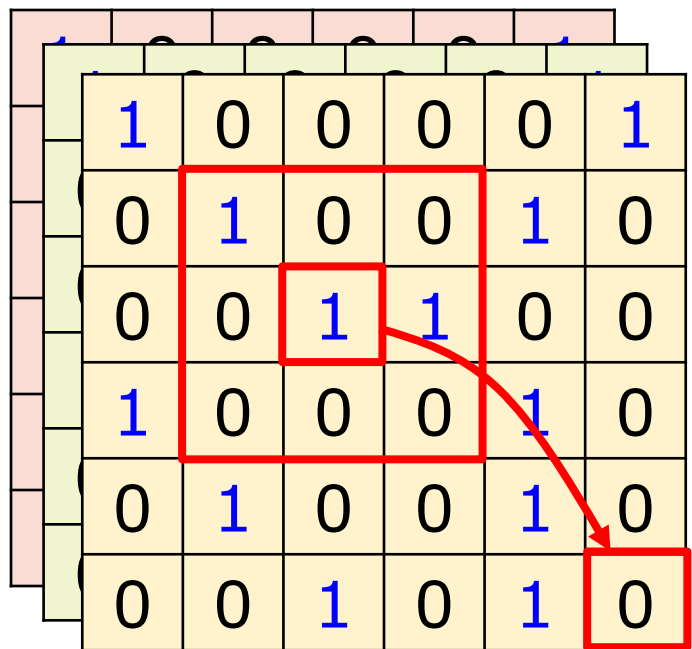
Self-Attention GAN



DEtection Transformer (DETR)



比较：自注意力与卷积神经网络



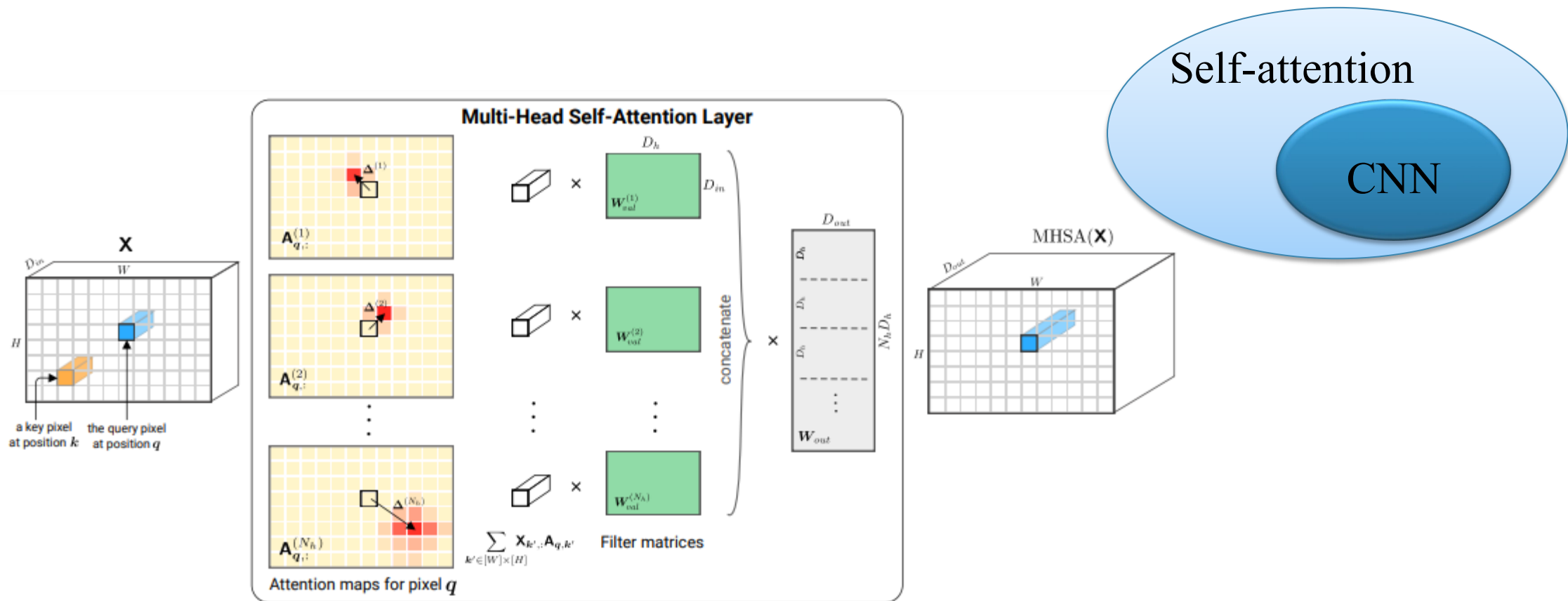
CNN: 可以看成一种限制在局部感受野的自注意力

➤ CNN是一种简化版的自注意力

自注意力: 有可学习感受野的CNN

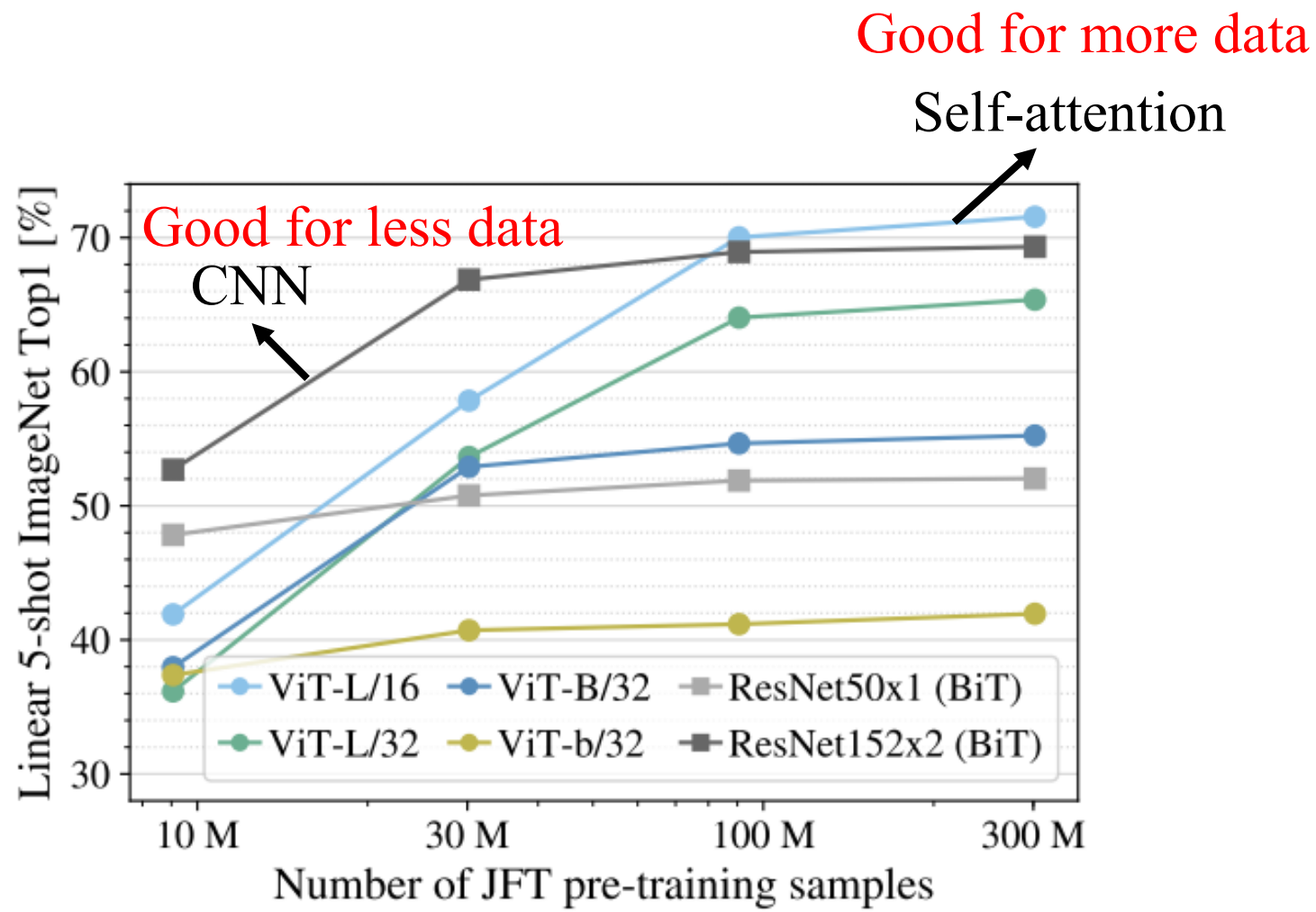
➤ 自注意力是一种复杂版本的CNN

比较：自注意力与卷积神经网络

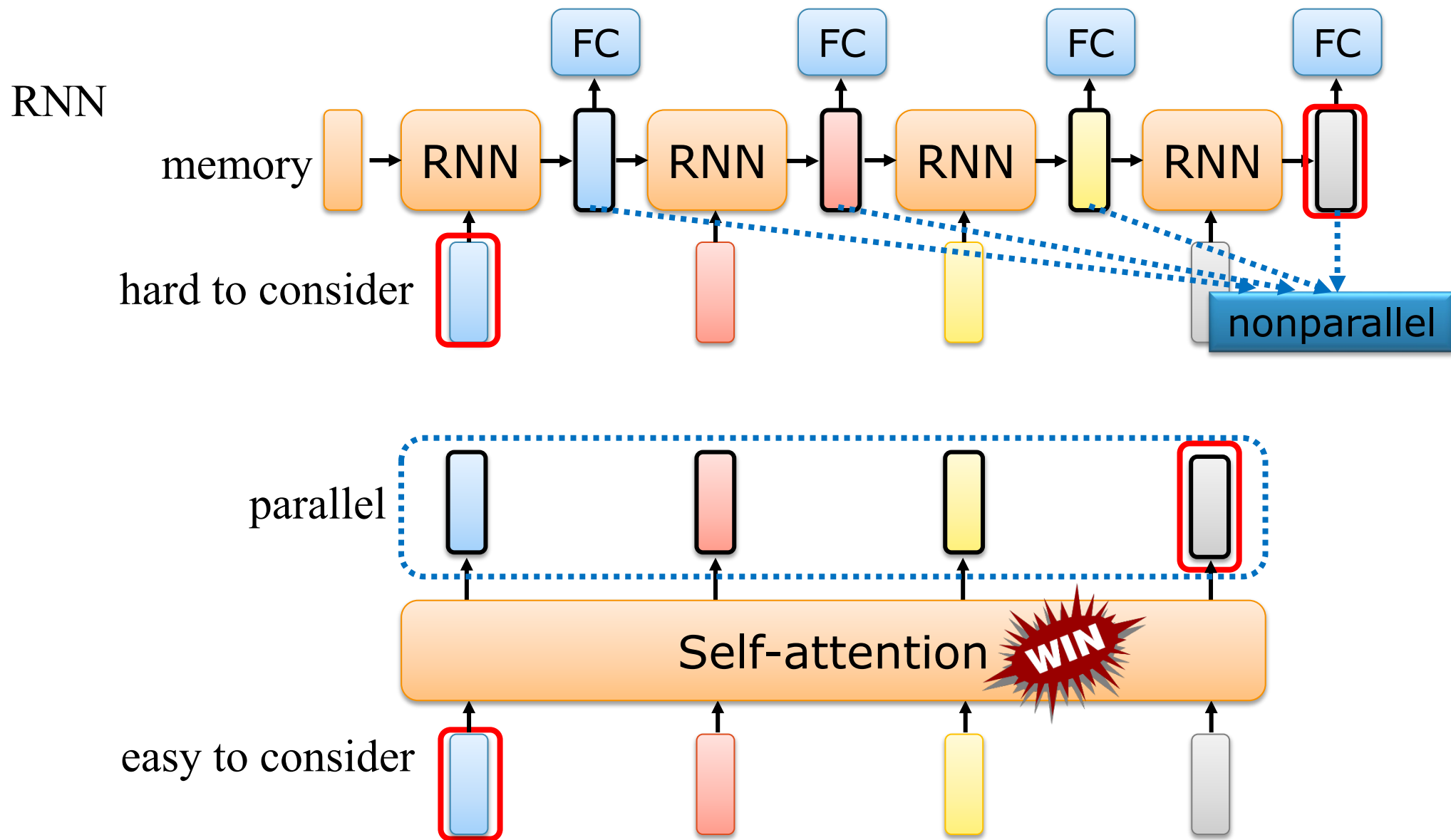


On the Relationship between Self-Attention and Convolutional Layers

比较：自注意力与卷积神经网络



比较：自注意力与循环神经网络



复杂度分析

模型	每层复杂度	序列操作数	最大路径长度
CNN	$O(kLd^2)$	$O(1)$	$O(\log_k(L))$
RNN	$O(Ld^2)$	$O(L)$	$O(L)$
自注意力	$O(L^2d)$	$O(1)$	$O(1)$

k 卷积核大小 L 序列长度 d 维度

Transformer



Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com	Noam Shazeer* Google Brain noam@google.com	Niki Parmar* Google Research nikip@google.com	Jakob Uszkoreit* Google Research usz@google.com
Llion Jones* Google Research llion@google.com	Aidan N. Gomez*[†] University of Toronto aidan@cs.toronto.edu	Łukasz Kaiser* Google Brain lukaszkaiser@google.com	
Illia Polosukhin*[‡] illia.polosukhin@gmail.com			

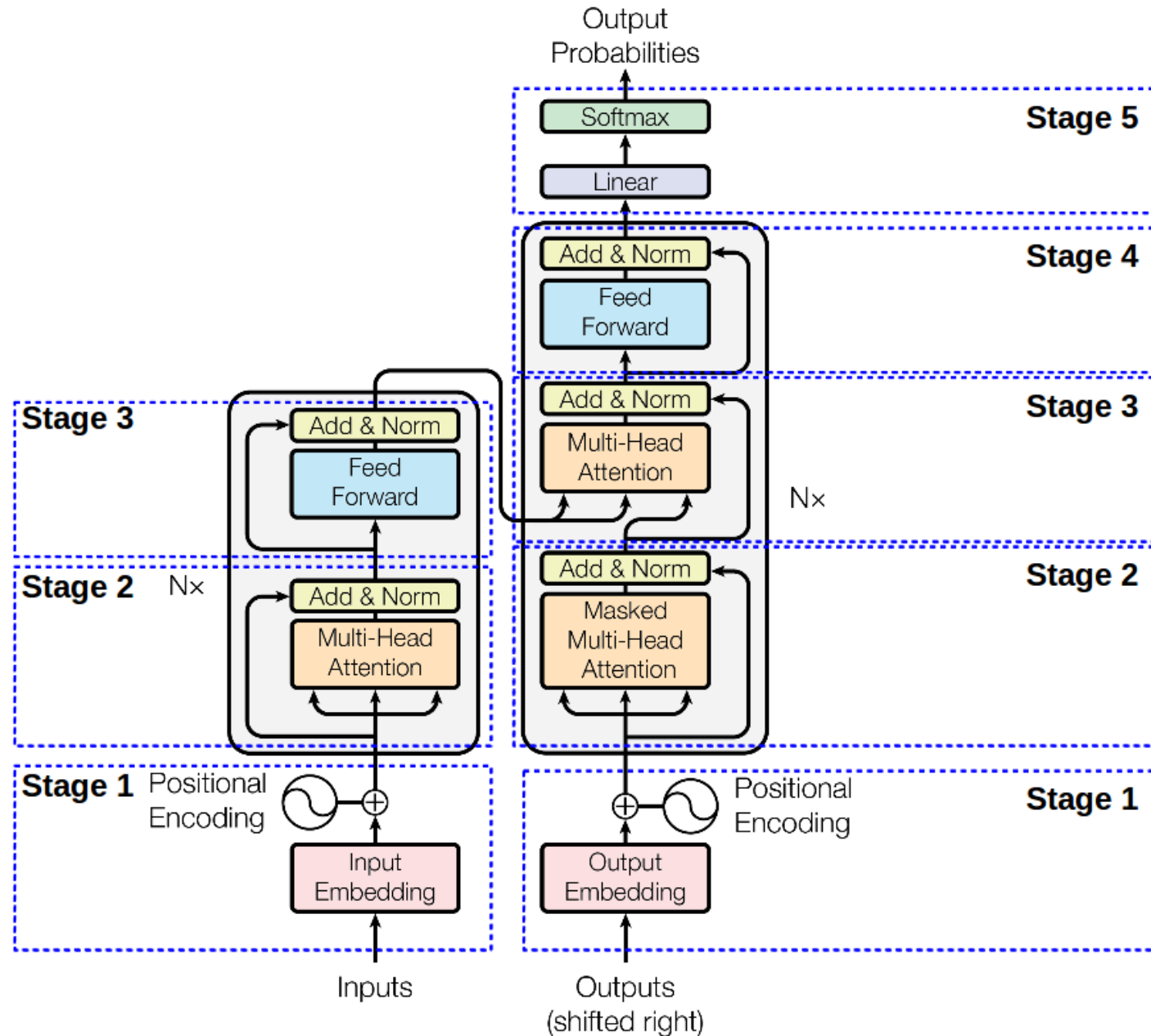
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions

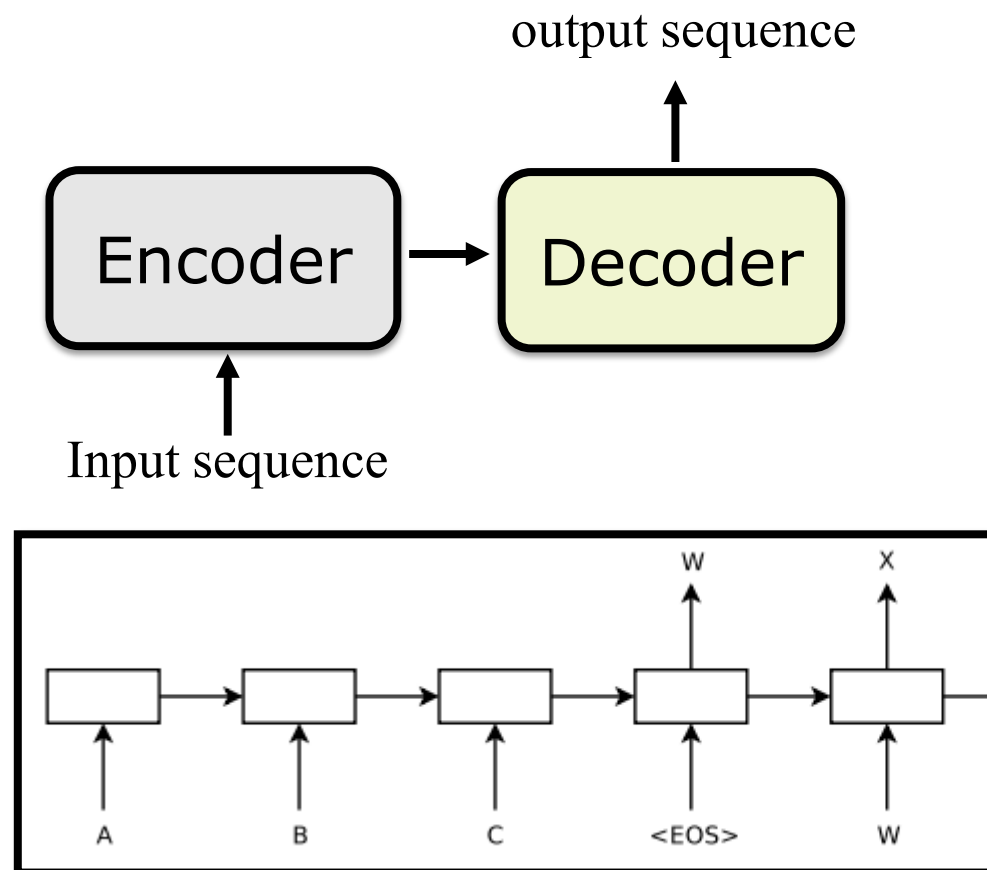


Transformer

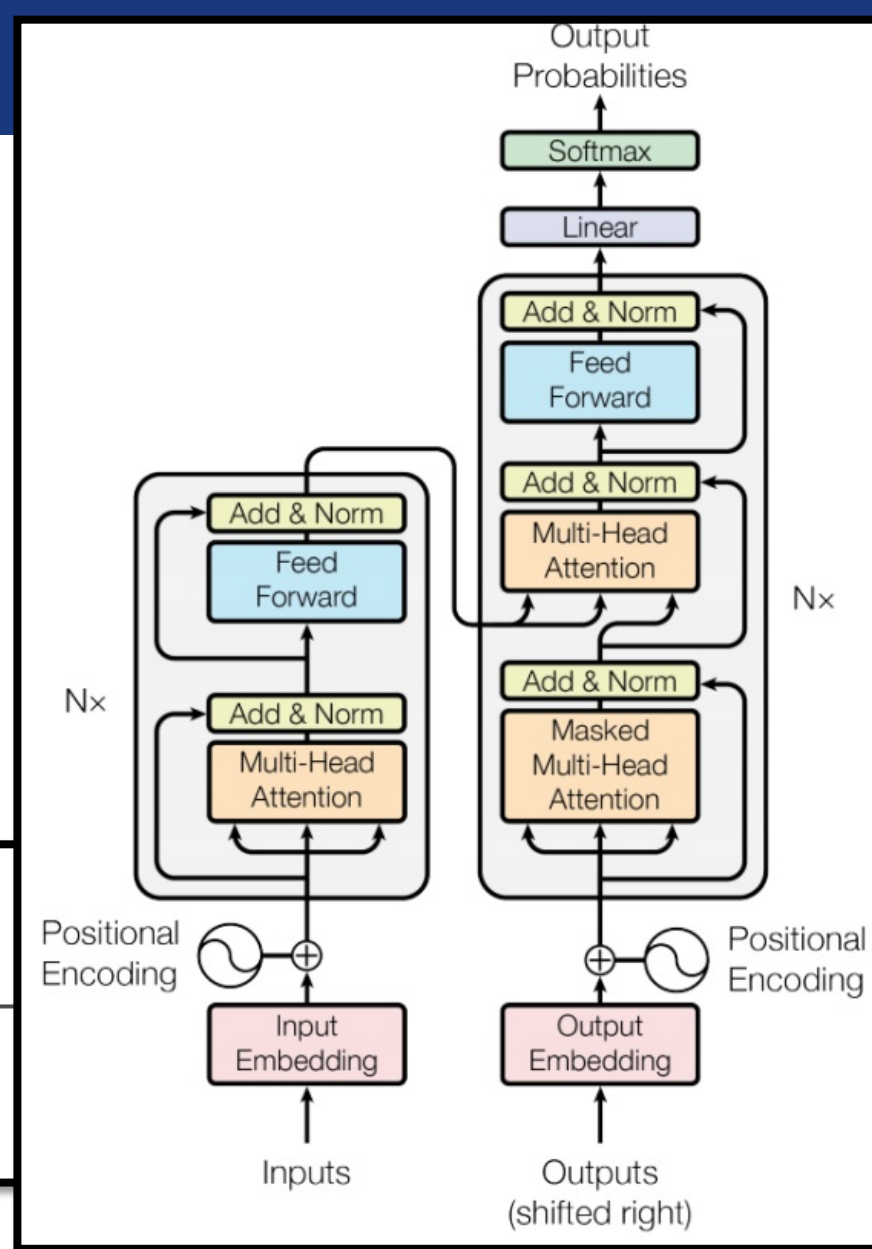
Transformer



Transformer是一种Seq2Seq模型



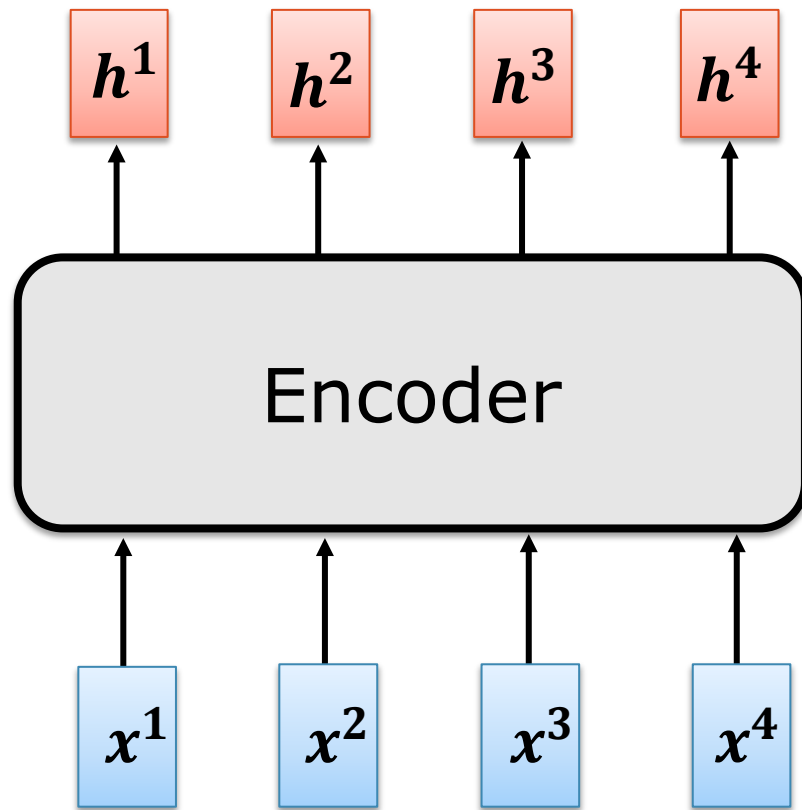
Sequence to Sequence Learning
with Neural Networks



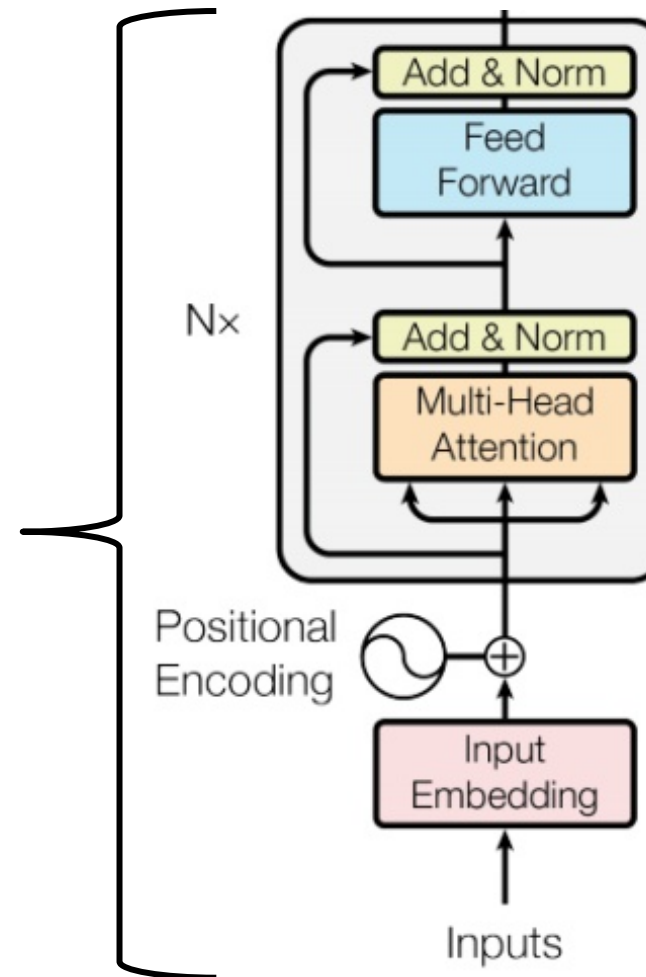
Transformer

Transformer Encoder

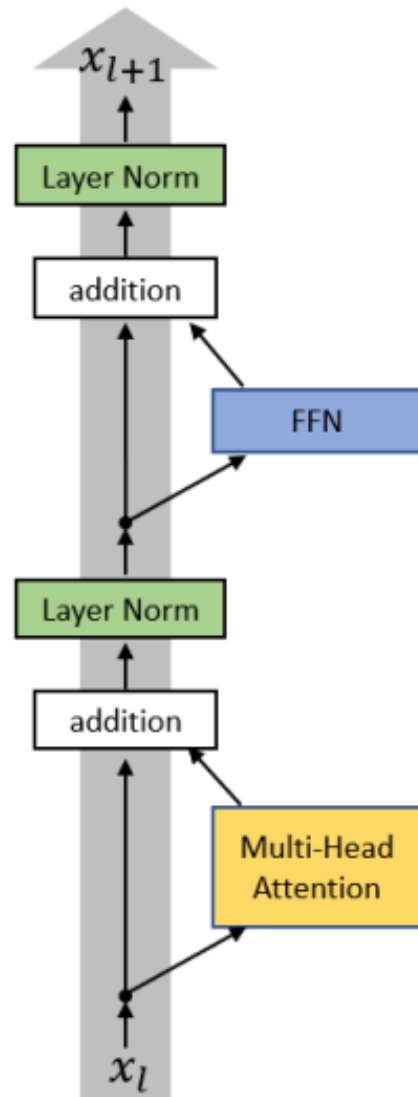
RNN, CNN, or ... ?



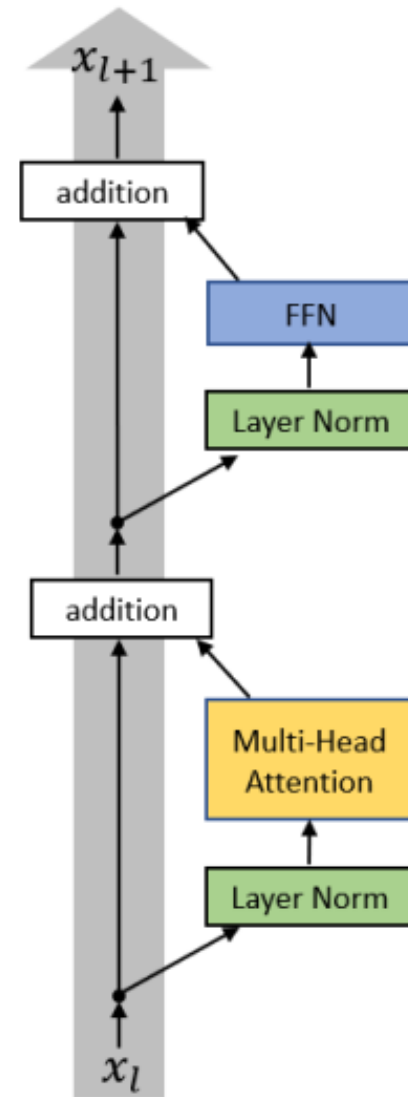
Transformer's Encoder



Transformer Encoder



(a)



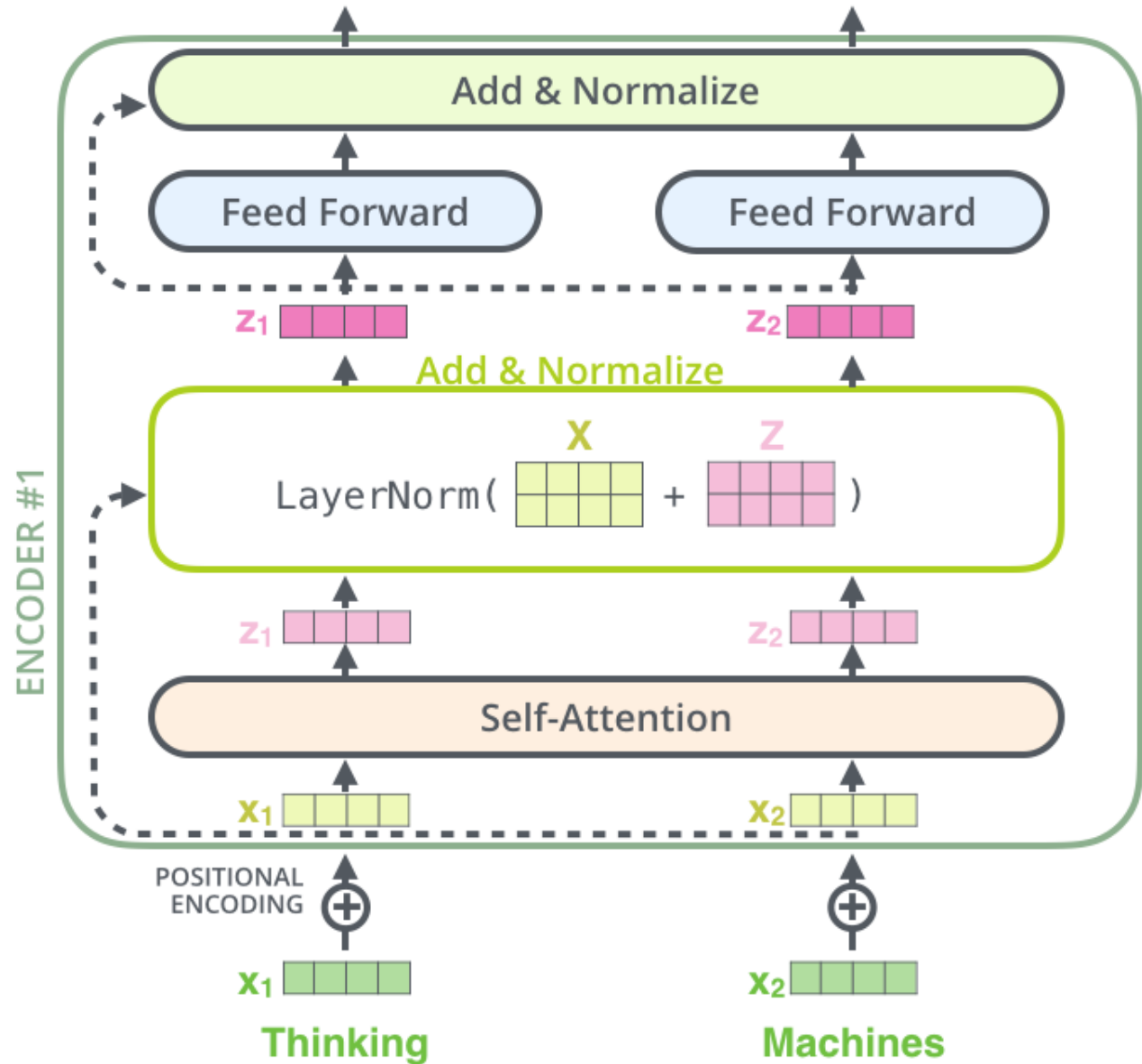
(b)

Transformer Encoder

- 基于自注意力的模型

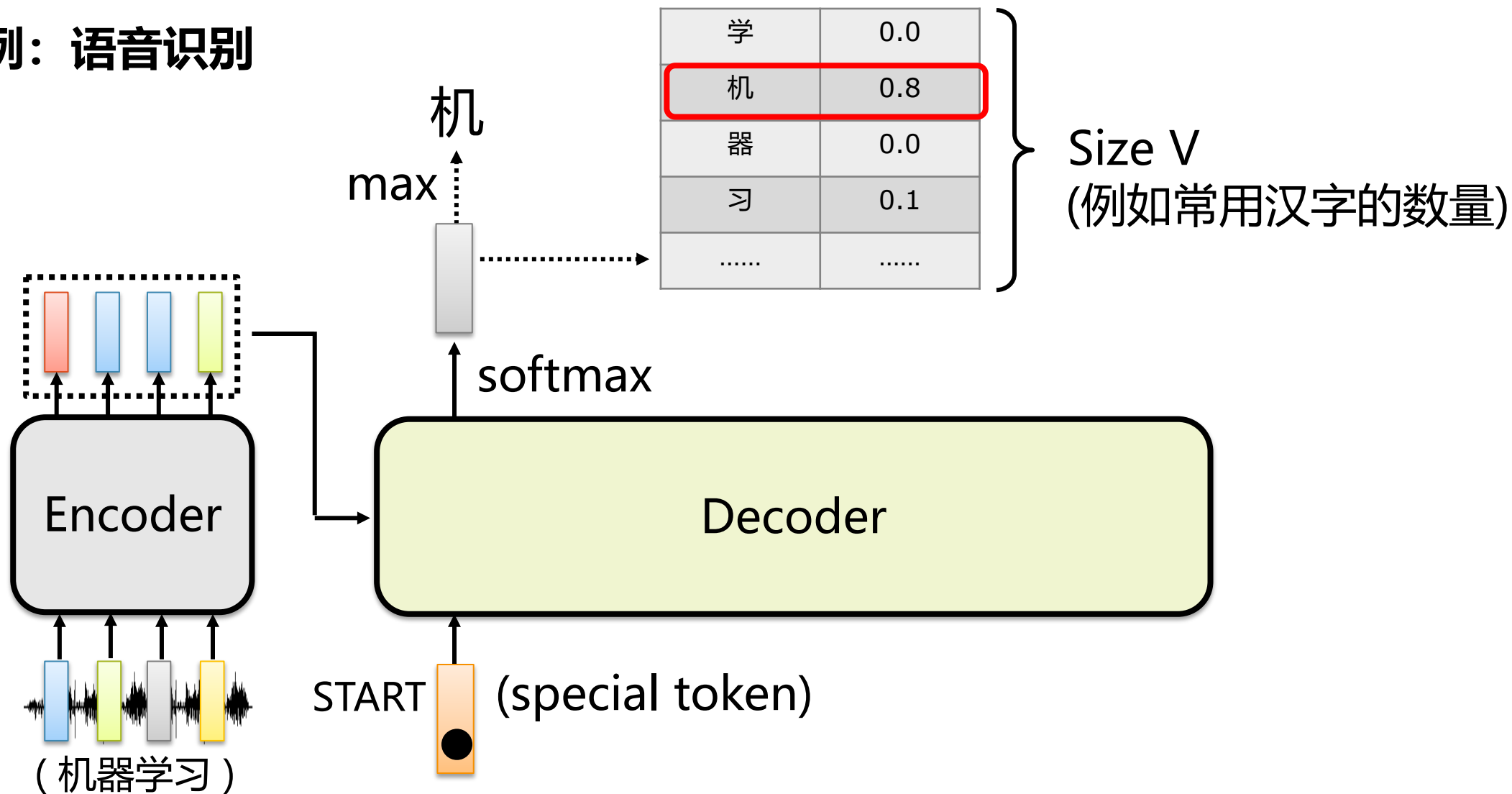
- 引入其它操作

- 位置编码
- 层归一化
- 直连边
- 逐位的FNN

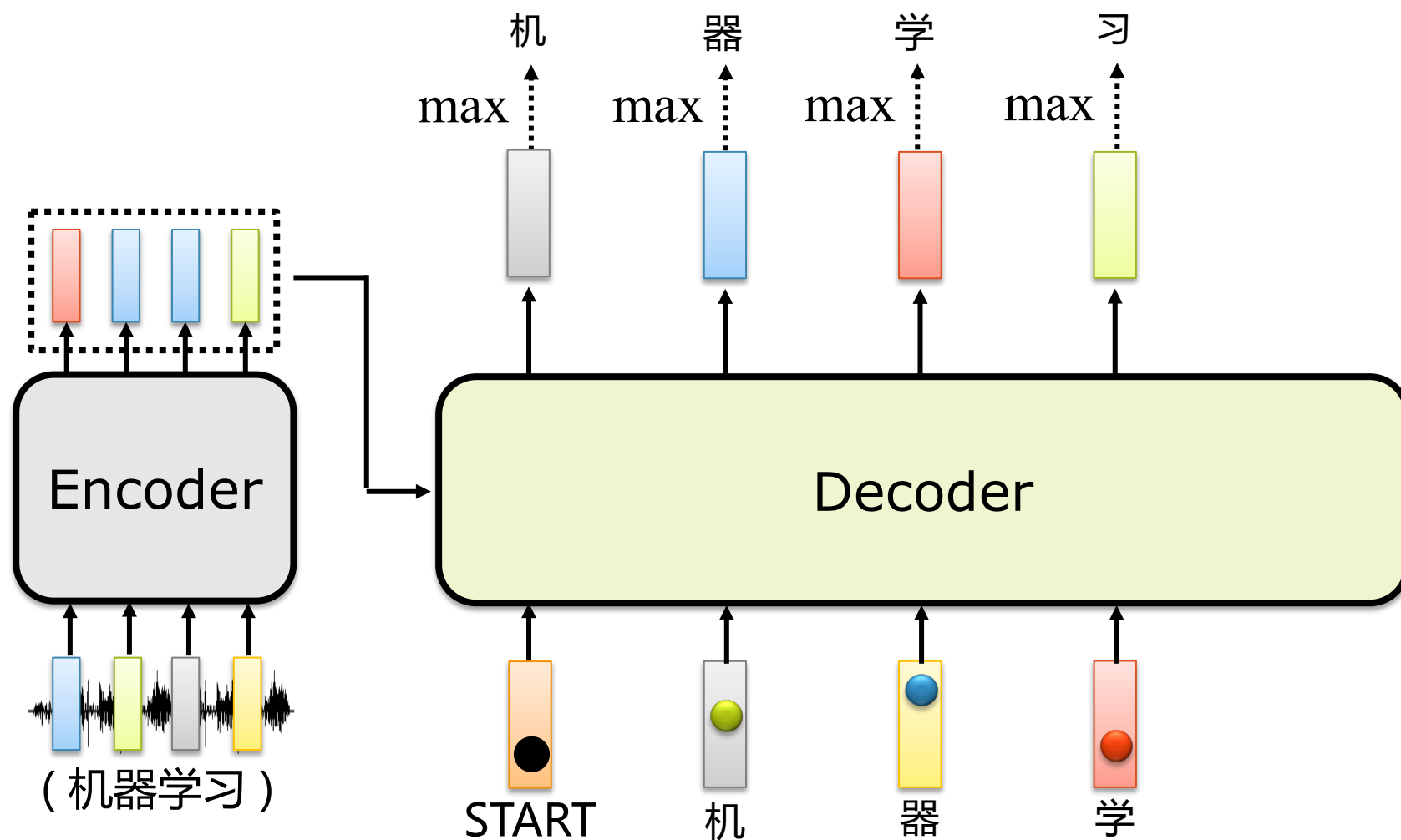


Transformer Decoder —— Autoregressive (AT)

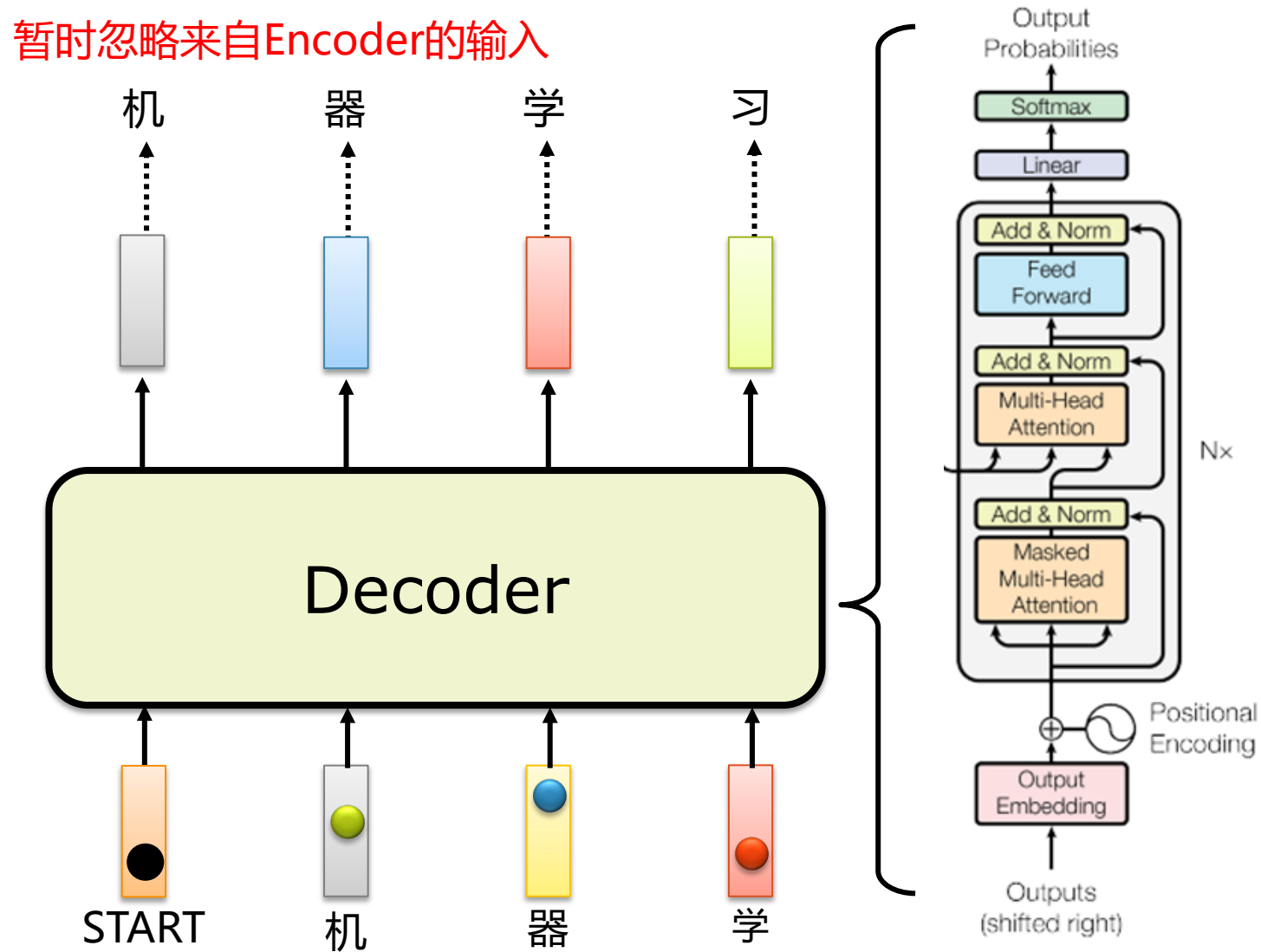
案例：语音识别



Transformer Decoder — Autoregressive (AT)

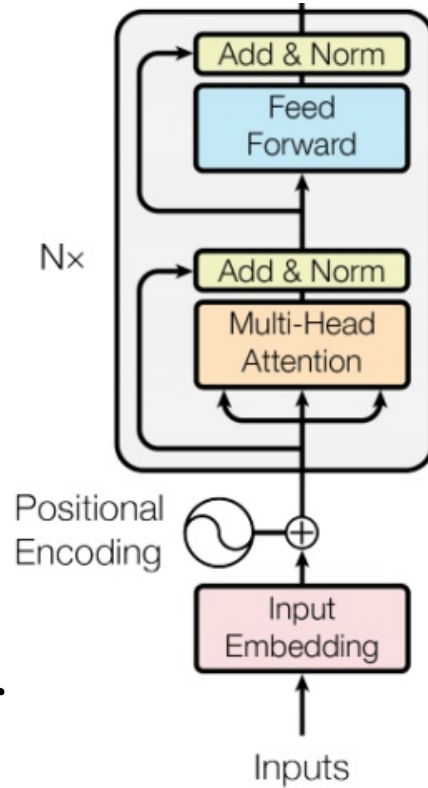


Transformer Decoder — Autoregressive (AT)

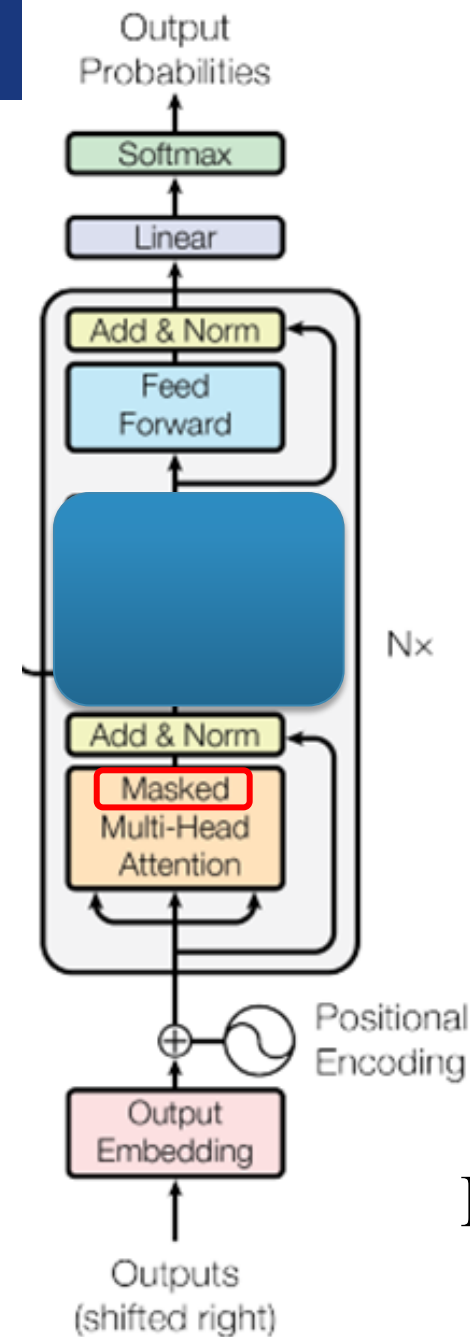


Encoder VS Decoder

Encoder

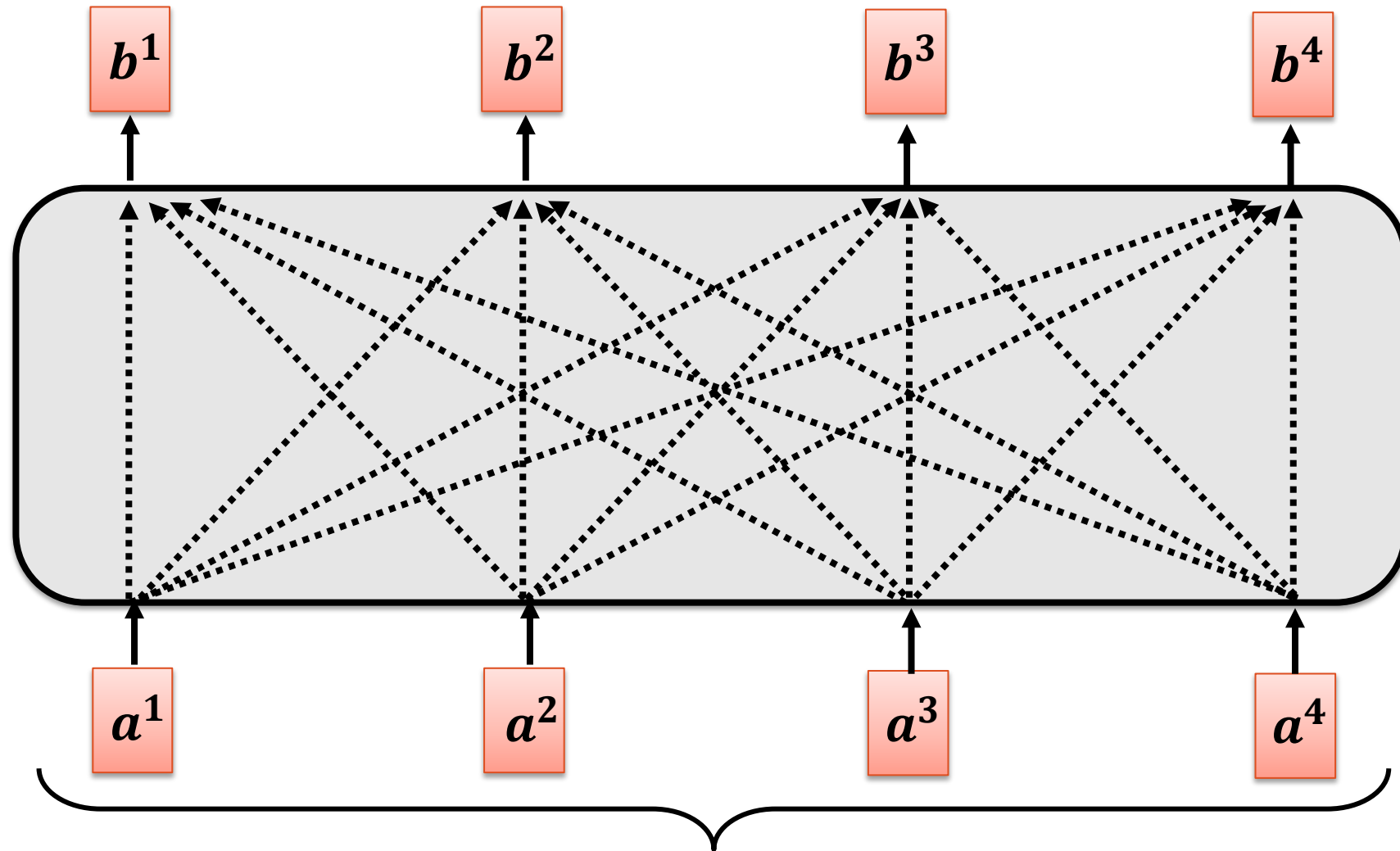


Decoder



Masked Self-attention

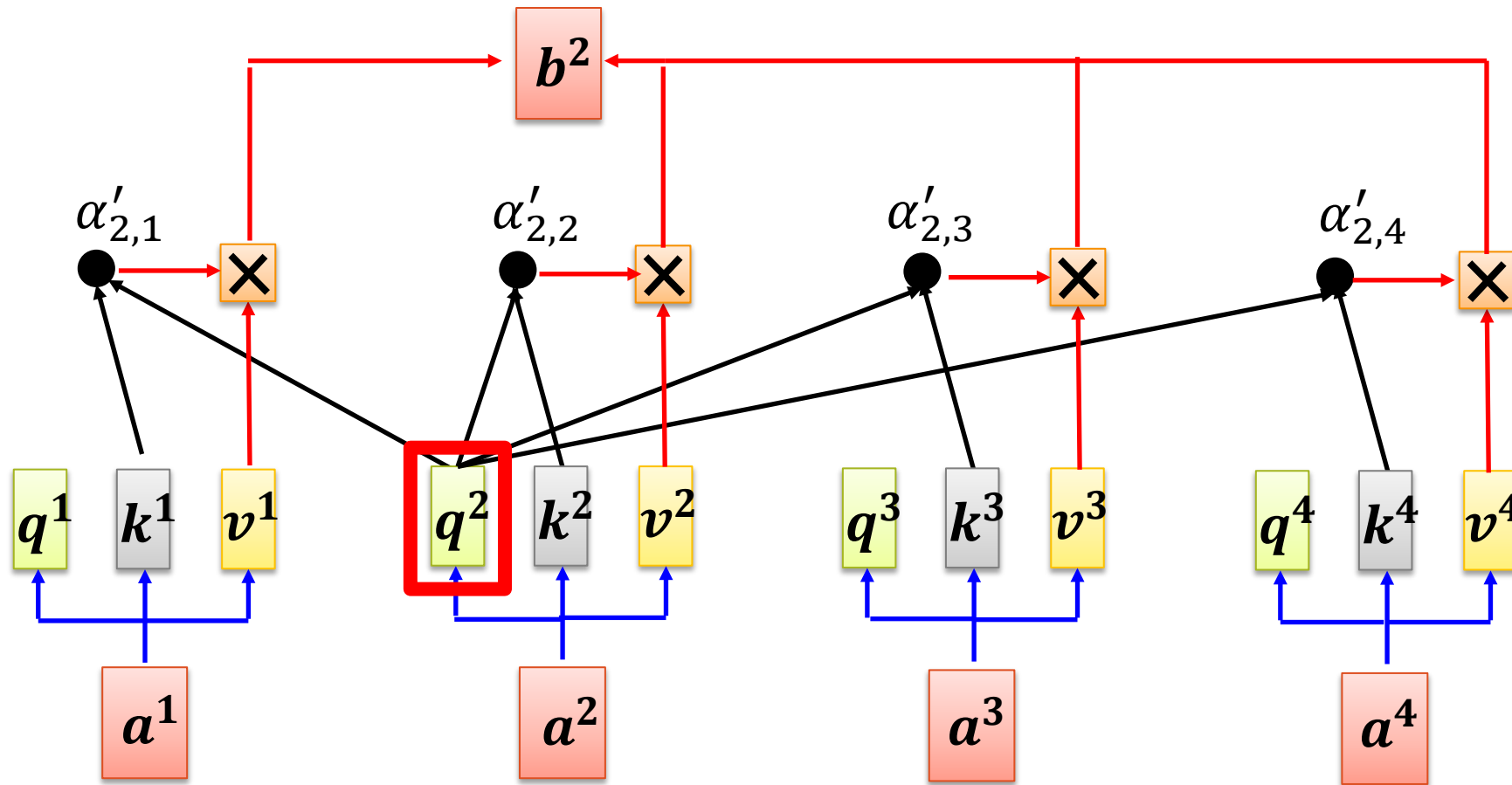
Self-attention $\longrightarrow$ Masked Self-attention



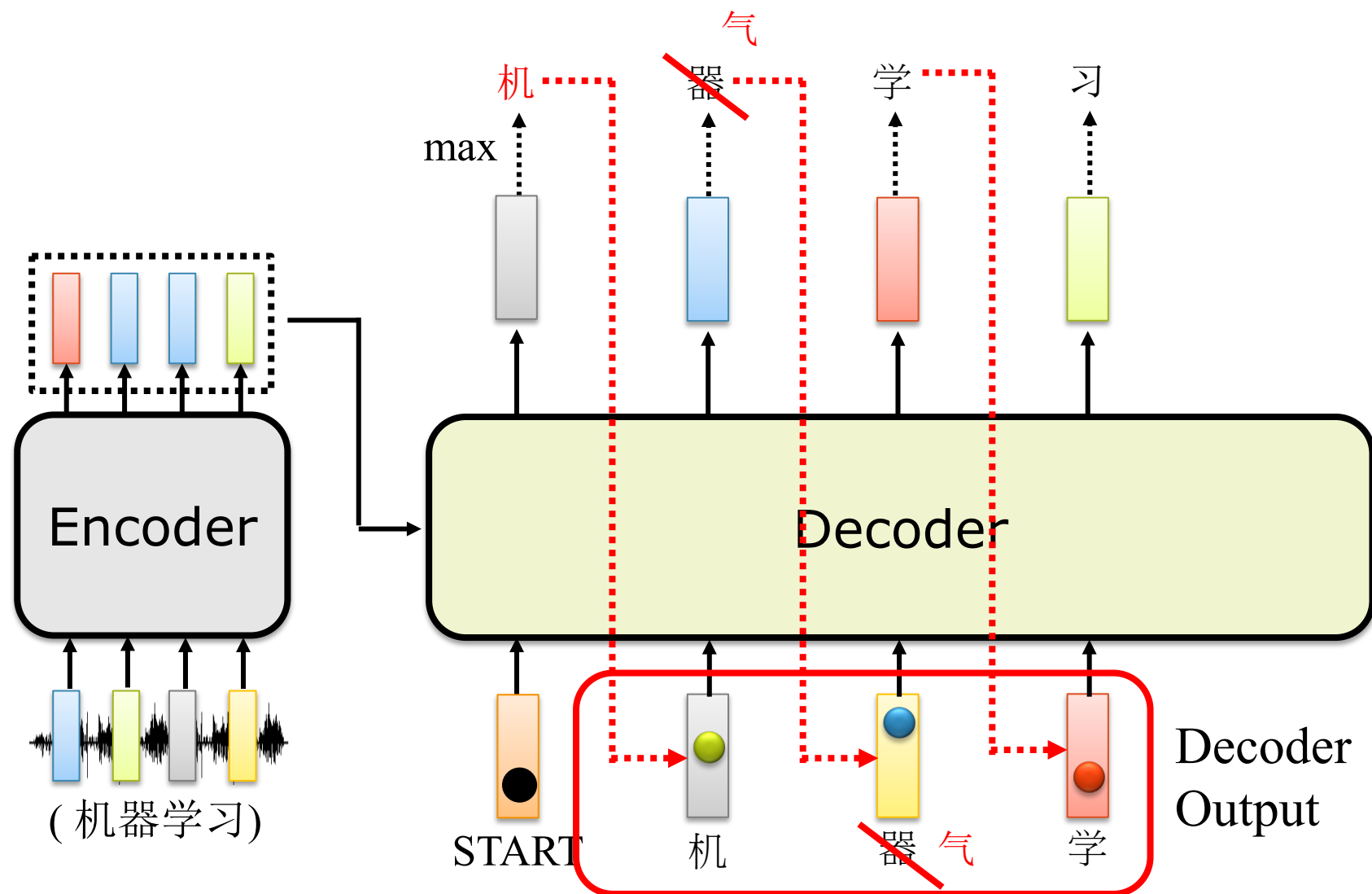
原始输入或来自隐层的输入

Masked Self-attention

Self-attention $\longrightarrow$ Masked Self-attention

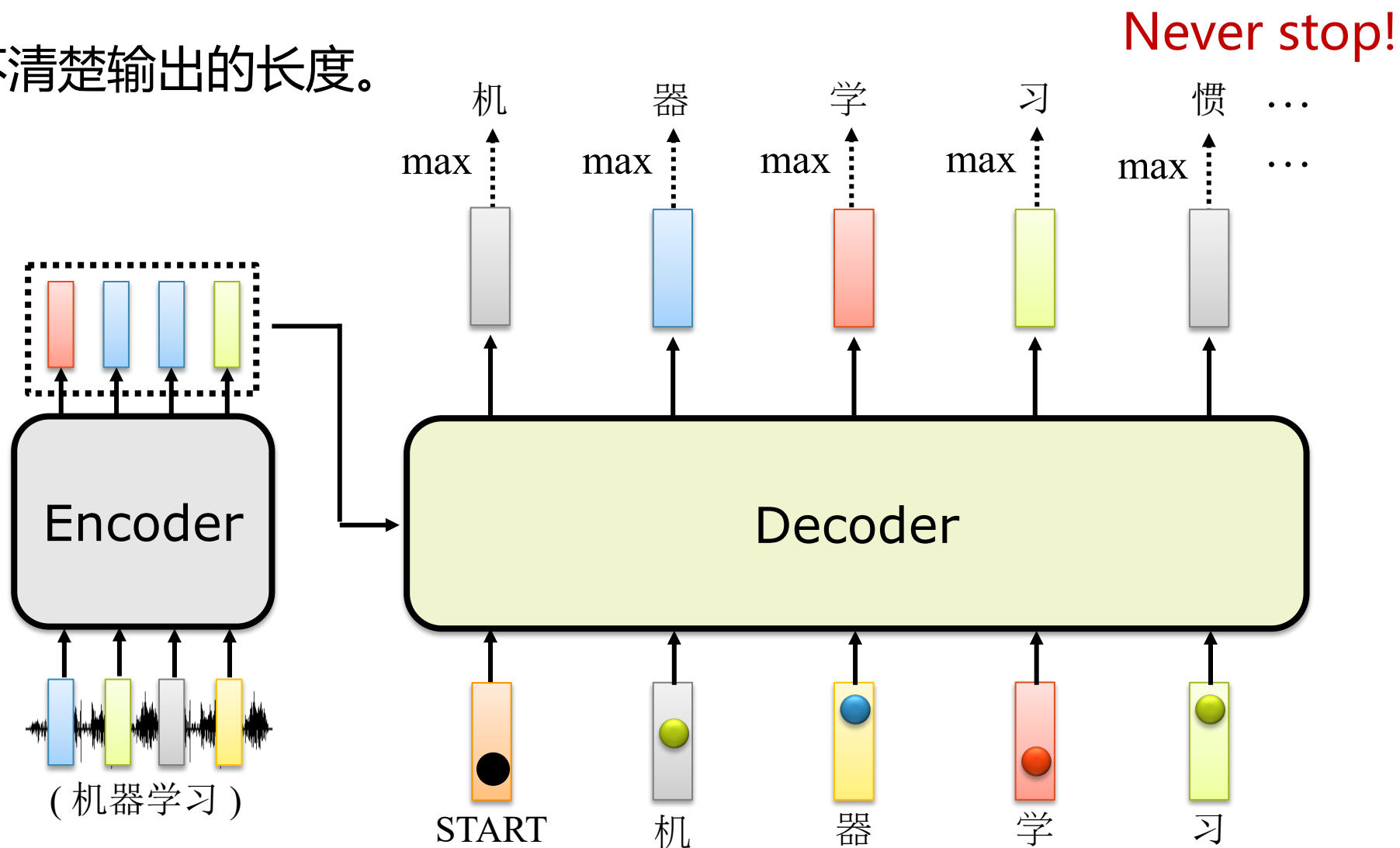


Transformer Decoder — Autoregressive (AT)



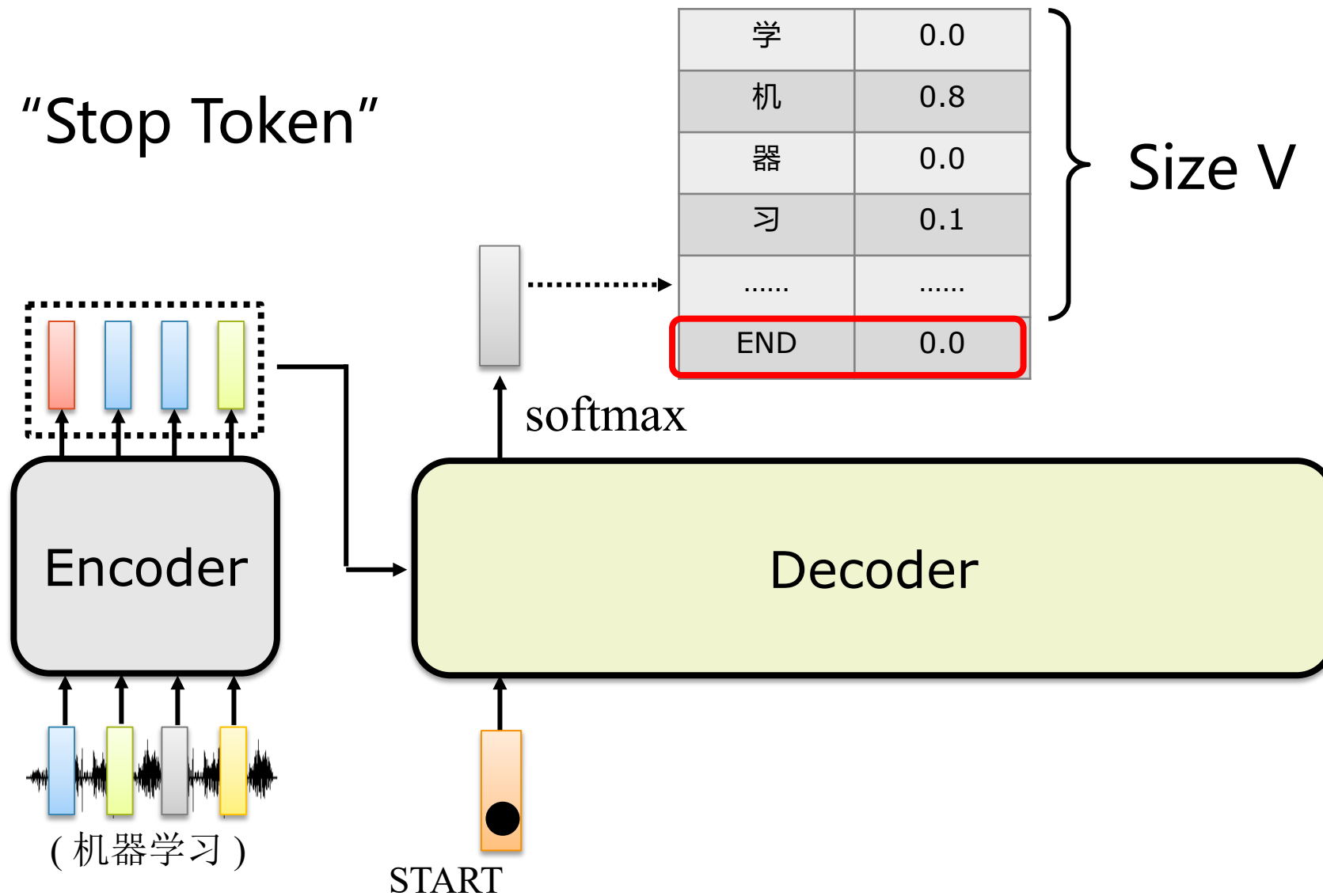
Transformer Decoder — Autoregressive (AT)

我们并不清楚输出的长度。

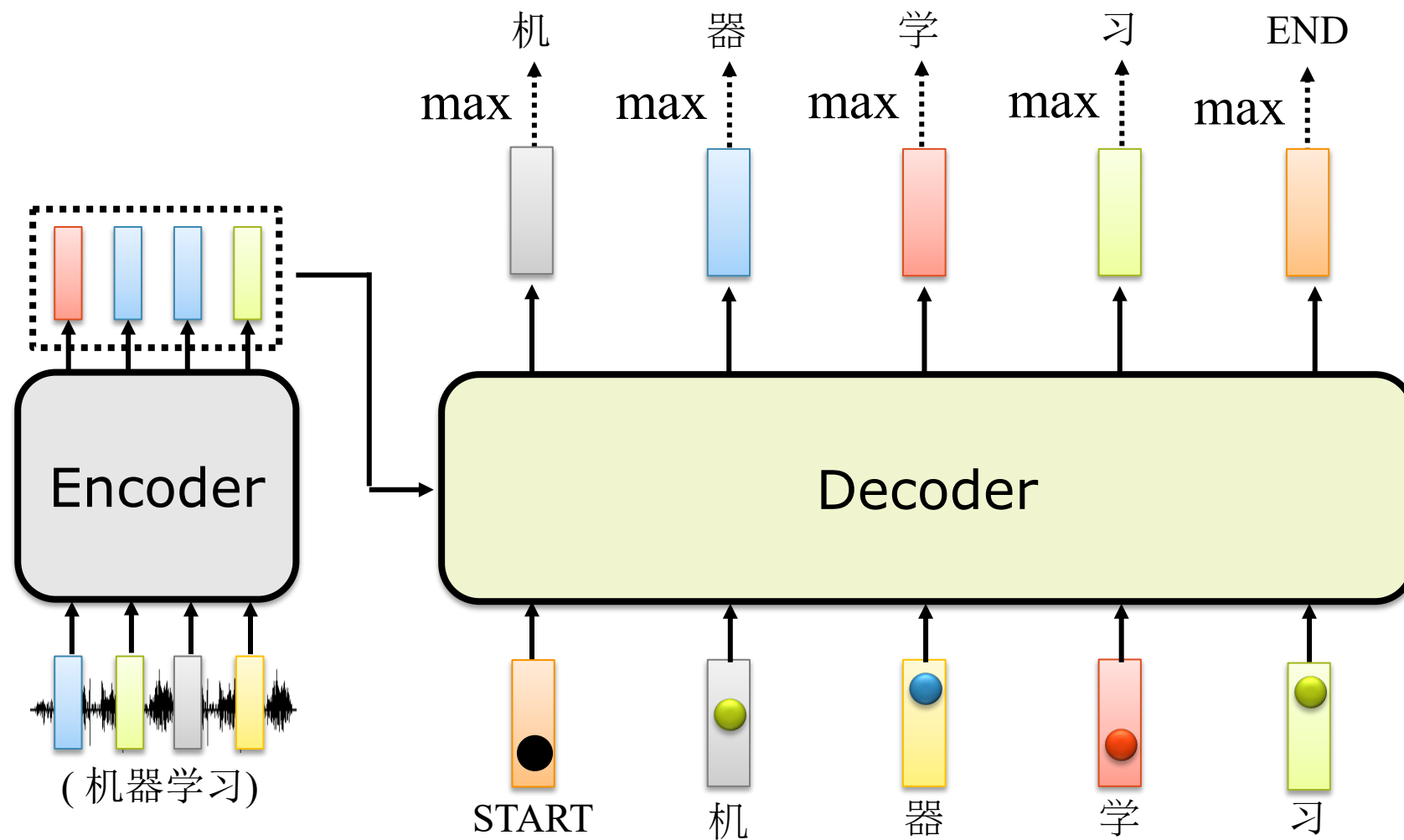


Transformer Decoder — Autoregressive (AT)

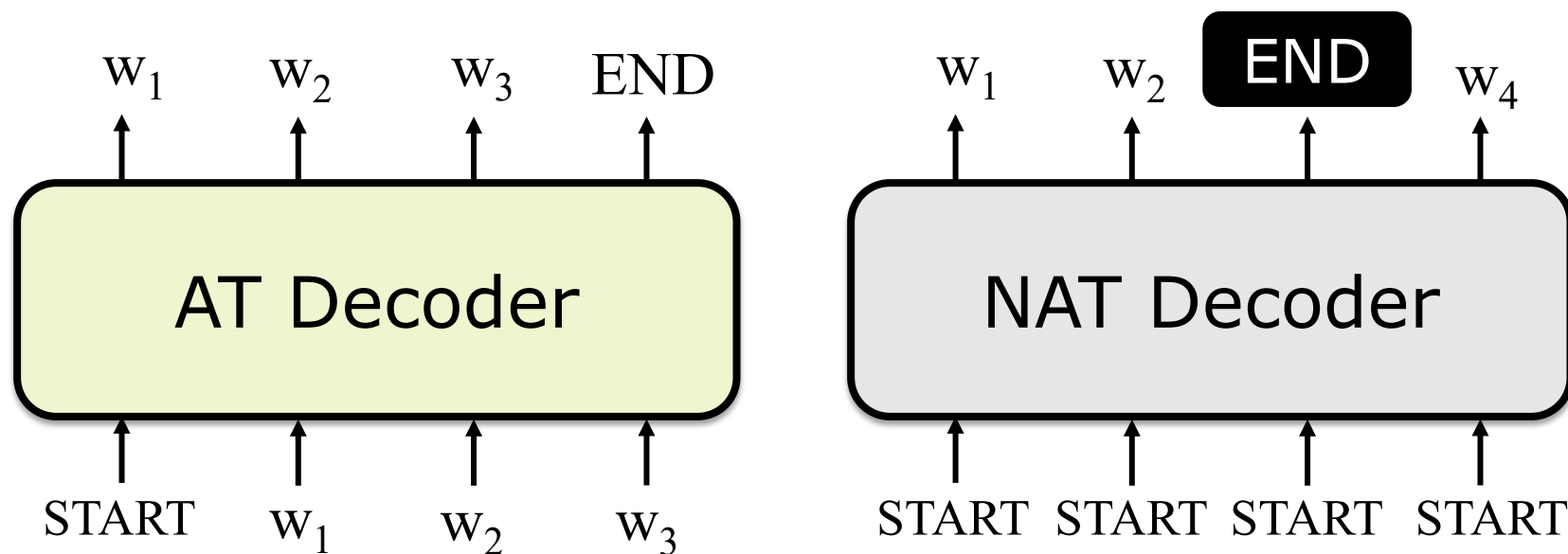
添加 “Stop Token”



Transformer Decoder — Autoregressive (AT)

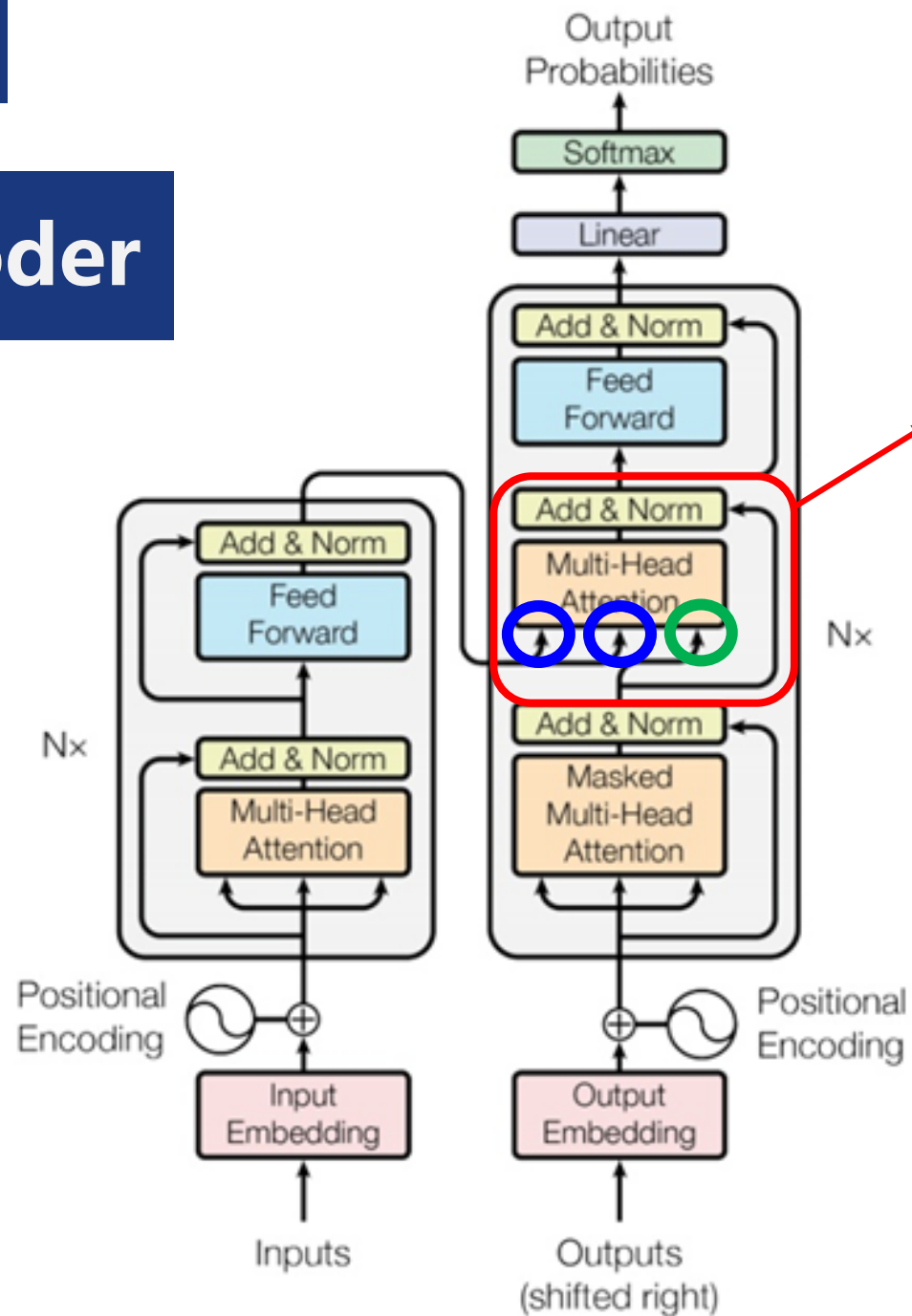


Transformer Decoder —— Non-Autoregressive (NAT)

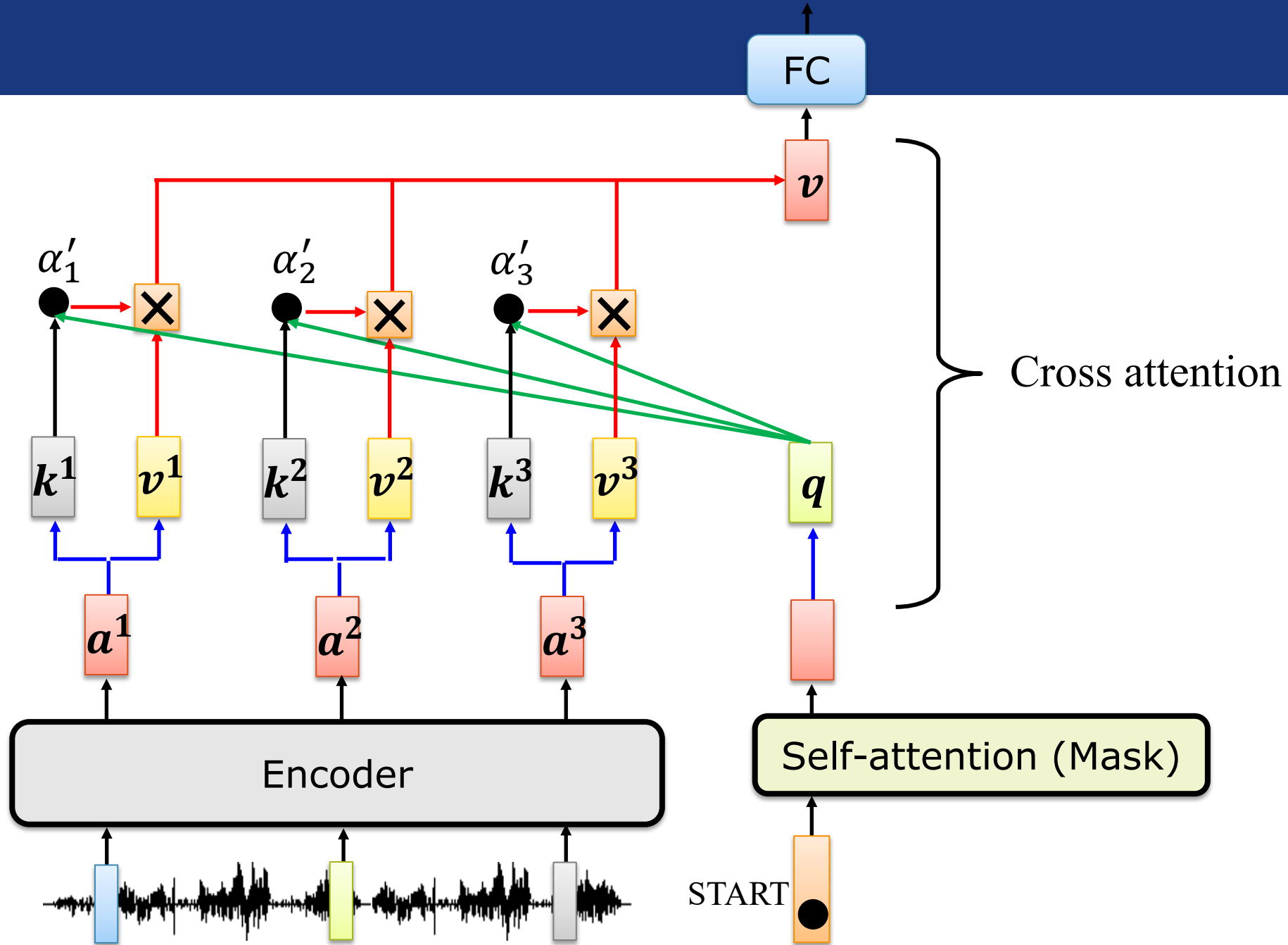


- NAT解码器一次生成整个序列；
- NAT怎样决定输出的长度？用另一个模型预测输出长度；或截取END之前的内容。
- NAT的优势：可并行（而AT只能逐个Token生成）；能够很好地控制输出长度；
- NAT的劣势：通常性能比AT差。

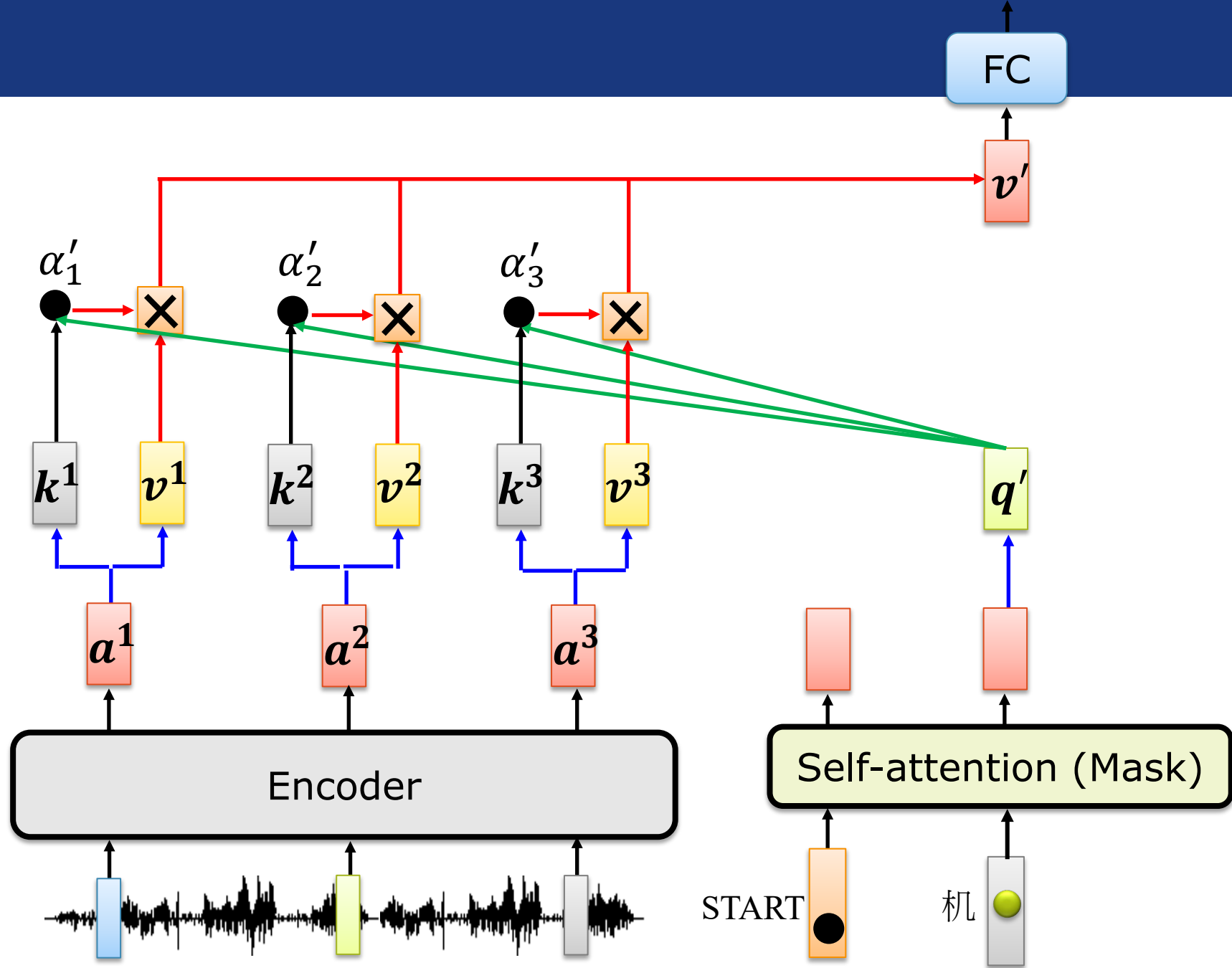
Encoder-Decoder



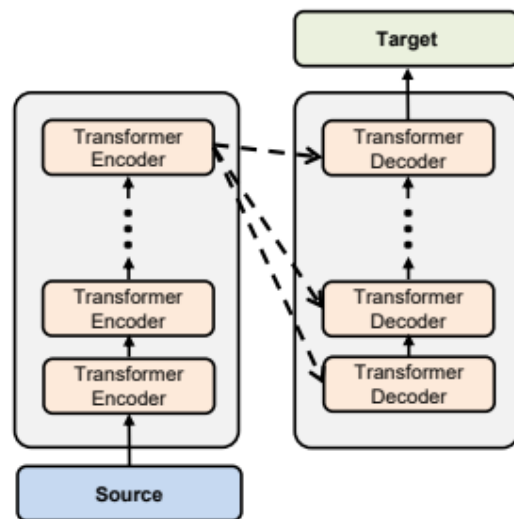
交叉注意力



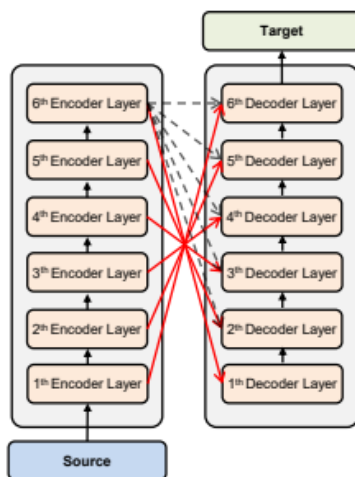
交叉注意力



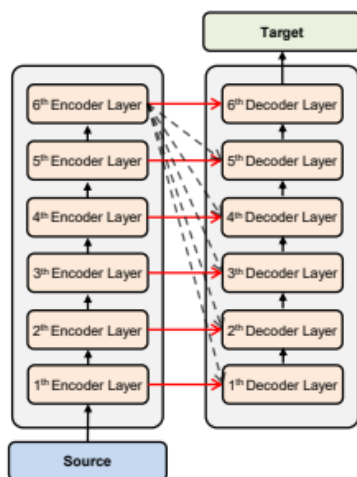
交叉注意力



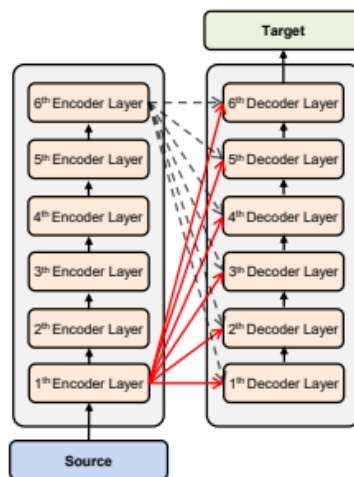
(a) Conventional Transformer



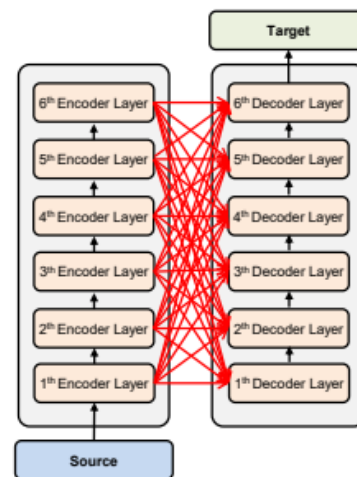
(a) Granularity Consistent Attention



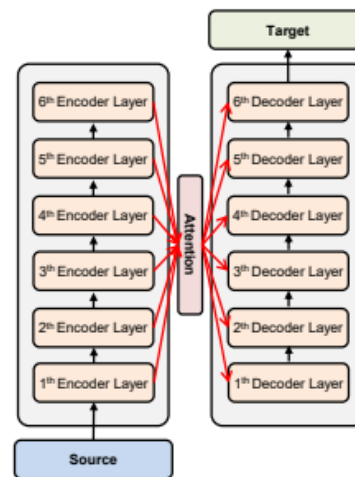
(b) Granularity Parallel Attention



(c) Fine-Grained Attention



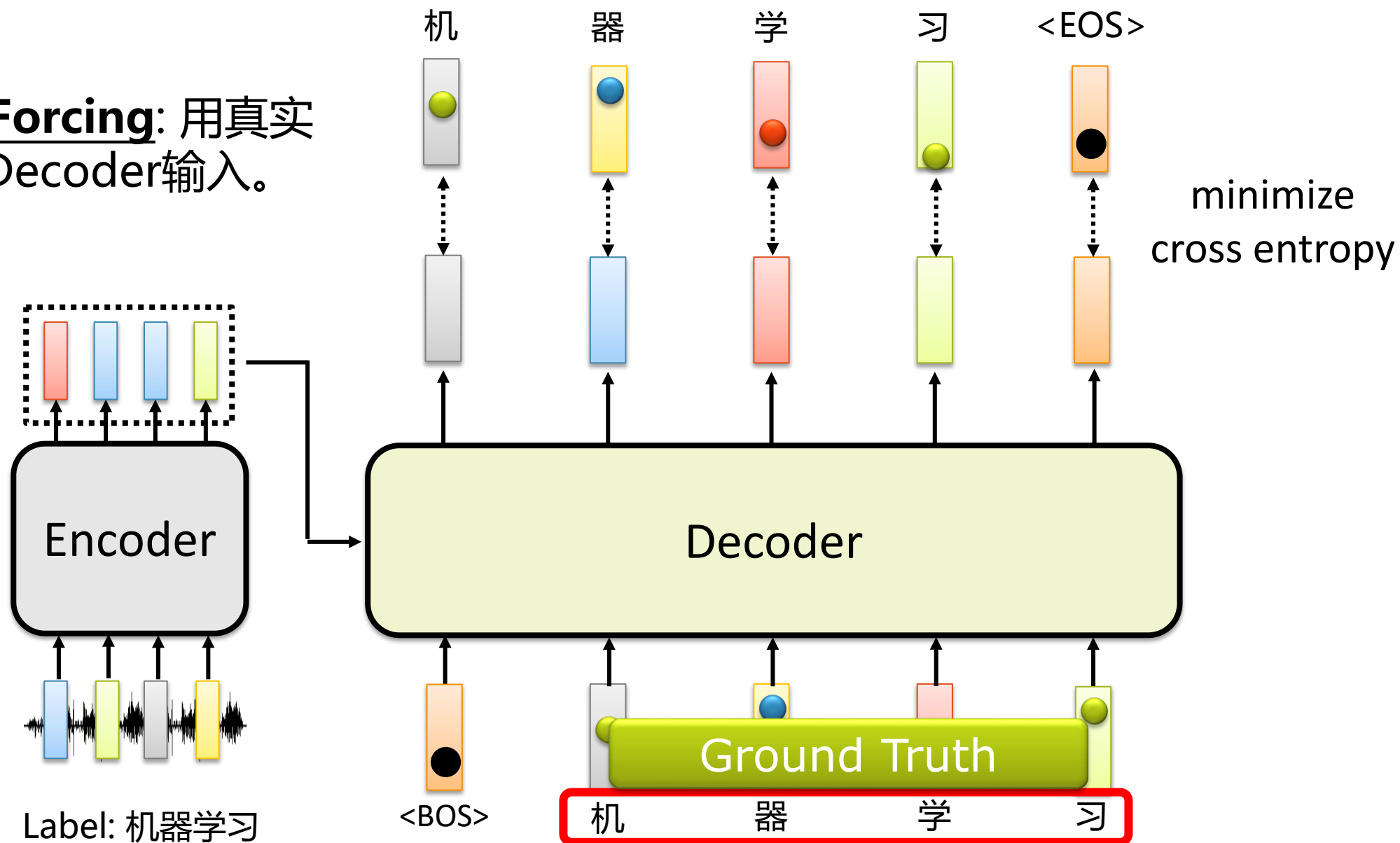
(d) Full Matching Attention



(e) Adaptive Matching Attention

Transformer的训练

Teacher Forcing: 用真实标签作为Decoder输入。



Transformer原理演示

BERT

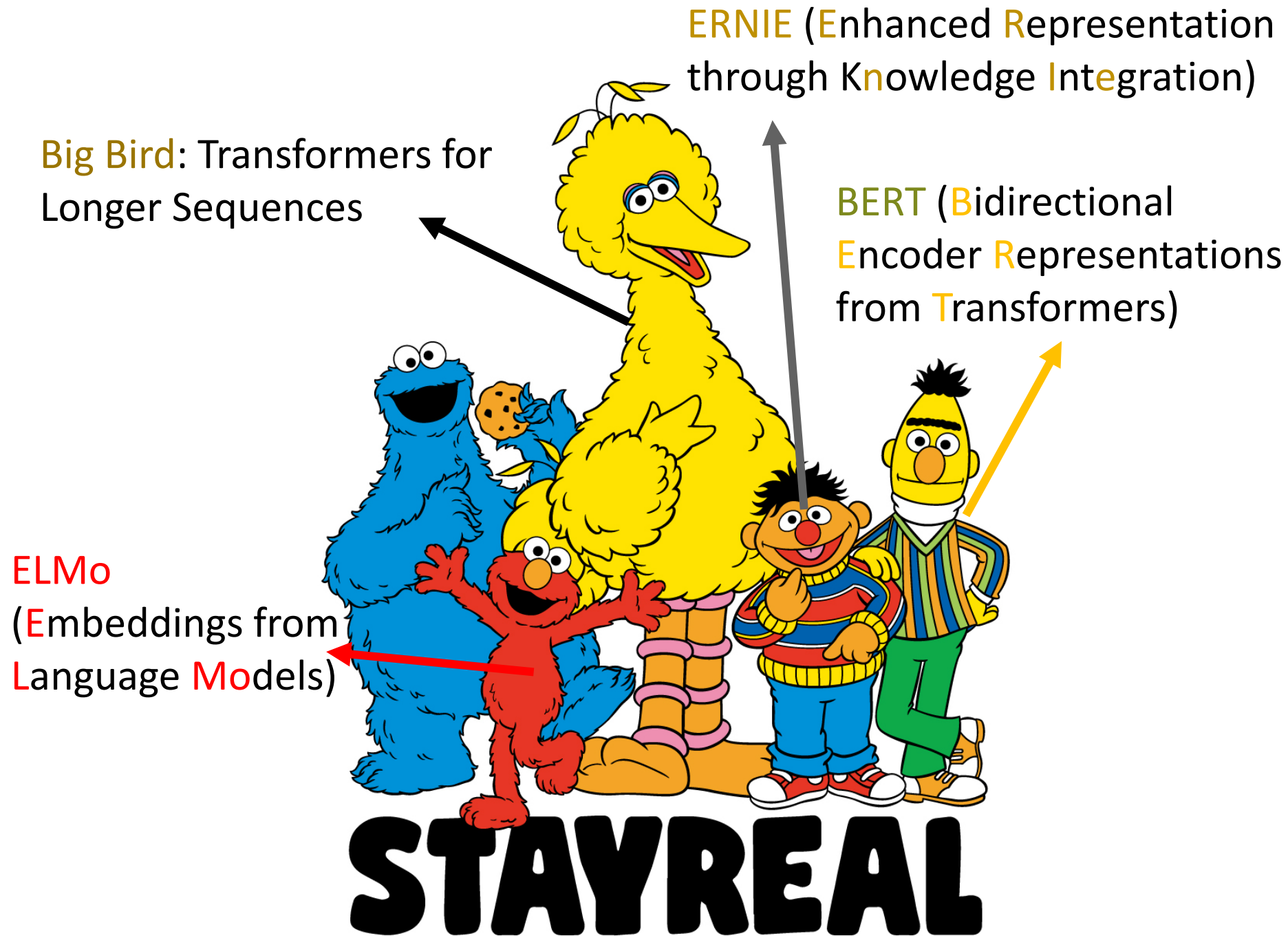


BERT





Bert



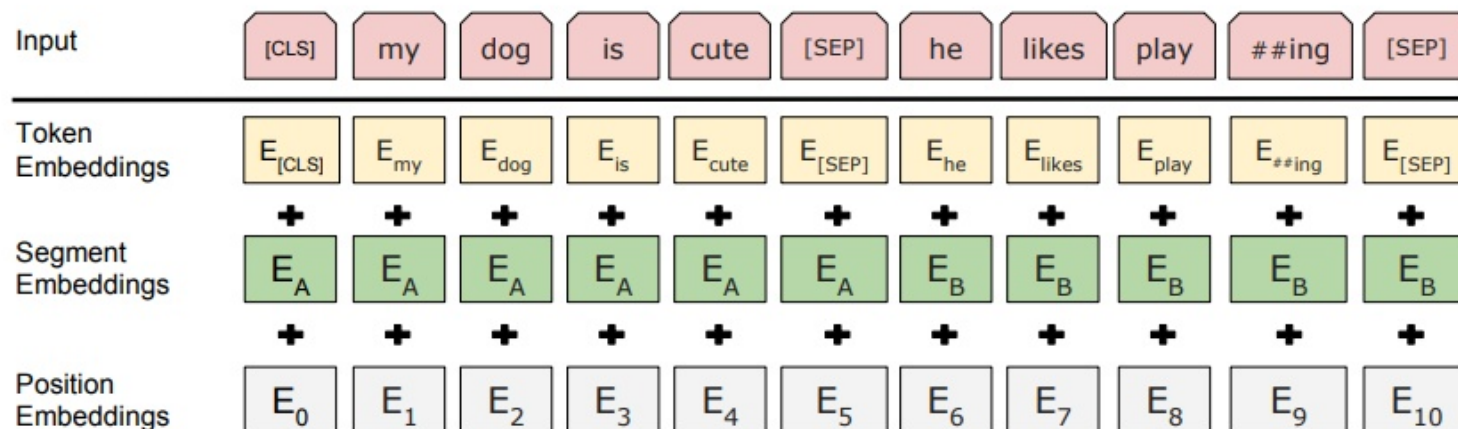
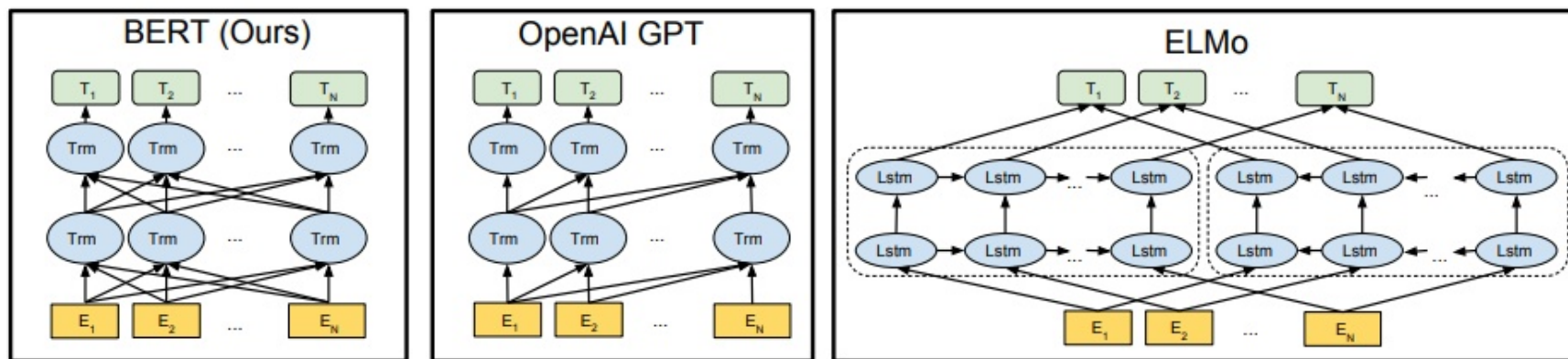
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova

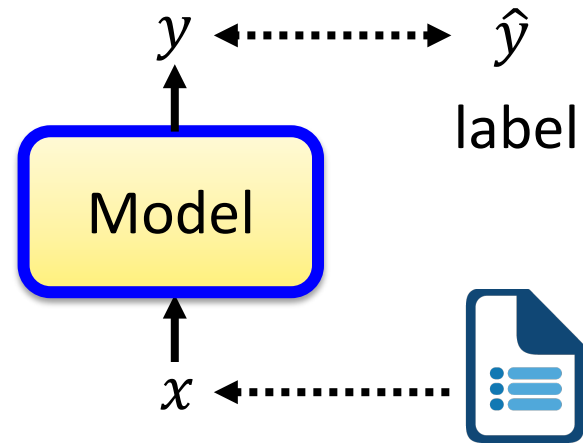
Google AI Language

{jacobdevlin, mingweichang, kentonl, kristout}@google.com

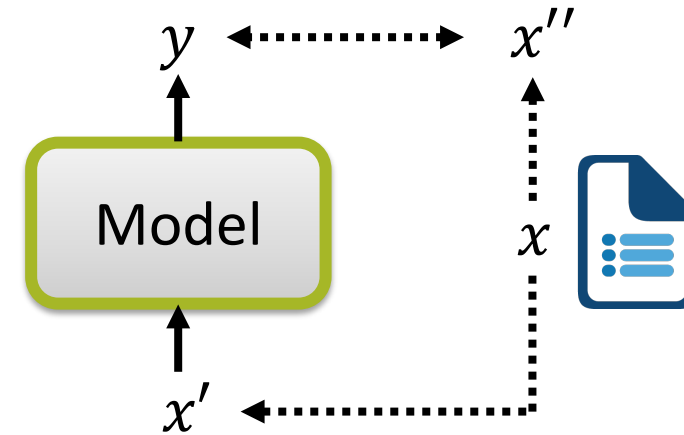
BERT = **B**idirectional
Encoder **R**epresentations
from **T**ransformers



Supervised



Self-supervised



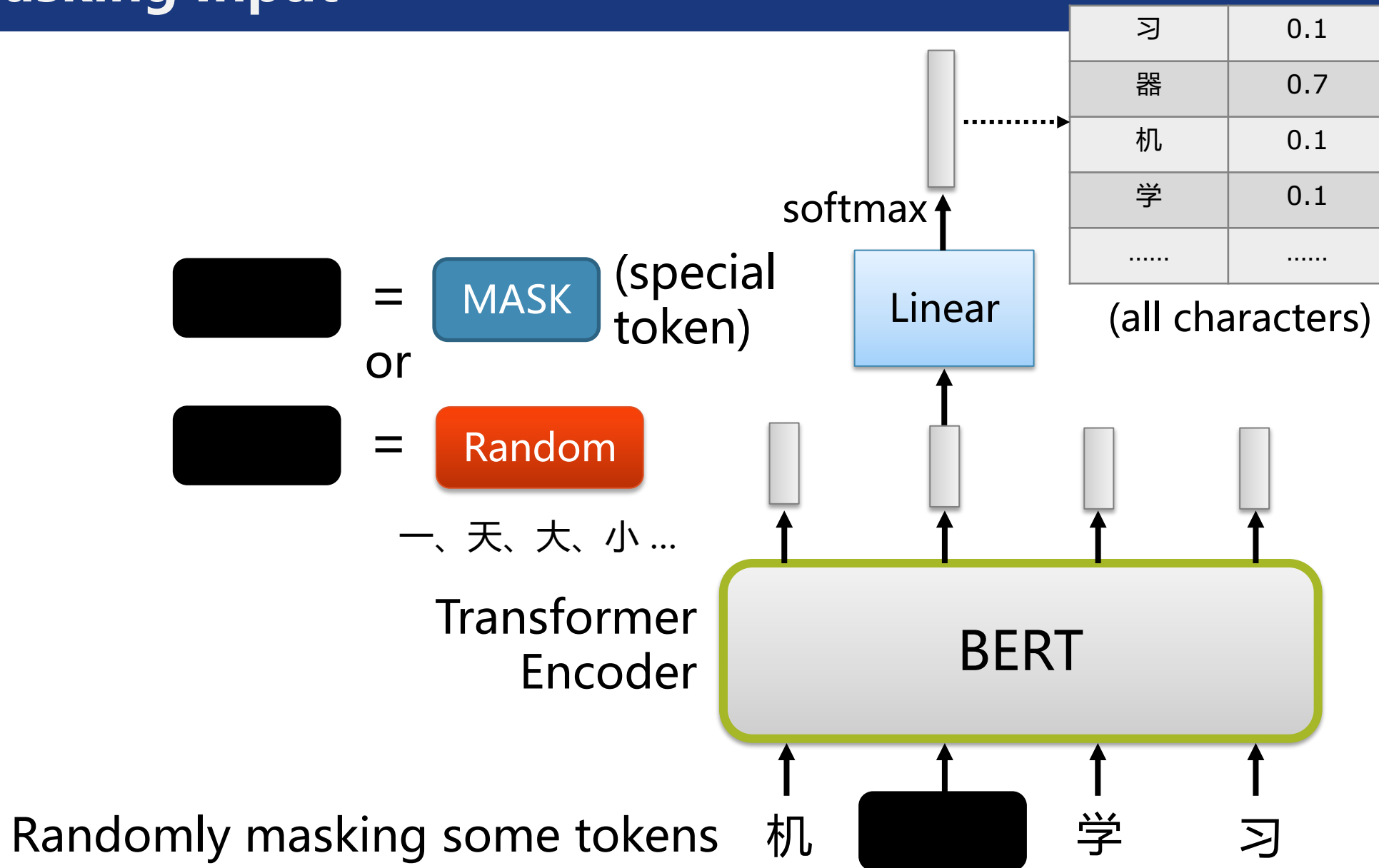
Yann LeCun

2019年4月30日 · 🌐

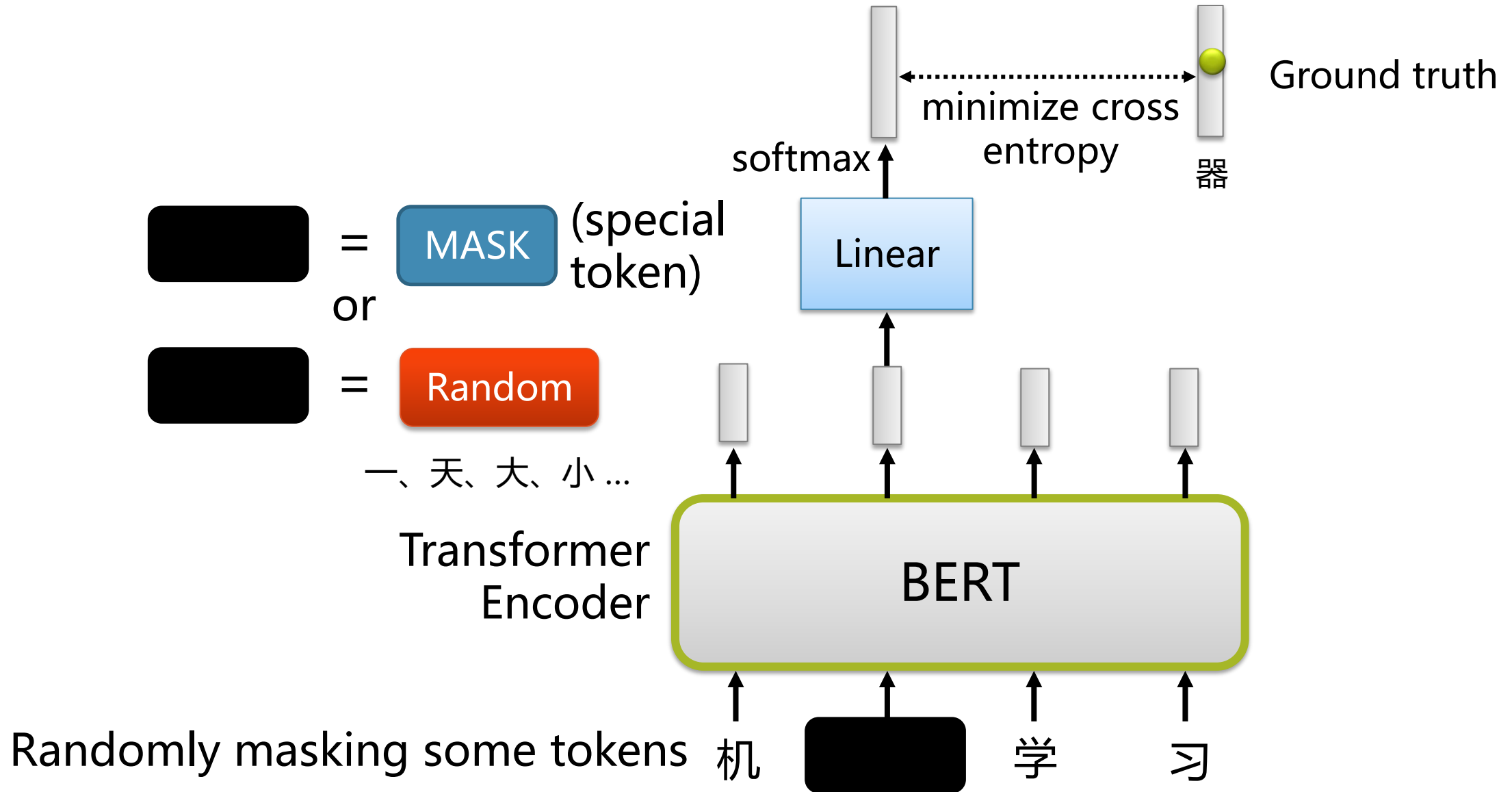
I now call it "self-supervised learning", because "unsupervised" is both a loaded and confusing term.

In self-supervised learning, the system learns to predict part of its input from other parts of it input. In other words a portion of the input is used as a supervisory signal to a predictor fed with the remaining portion of the input.

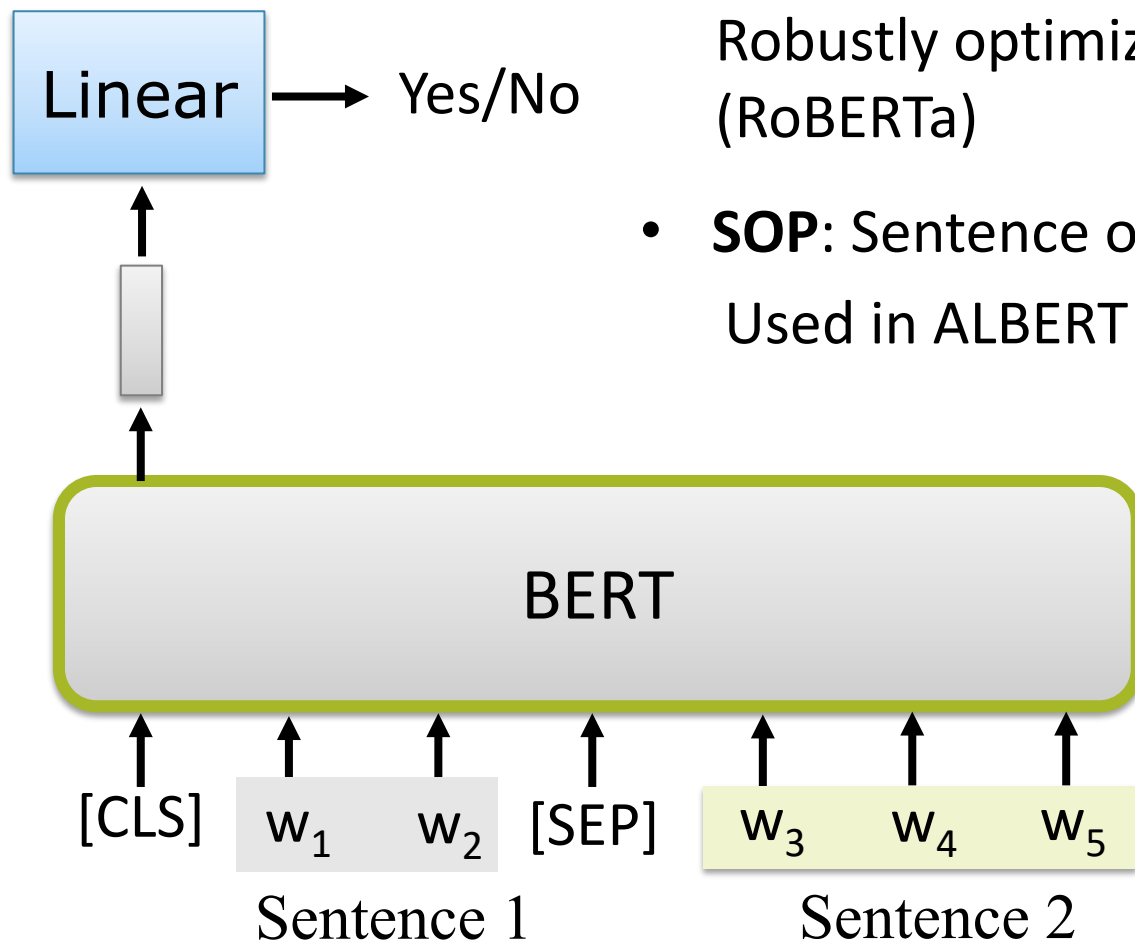
Masking Input



Masking Input



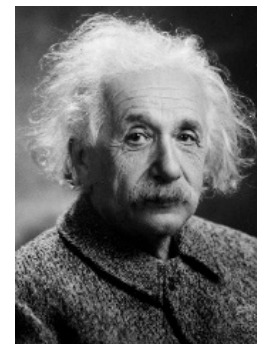
“下一句” 预测



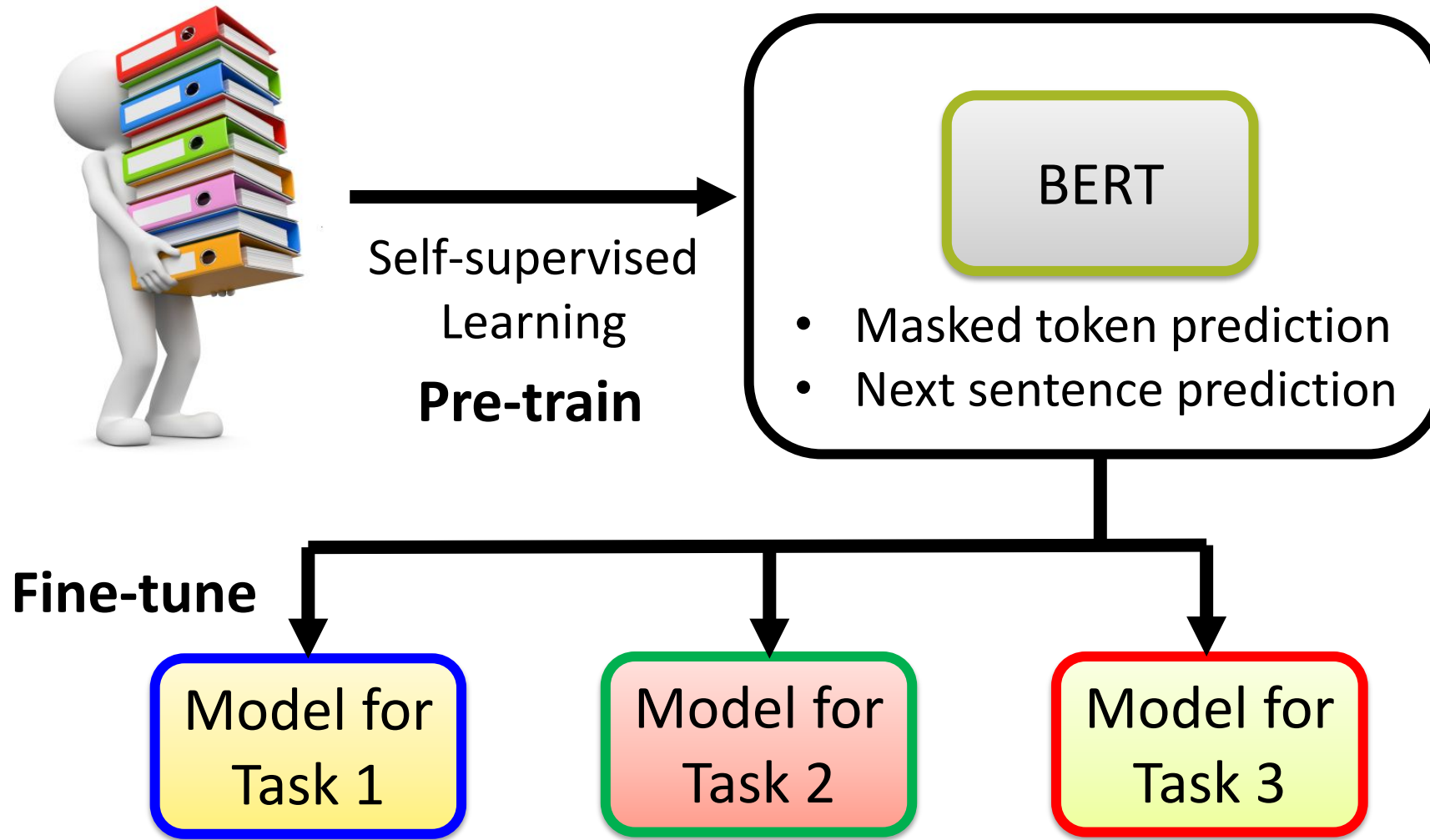
- This approach is not helpful.

Robustly optimized BERT approach
(RoBERTa)

- **SOP**: Sentence order prediction——句子顺序预测
Used in ALBERT



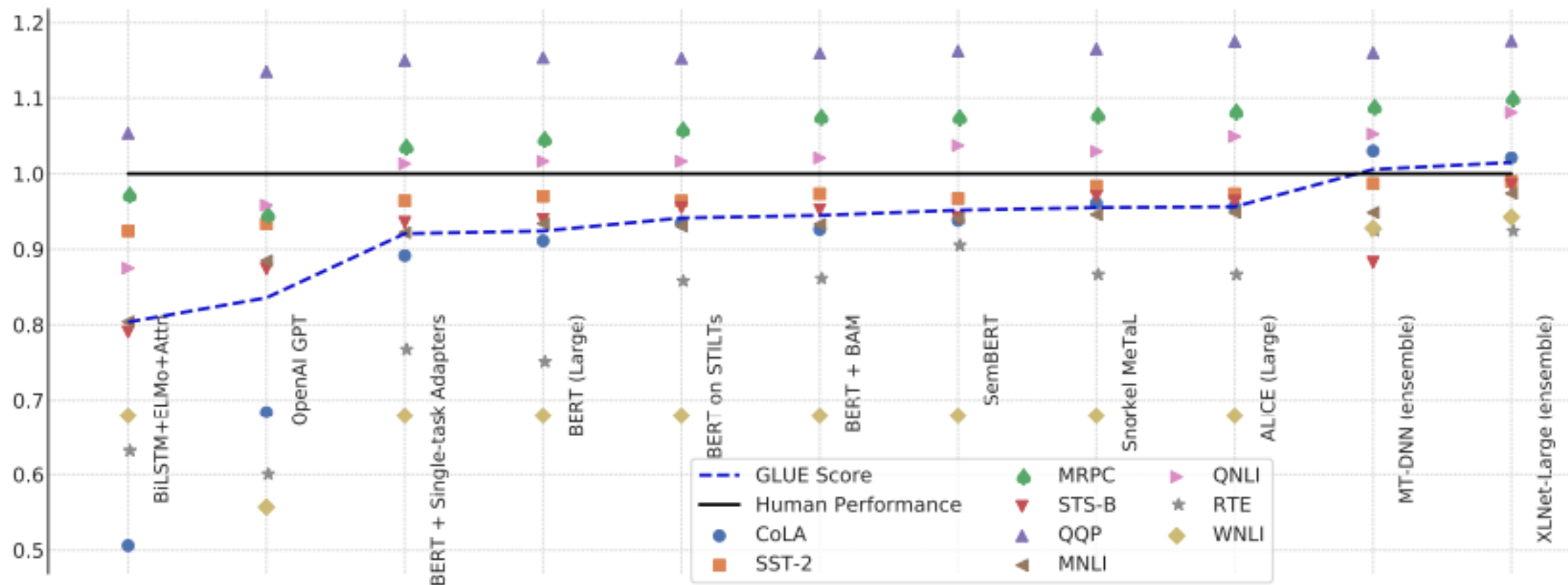
BERT模型的使用



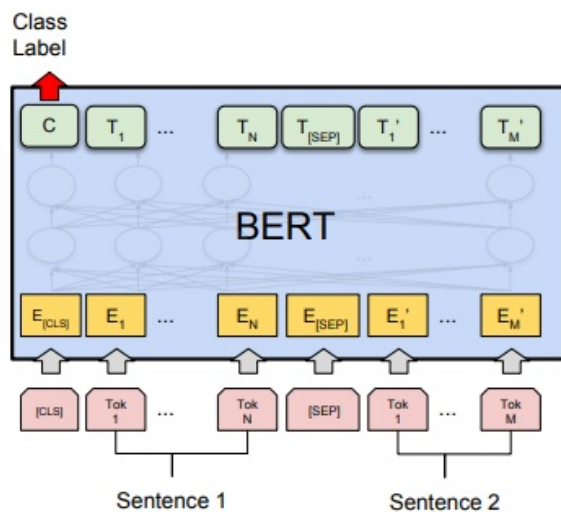
评价标准: General Language Understanding Evaluation (GLUE)

- **Corpus of Linguistic Acceptability (CoLA)**
- **Stanford Sentiment Treebank (SST-2)**
- **Microsoft Research Paraphrase Corpus (MRPC)**
- **Quora Question Pairs (QQP)**
- **Semantic Textual Similarity Benchmark (STS-B)**
- **Multi-Genre Natural Language Inference (MNLI)**
- **Question-answering NLI (QNLI)**
- **Recognizing Textual Entailment (RTE)**
- **Winograd NLI (WNLI)**

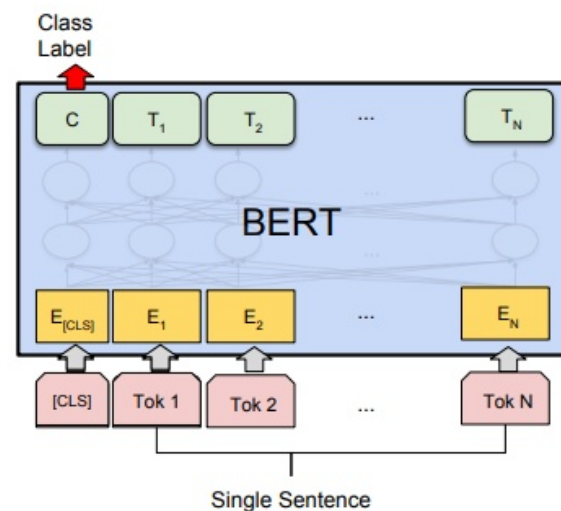
• GLUE scores



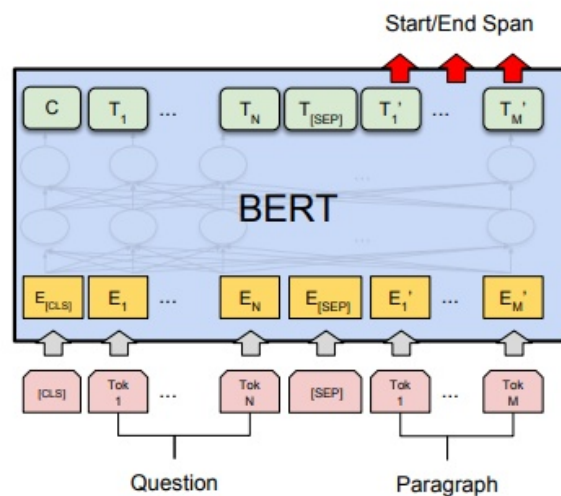
BERT模型的使用



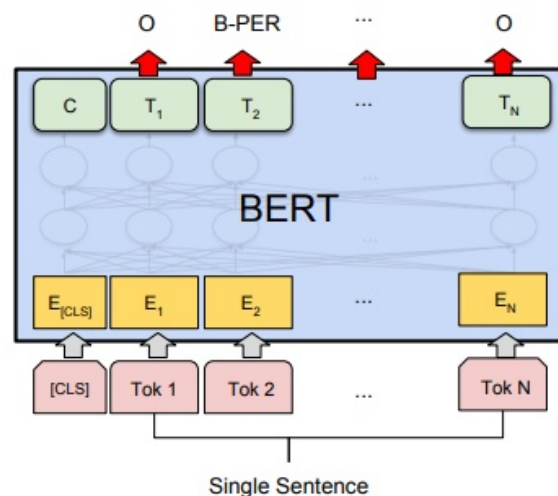
(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



(b) Single Sentence Classification Tasks:
SST-2, CoLA

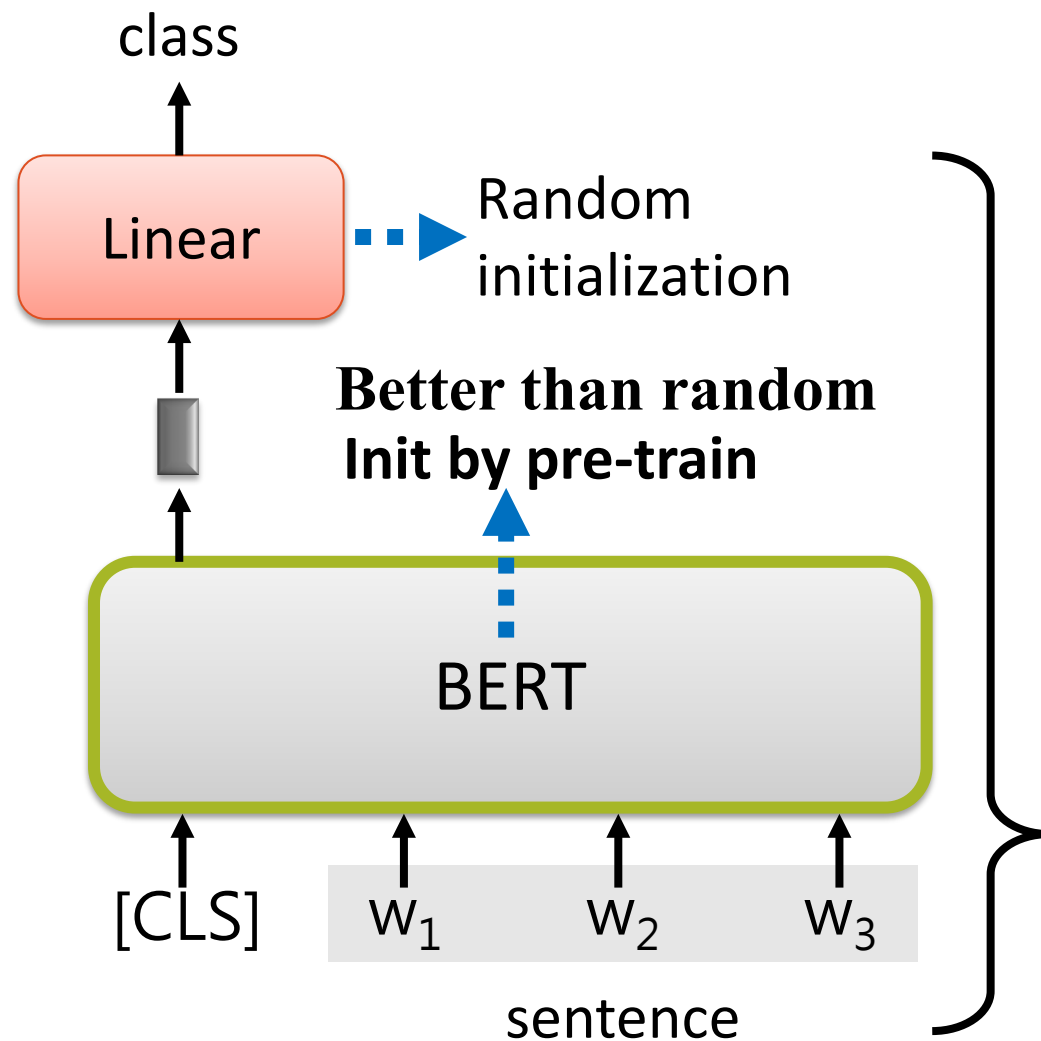


(c) Question Answering Tasks:
SQuAD v1.1



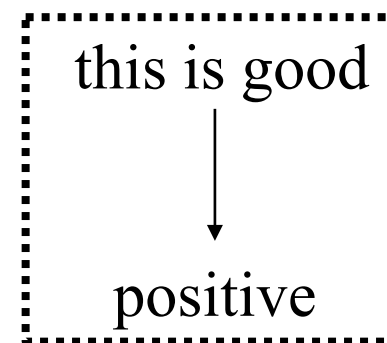
(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

BERT模型的使用：句子分类



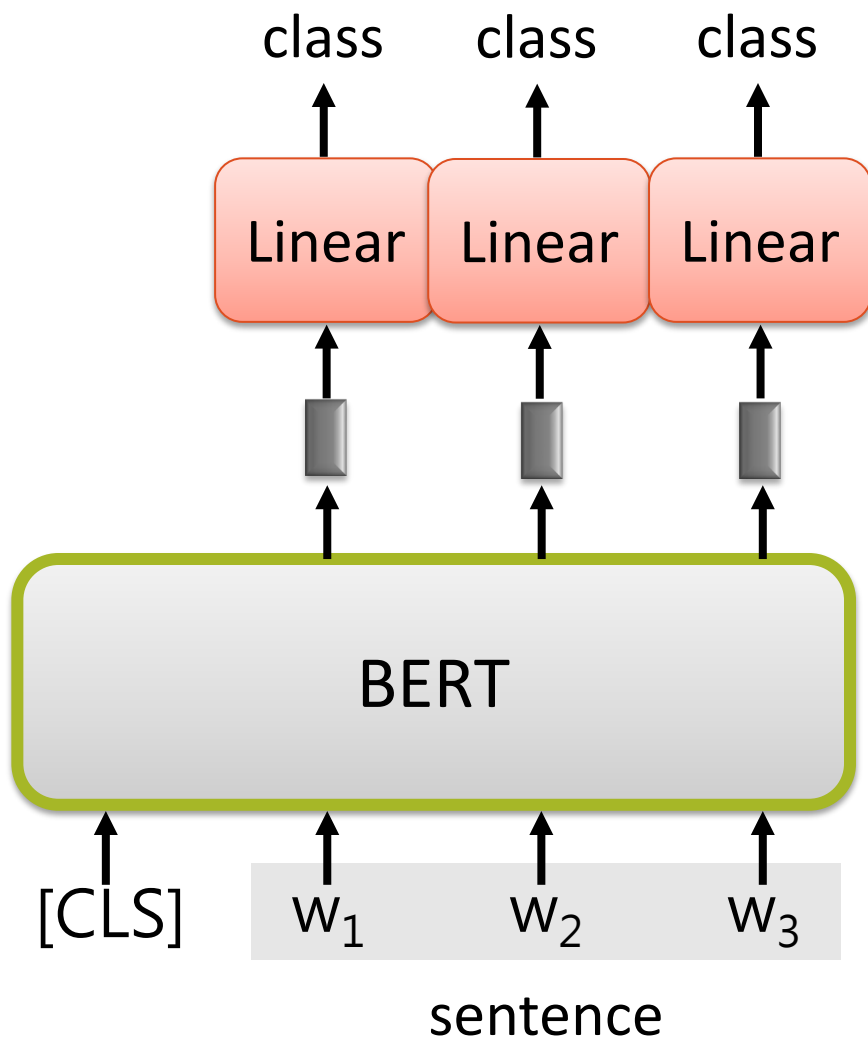
输入: 句子
输出: 句子的类别

Example:
Sentiment analysis



This is the model
to be learned.

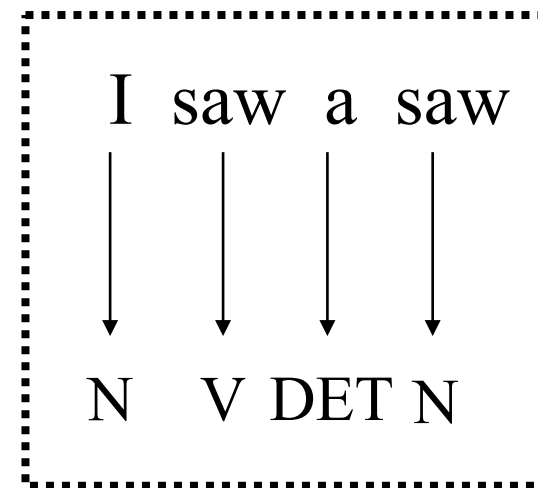
BERT模型的使用：单词标注



输入: 句子

输出: 与句子等长的序列

Example:
POS tagging

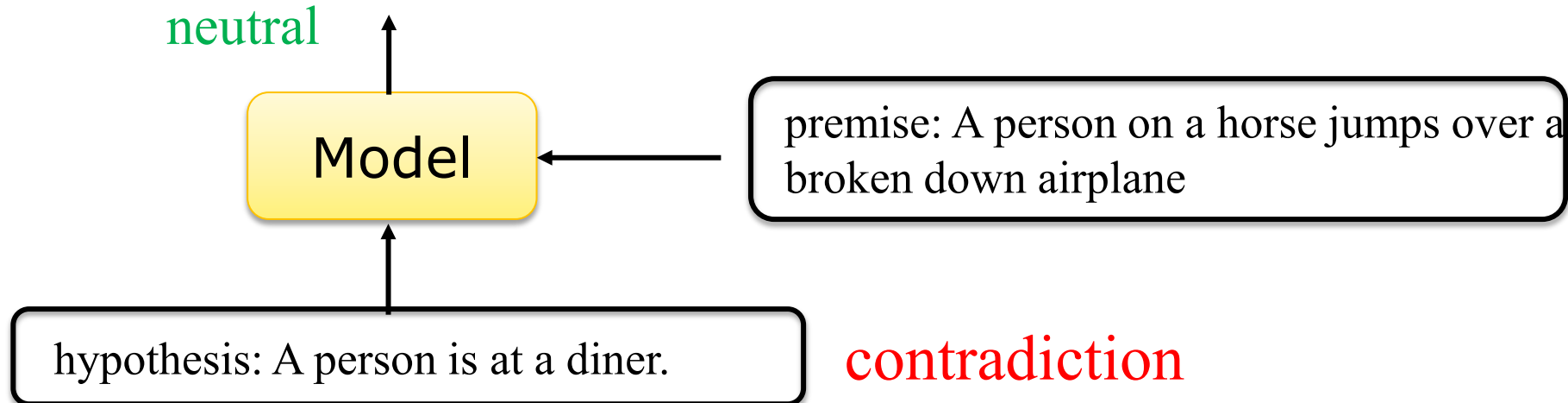


BERT模型的使用：语句对的分类

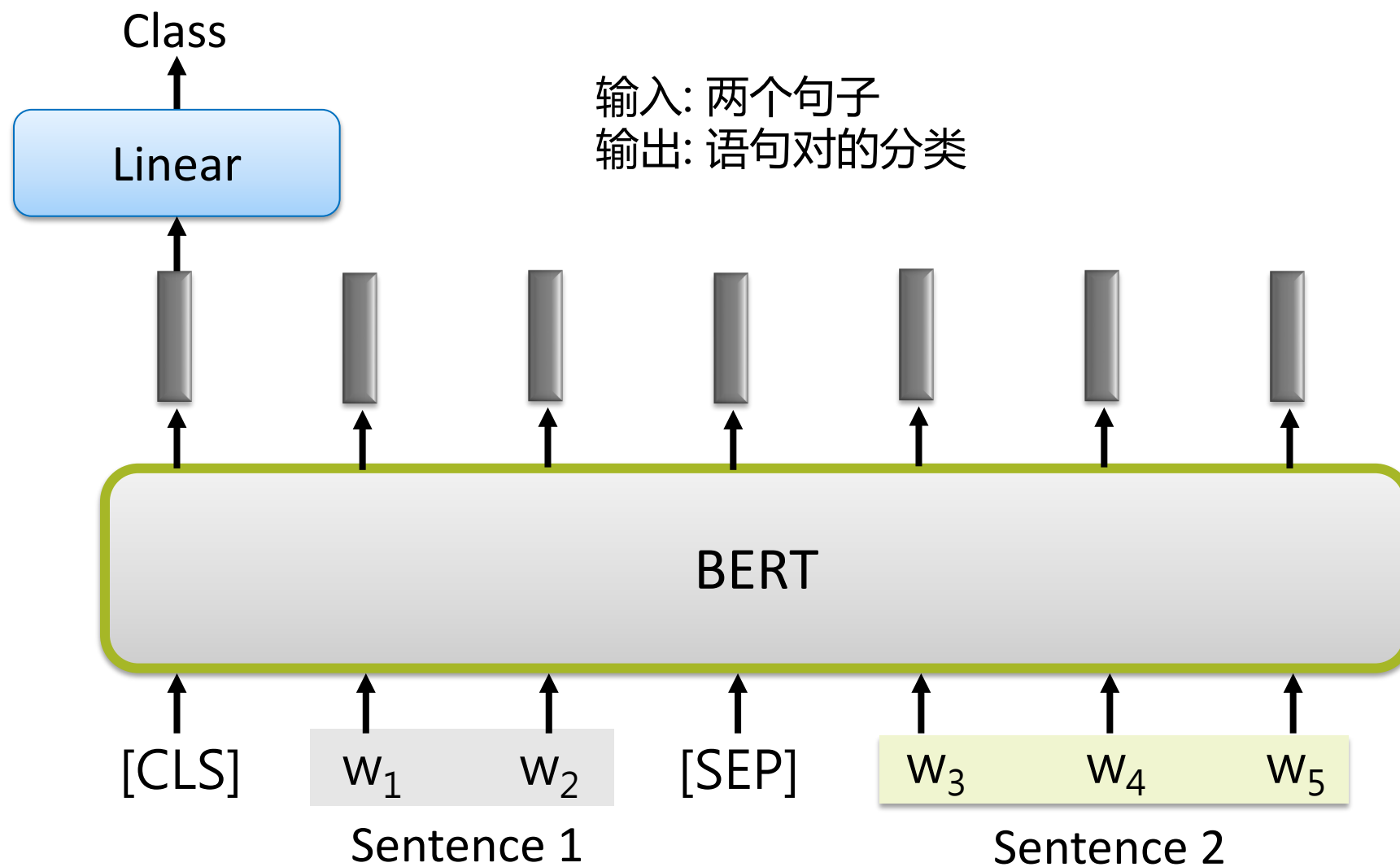
输入: 两个句子
输出: 语句对的分类

contradiction
entailment
neutral

Example:
Natural Language Inference (NLI)



BERT模型的使用：语句对的分类

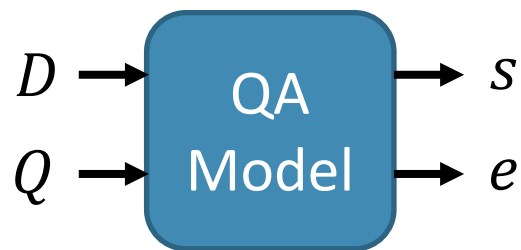


- Extraction-based

Question Answering (QA)

Document: $D = \{d_1, d_2, \dots, d_N\}$

Query: $Q = \{q_1, q_2, \dots, q_M\}$



output: two integers (s, e)

Answer: $A = \{d_s, \dots, d_e\}$

In meteorology, precipitation is any product of the condensation of 17 spheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain 77 at 79 locations are called "showers".

What causes precipitation to fall?

gravity

$s = 17, e = 17$

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

graupel

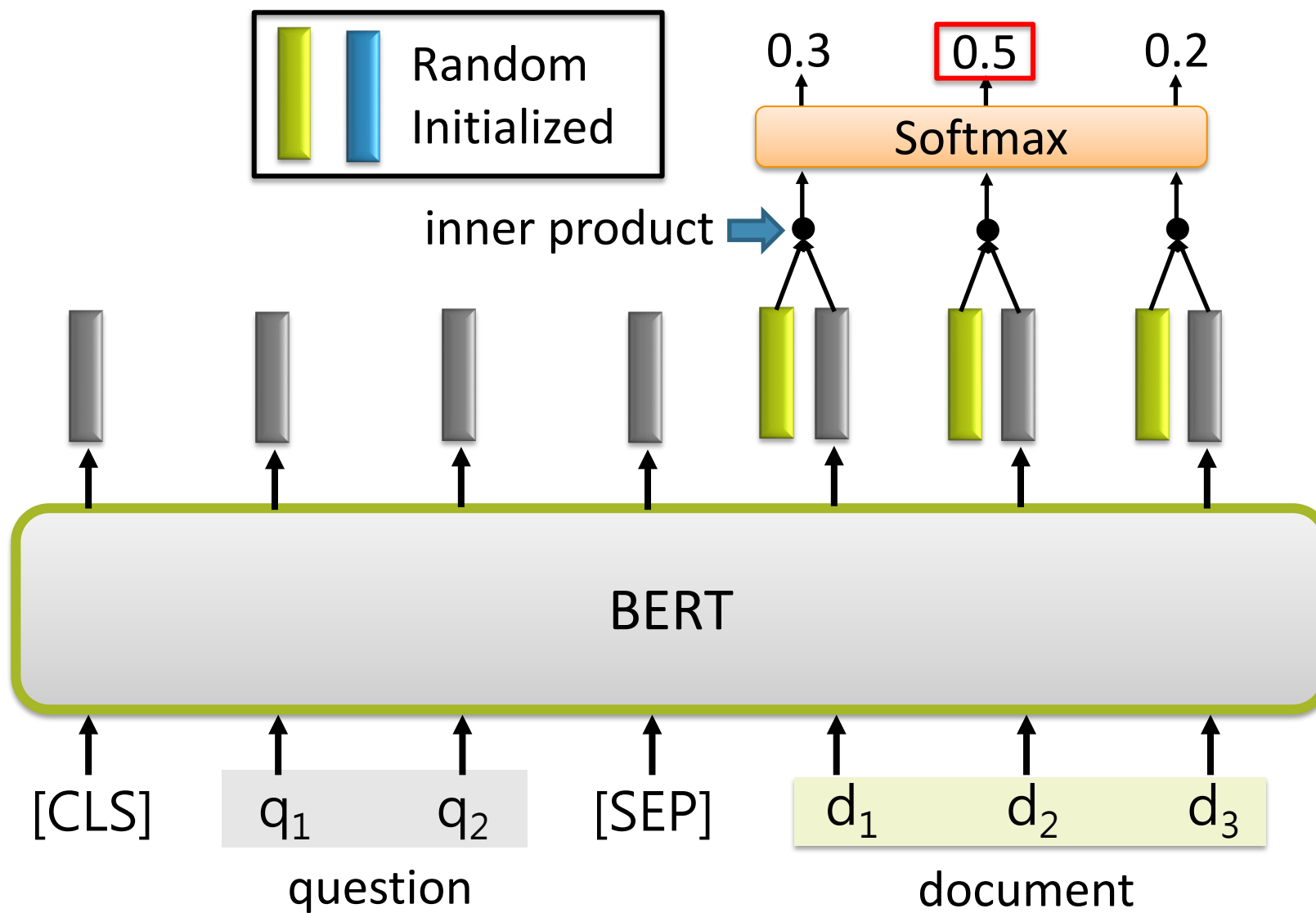
Where do water droplets collide with ice crystals to form precipitation?

within a cloud

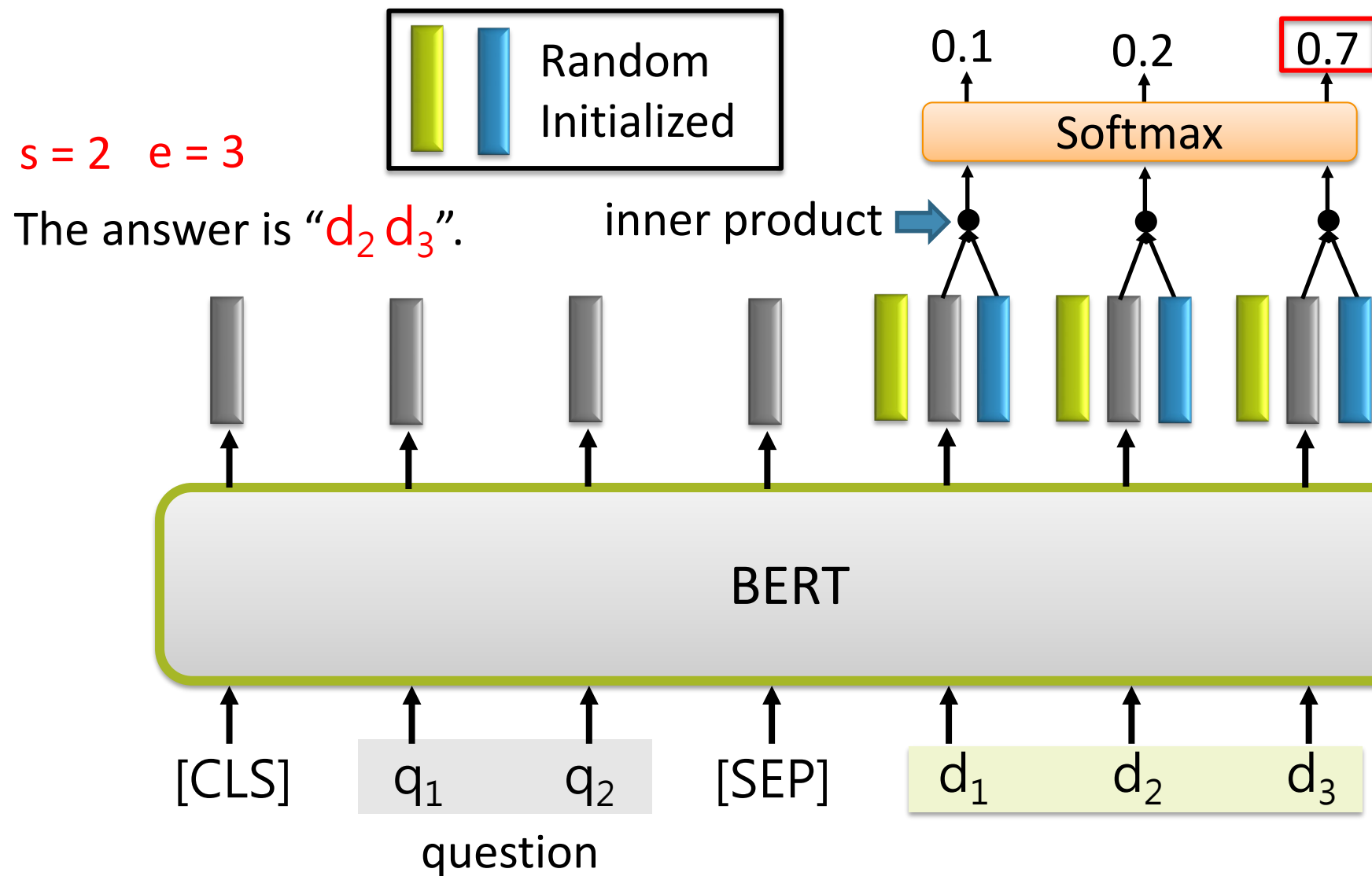
$s = 77, e = 79$

BERT模型的使用：问答

$s = 2$

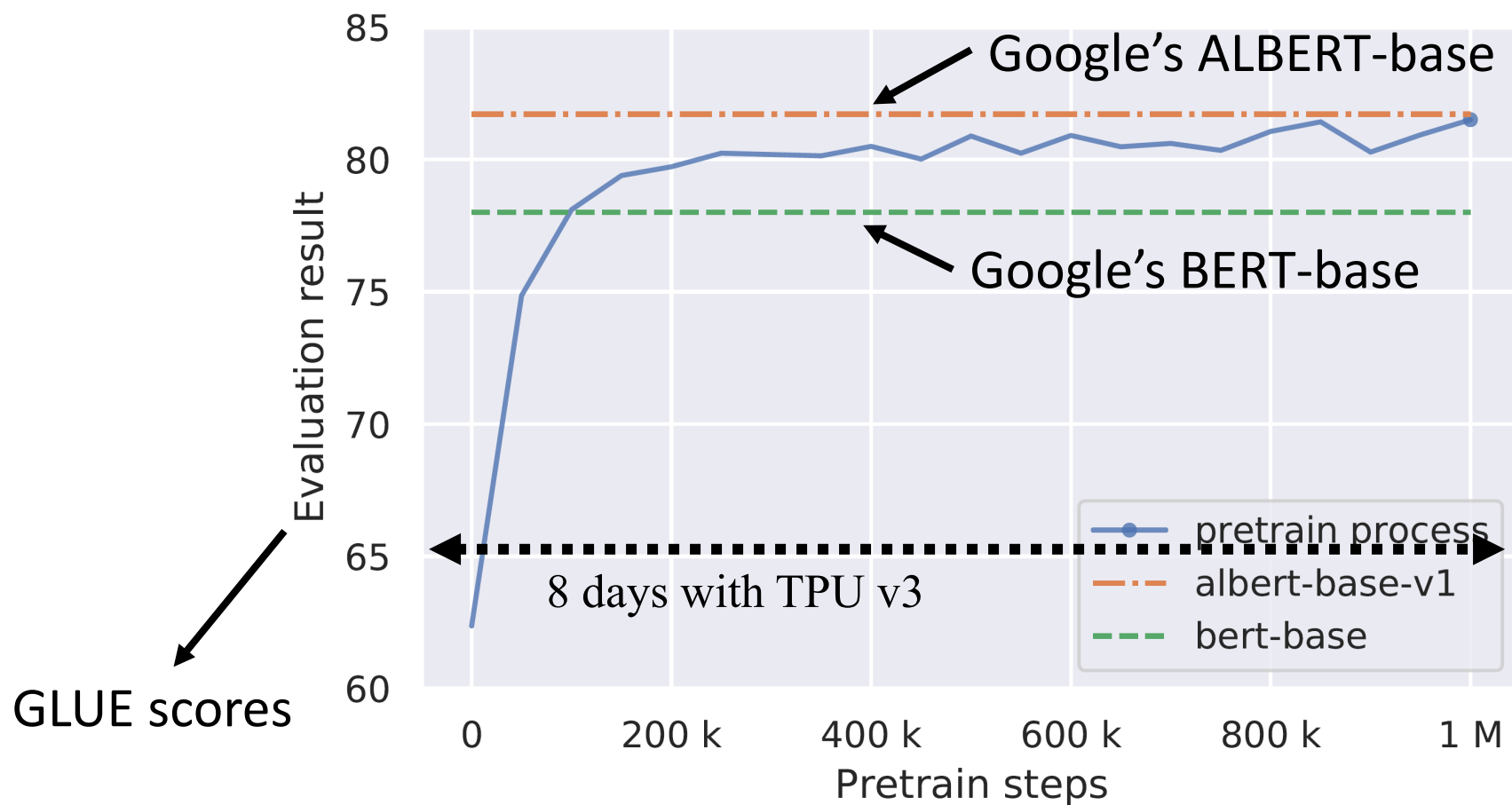


BERT模型的使用：问答

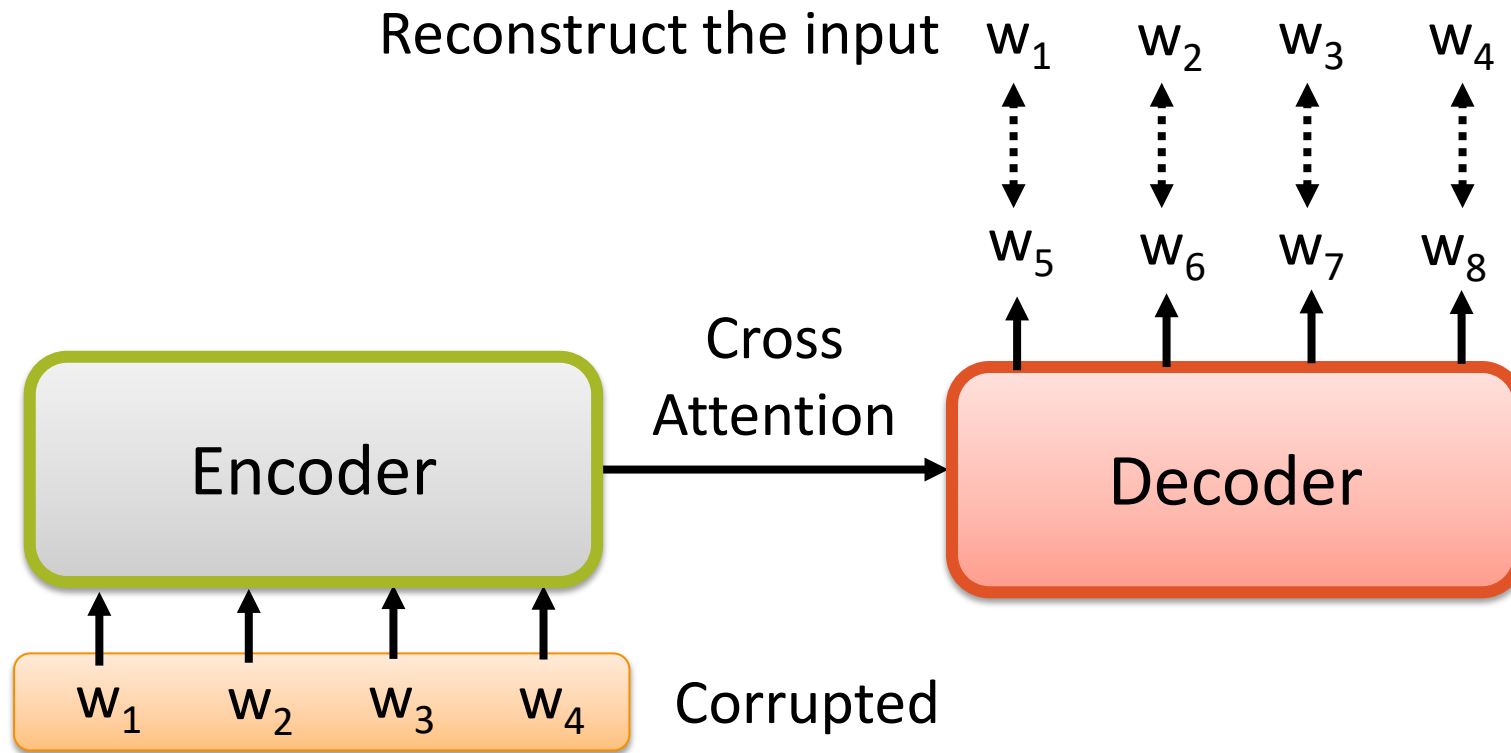


BERT模型的训练

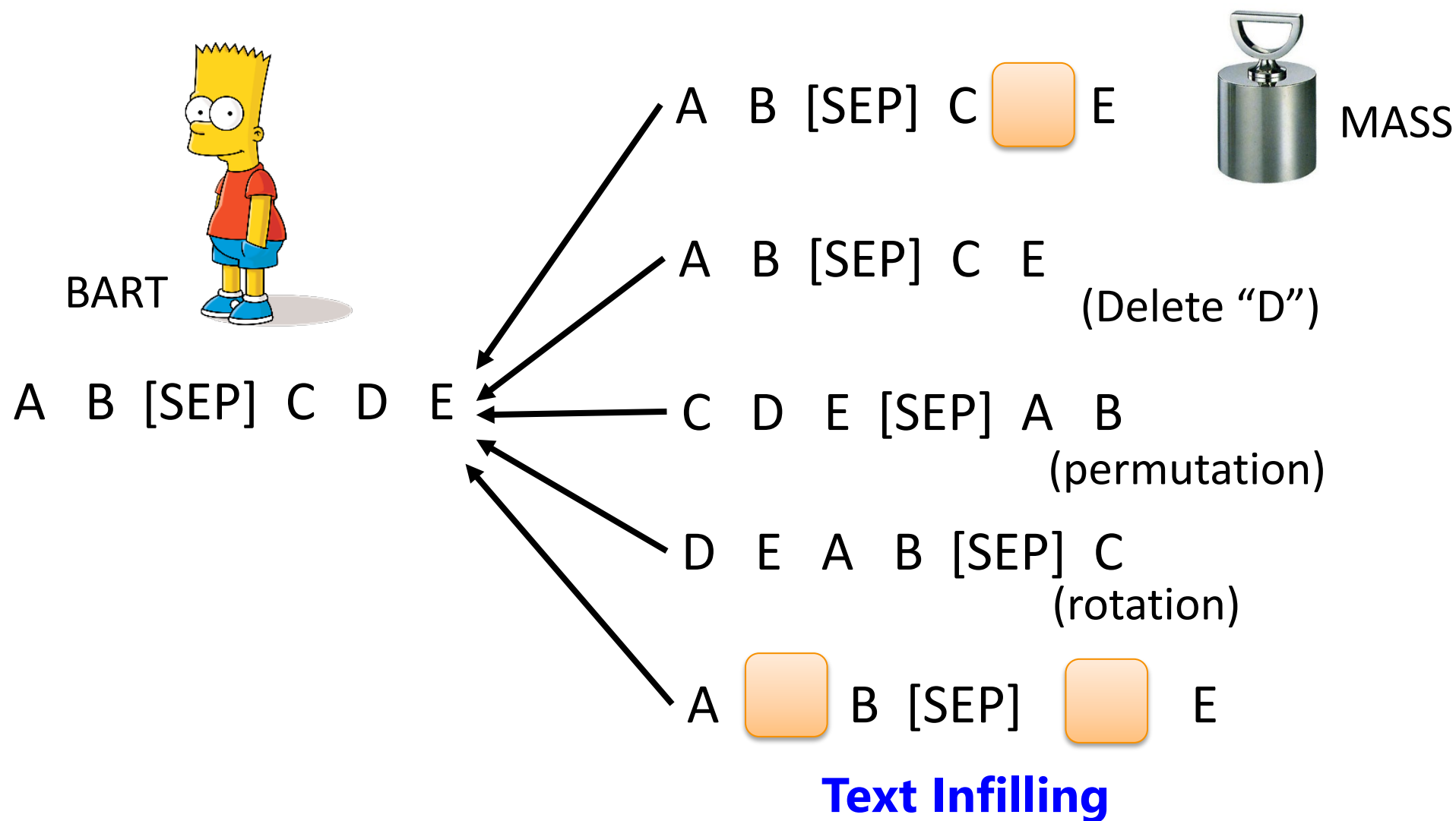
最初版本BERT的训练使用了 > 30亿词的语料。



进行Seq2seq模型的自监督训练



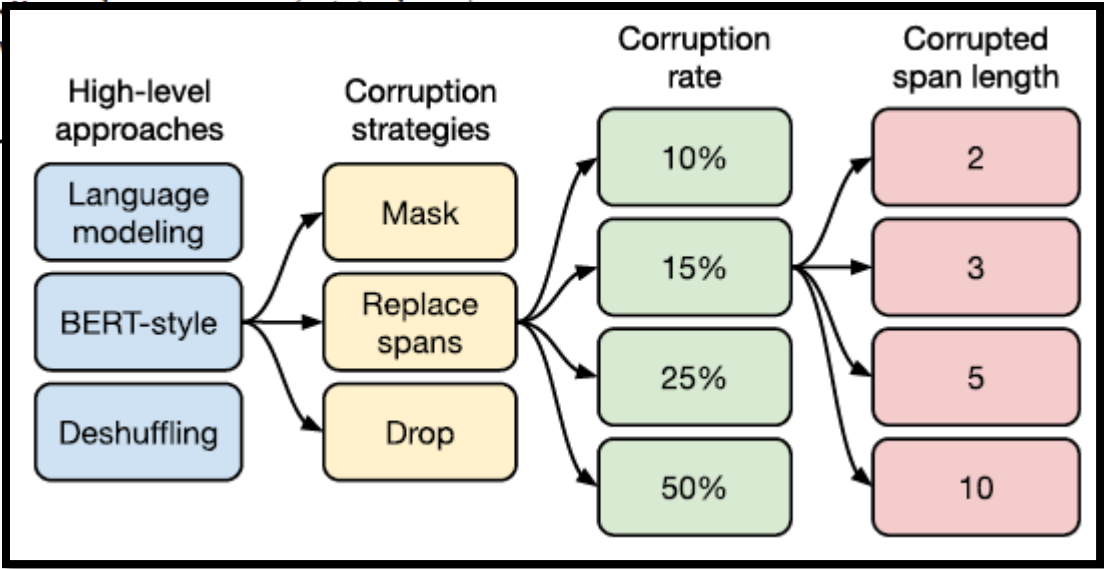
进行Seq2seq模型的自监督训练



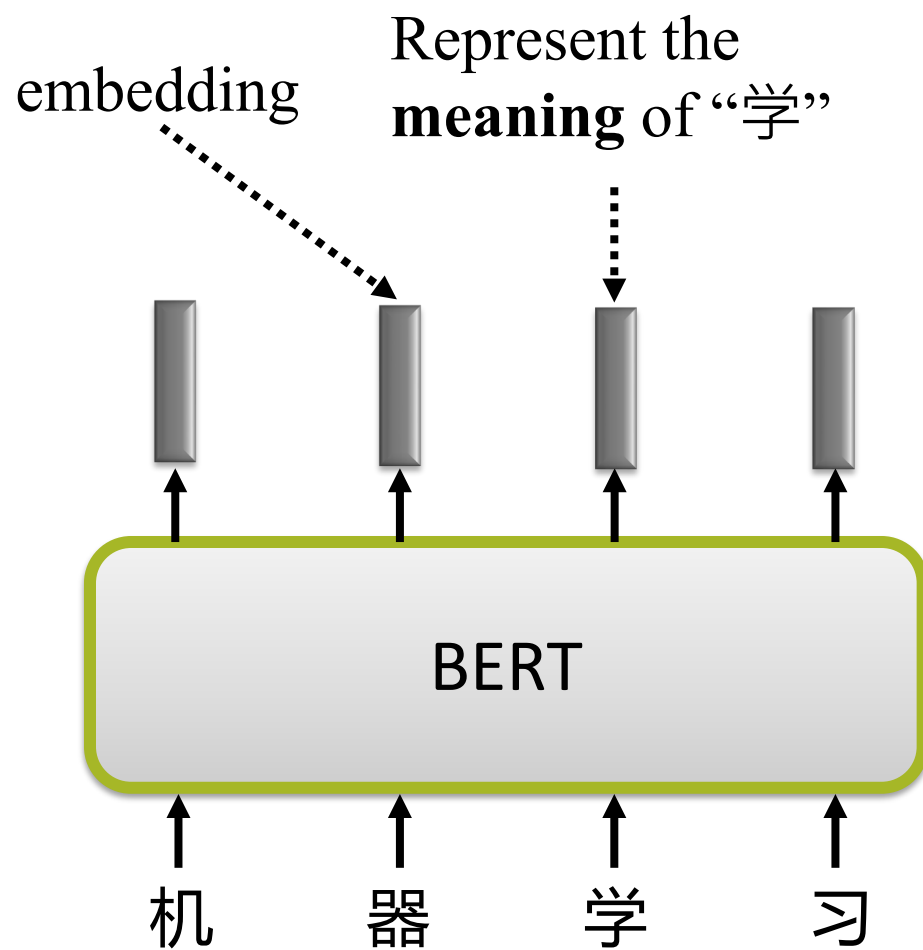
进行Seq2seq模型的自监督训练

- Transfer Text-to-Text Transformer (T5)
- Colossal Clean Crawled Corpus (C4语料库——7TB)

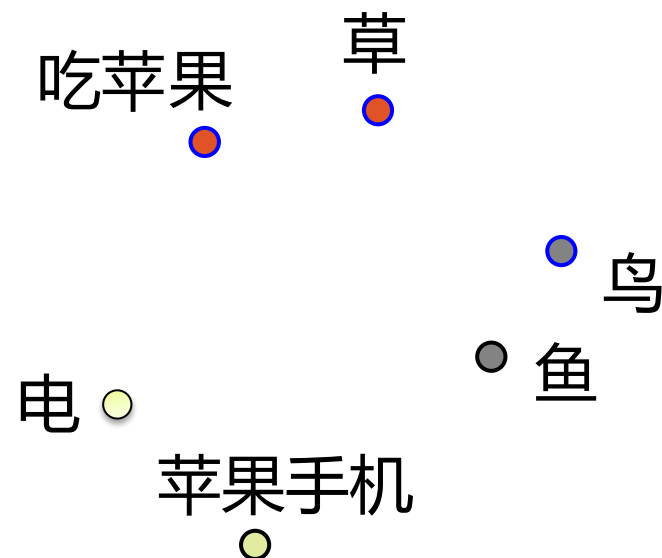
Objective	Inputs	Targets
Prefix language modeling	Thank you for inviting	me to your party last week .
BERT-style	Thank you <M> <M> me to your party apple week .	(original text)
Deshuffling	party me for your to . last fun you inviting week Thank	(original text)
I.i.d. noise, mask tokens	Thank you <M> <M> me to your party	
I.i.d. noise, replace spans	Thank you <X> me to your party <Y>	
I.i.d. noise, drop tokens	Thank you me to your party week .	
Random spans	Thank you <X> to <Y> week .	



为什么BERT有效?

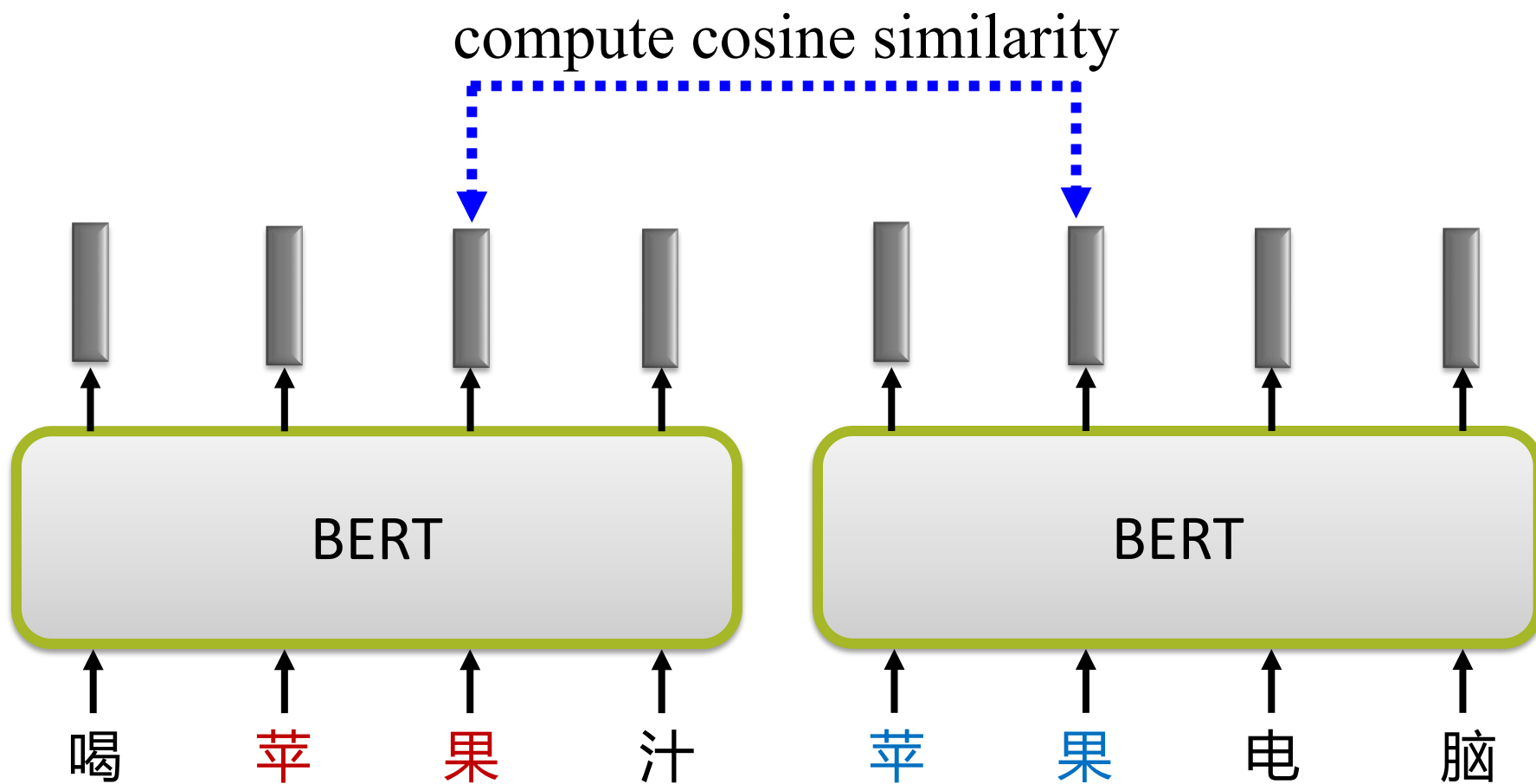


The tokens with similar meaning have similar embedding.



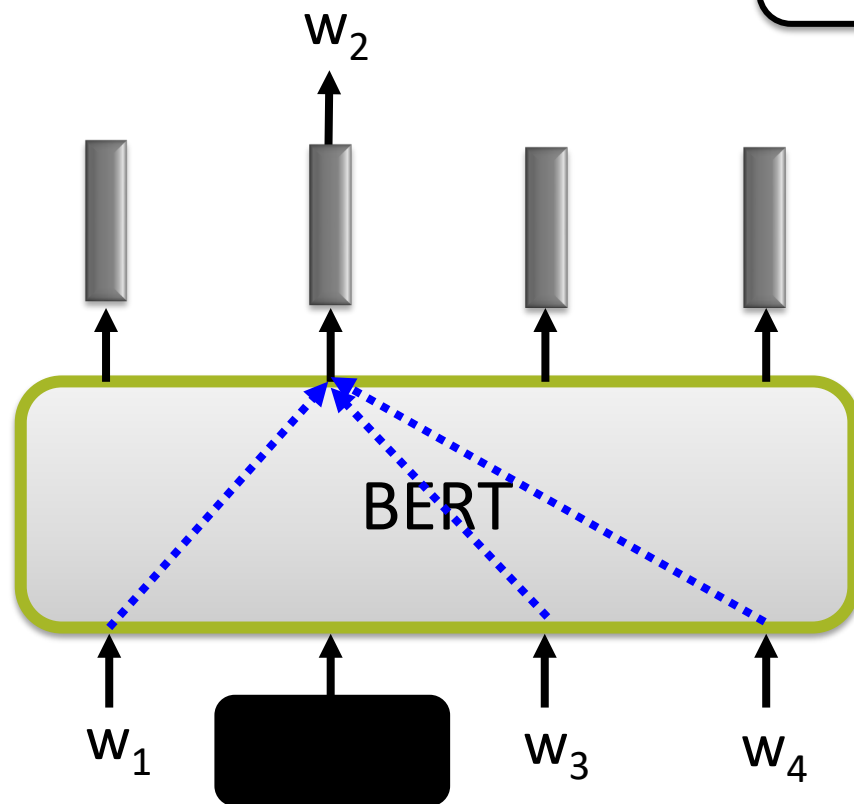
Context is considered.

为什么BERT有效?

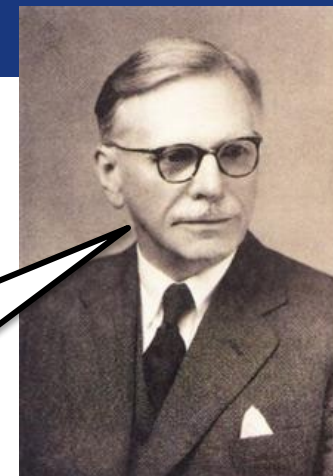


为什么BERT有效?

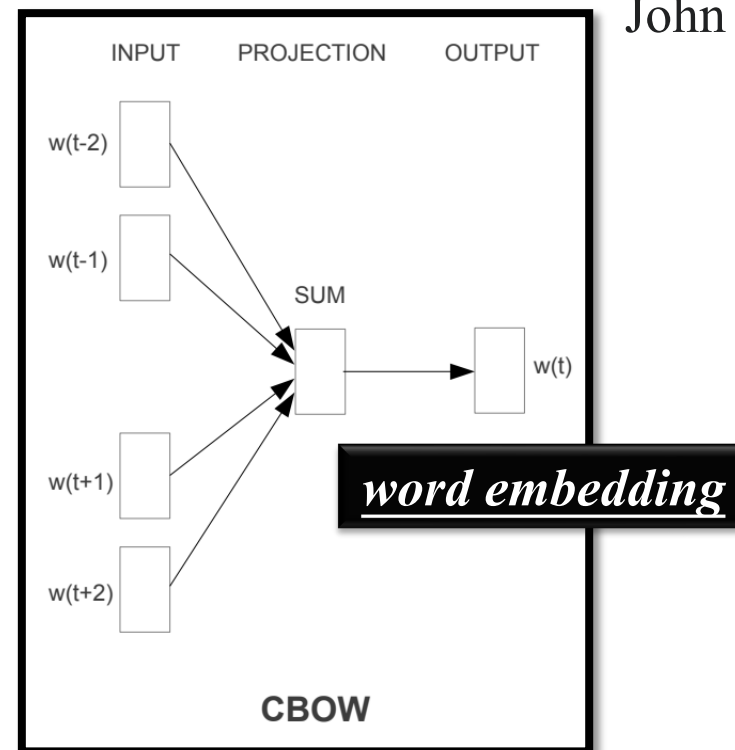
Contextualized word
embedding



You shall know a word by
the company it keeps



John Rupert Firth



为什么BERT有效?

- Applying BERT to protein, DNA, music classification



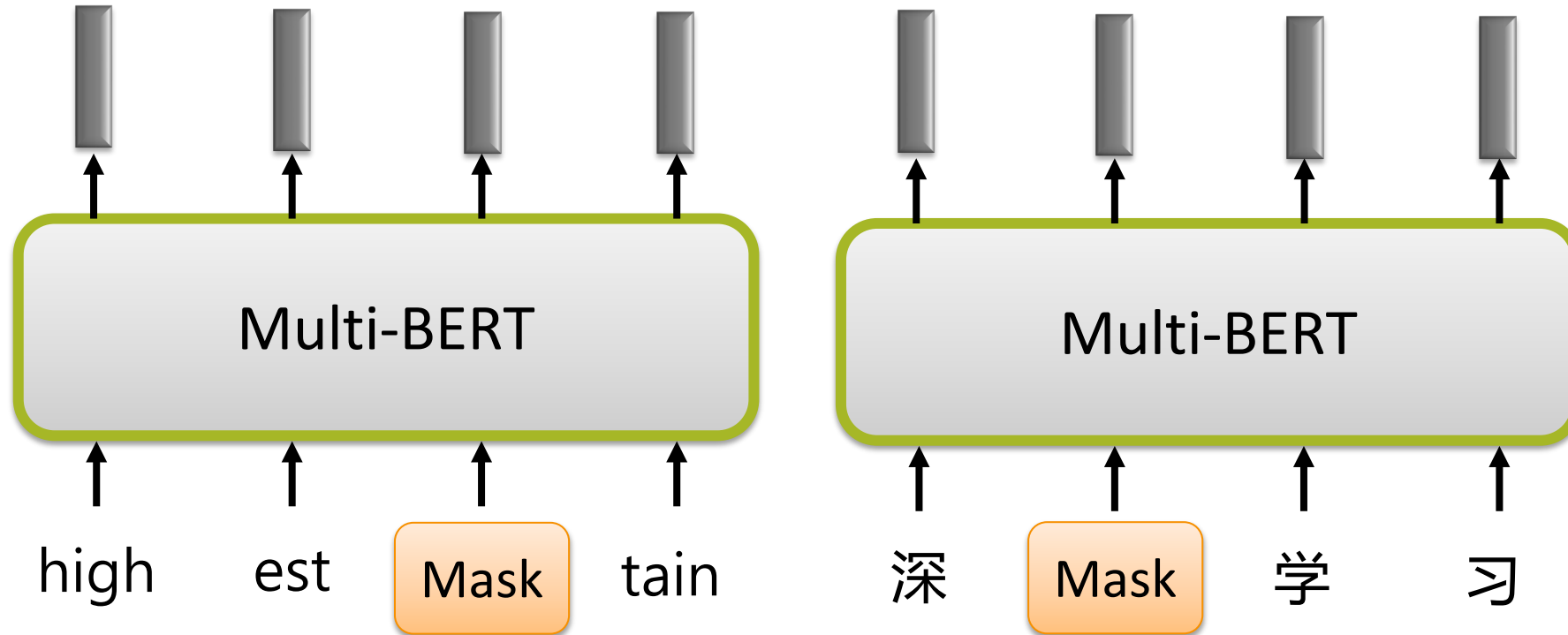
EI	CCAGCTGCATCACAGGAGGCCAGCGAGCAGGGTCTC
EI	AGACCCGCCGGGAGGCGGAGGACCTGCAGGGGTGA
IE	AACGTGGCCTCCTTGTGCCCTTCCCCACAGTGCCCC
IE	CCACTCAGCCAGGCCCTTCTTCTCCTCCAGGTCCCC
IE	CCTGATCTGGGTCTCCCCCTCCCACCCTCAGGGAGC
IE	AGCCCTCAACCCTTCTGTCTCACCCCTCCAGCCTAA
IE	CCACTCAGCCAGGCCCTTCTTCTCCTCCAGGTCCCC
N	CTGTGTTCAACCACATCAAGCGCCGGGACATCGTGCT
N	GTGTTACCGAGGGGCATTTCTAACAGTCTTCTTACTA
N	TCTGAGCTCTGCATTTGTCTATTCTCCAGCTGACCC
class	DNA sequence

为什么BERT有效?

- Applying BERT to protein, DNA, music classification

	Protein			DNA				Music
	localization	stability	fluorescence	H3	H4	H3K9ac	Splice	composer
specific	69.0	76.0	63.0	87.3	87.3	79.1	94.1	-
BERT	64.8	74.5	63.7	83.0	86.2	78.3	97.5	55.2
re-emb	63.3	75.4	37.3	78.5	83.7	76.3	95.6	55.2
rand	58.6	65.8	27.5	75.6	66.5	72.8	95	36

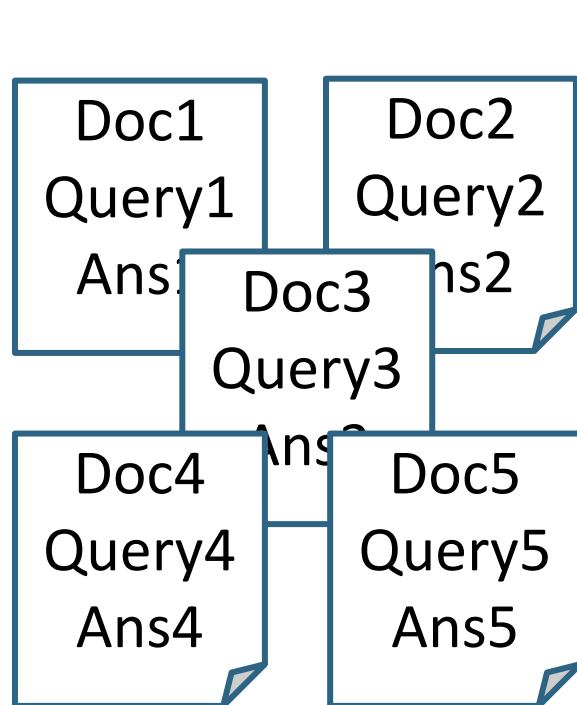
Multi-lingual BERT



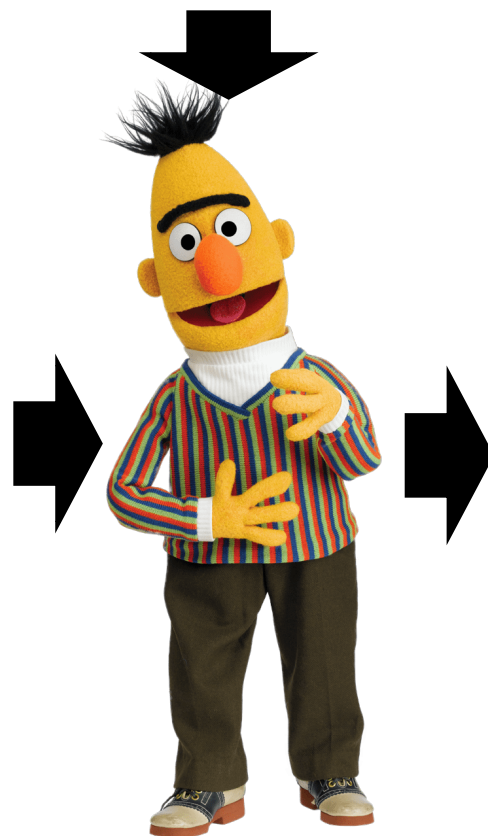
Training a BERT model by many different languages.

Multi-lingual BERT

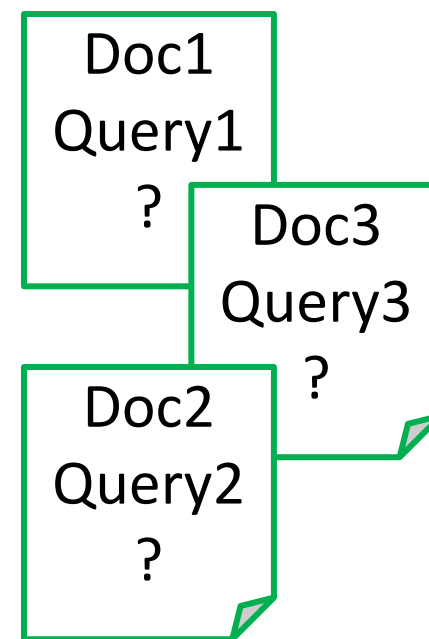
Training on the sentences of 104 languages



Train on **English** QA
training examples



Multi-BERT



Test on **Chinese**
QA test

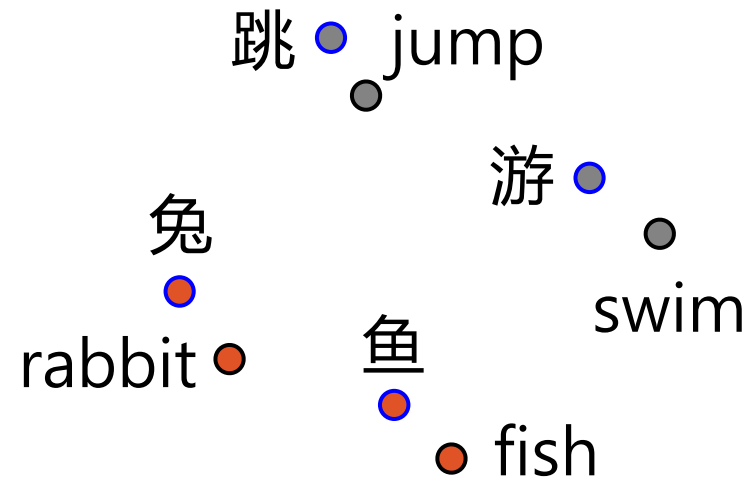
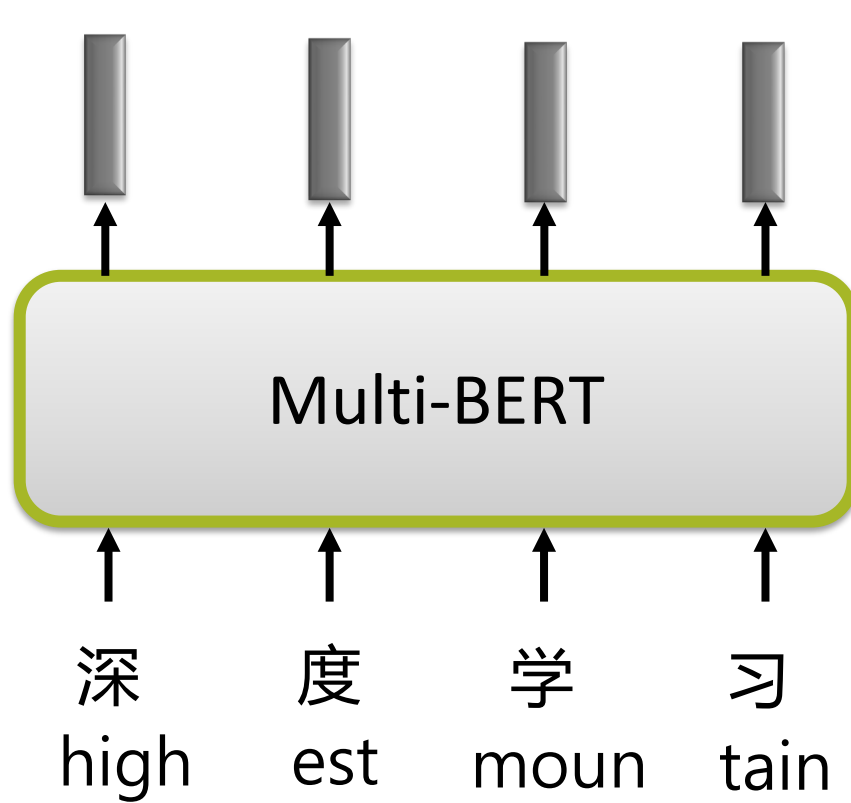
Multi-lingual BERT

- English: SQuAD, Chinese: DRCD

Model	Pre-train	Fine-tune	Test	EM	F1
QANet	none	Chinese	Chinese	66.1	78.1
BERT	Chinese	Chinese		82.0	89.1
	104 languages	Chinese		81.2	88.7
		English		63.3	78.8
		Chinese + English		82.6	90.1

F1 score of Human performance is 93.30%

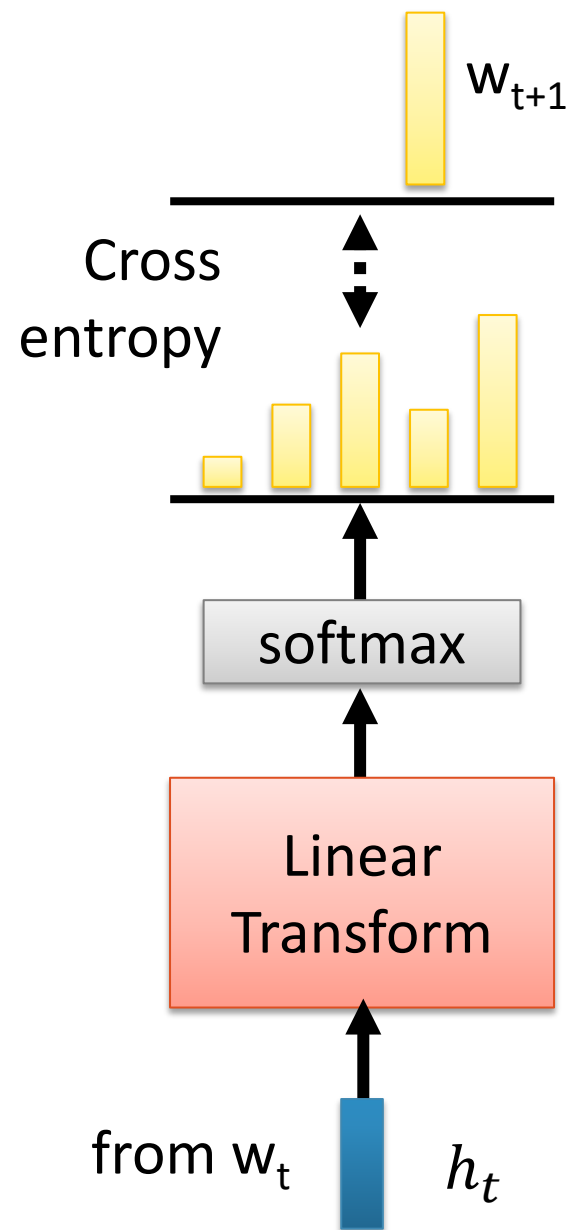
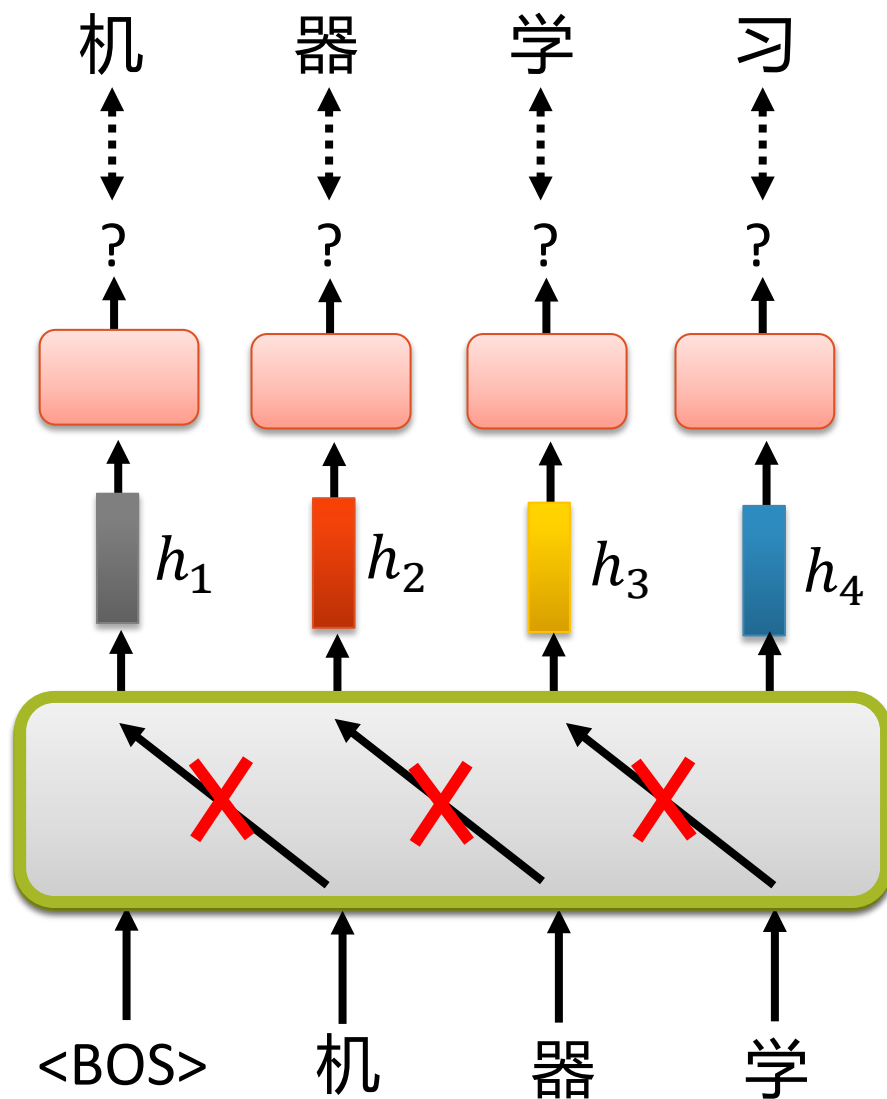
Multi-lingual BERT



GPT



预测下一Token



Training data:
“机器学习”

预测下一Token

SYSTEM PROMPT
(HUMAN-WRITTEN)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

MODEL
COMPLETION
(MACHINE-
10 TRIES)

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.



In-context learning

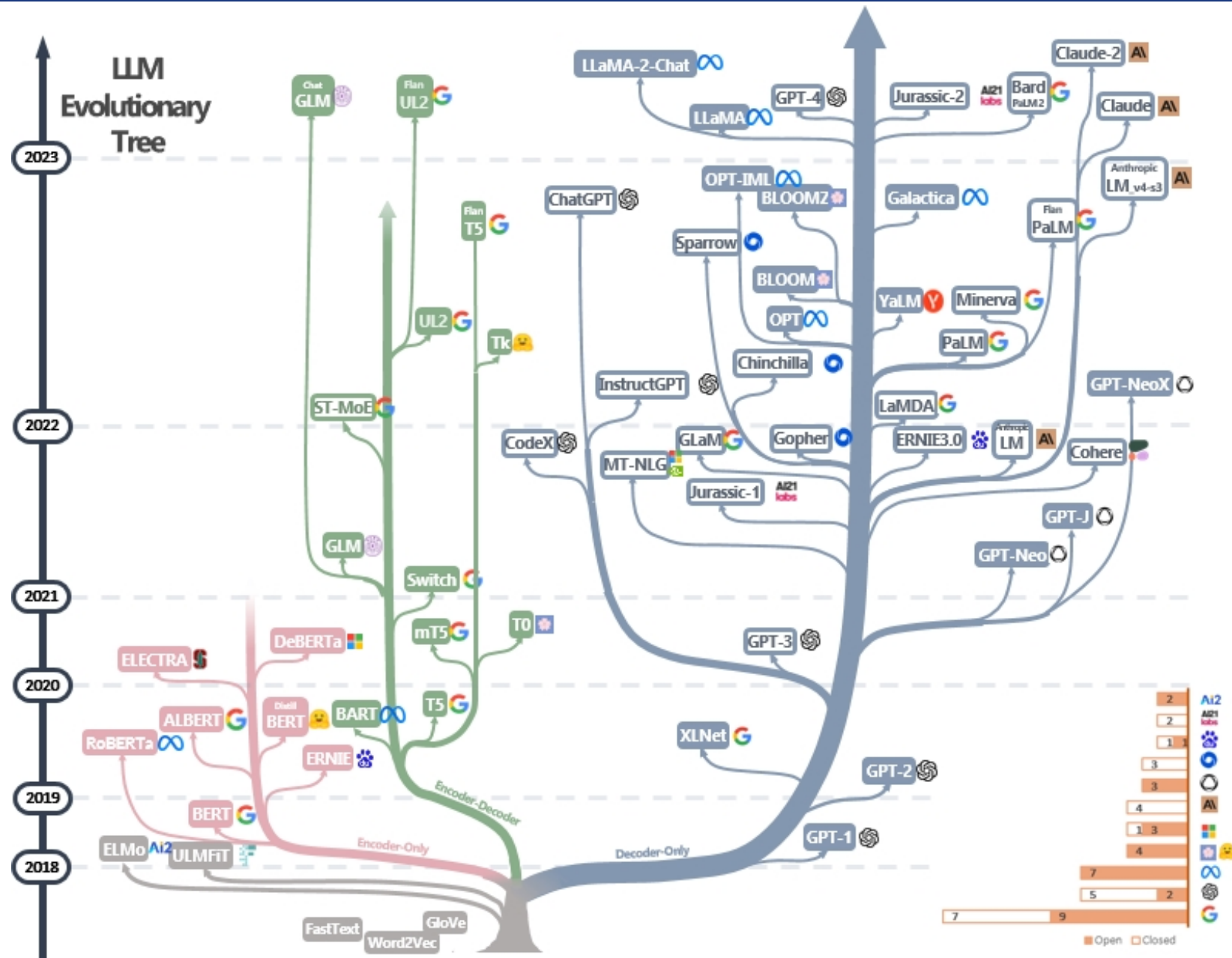
“Few-shot” Learning

(no gradient
descent)

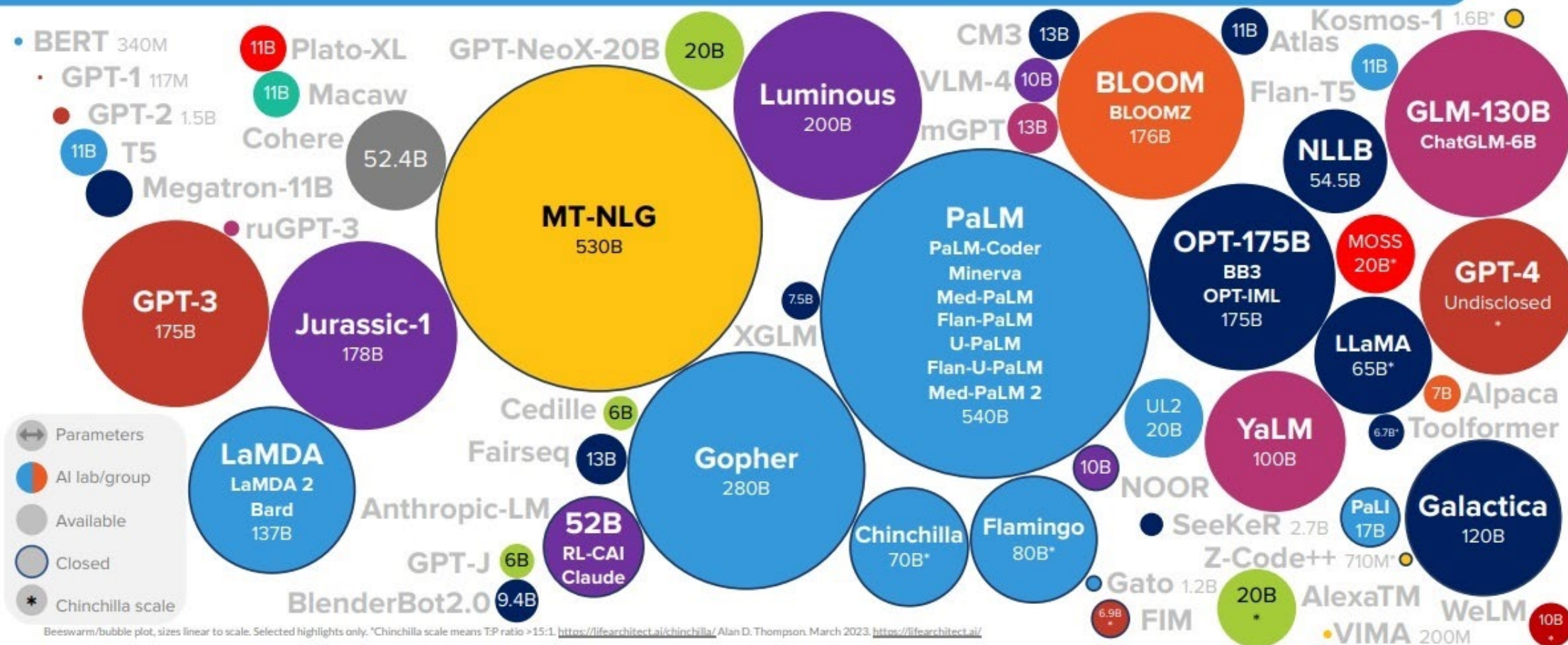
The diagram shows a light blue rounded rectangle containing a sequence of five lines. To the right of the rectangle, three labels with arrows point to specific lines: 'task description' points to line 1, 'examples' points to lines 2, 3, and 4, and 'prompt' points to line 5. The text inside the rectangle is as follows:

Line Number	Text	Annotation
1	Translate English to French:	task description
2	sea otter => loutre de mer	examples
3	peppermint => menthe poivrée	
4	plush girafe => girafe peluche	
5	cheese =>	prompt

大语言模型



LANGUAGE MODEL SIZES TO MAR/2023



Q&A